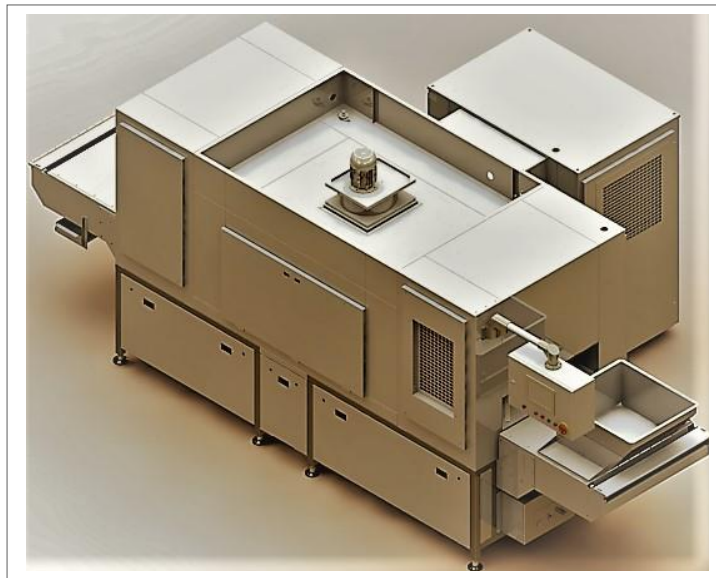




QMTI PROCESSING EQUIPMENT

TYPE : GP-15

Installation and Operation Manual



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Preface

This publication describes the Installation and Operations instructions for QMTI GP-15 machine. This section provides information on:

- The objectives of this publication
- Intended audience
- Prerequisites for using this publication
- Conventions and symbols used in this publication

DOCUMENT OBJECTIVES

The purpose of this document is to provide sequential instructions on proper installation, operation of the QMTI GP-15 machine. This also provides information on the technical specifications, and safety.

This document serves as a reference guide for safety, technical drawings, and dos and don'ts for operating the QMTI GP-15 machine.

AUDIENCE

This publication is meant for all the Engineers, Operators, and other staff of the company who will use the QMTI GP-15 machine.

PREREQUISITES

Prior experience in electrical and Food industry and knowledge about Food processing and drying process can be an advantage.

DOCUMENT CONVENTIONS

Item	Description
bold	Emphasis
<i>italics</i>	Cross References
Warning	Warning - A note of caution
	Used to indicate a Warning (above), except when the Warning is placed in a table
<u>Note</u>	Note - Additional information/Information requiring attention
	Used to indicate a Note (above), except when the Note is placed in a table
	Tip
	Yellow triangle signs indicate warning/danger symbols
	Red circle signs indicate forbidden/not to do symbols
	Blue circular signs indicate mandatory requirements

ABBREVIATIONS

This document uses the terms Machine and GP-15 machine or QMTI GP-15 machine interchangeably. In addition, the information presented in this User Manual uses the following abbreviations:

Abbreviation	Description	Abbreviation	Description

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CHAPTER 1. Introduction

This manual describes the QMTI GP-15 machine. The QMTI GP-15 machine is just one of a whole family of such devices supplied to industry for the processing, or removal of excess moisture from a wide range of manufacturers own products.

QMTI GP-15 works on the principle of dielectric heating and in this case the feeding of products through the GP-15, subjects them to a **direct** heating process in the form of a radio frequency energy source.

The HMI and PLC, provides the operator of GP-15 equipment with a powerful control system, which is also simple in use and hence easy to learn.

INTENDED USE

The machine described in this manual uses the principle of dielectric heating for the processing of food products which is to be presented to the machine as per the technical specifications agreed upon by the manufacturer and the user.

RESTRICTED USE

The supplied machine is not to be used for any other application unless specifically agreed upon between the manufacturer and the user

QMTI GP™ MACHINE HEATING PROCESS

The temperature of any material may be increased either by **direct** or **indirect** heating. The traditional process whereby heat is generated remotely and then transferred to the material by conduction, convection, or radiation, are all examples of **Indirect heating**.

Direct methods of heating are those where the heat is generated within the material itself, with radio frequency and microwave methods being typical examples. This method of heating is also referred to as **dielectric heating**.

Most non-metallic substances (or dielectric materials) undergo a uniform rise in temperature when subjected to a high-frequency electromagnetic field. Each time the direction of the applied RF field reverses, so the polarisation of the individual molecules within the substance changes. This constant molecular movement causes friction, which in turn leads to a rise in temperature. As may be expected, the higher the frequency of the electromagnetic field the faster the molecules process, thus creating more friction and consequently more heat is produced.

The frequency of the RF fields generated by devices for dielectric heating, are centred on 27.12 Mhz. These frequencies are effectively reserved for industrial, scientific, and medical use, to avoid possible interference with other legitimate users of the frequency spectrum. (All radio transmissions are governed by the Department of Trade & Industry, and specific frequencies are allocated to users dependant on their classification and requirements).

Most substances differ in their reaction to direct heating, with water for example being highly receptive to the application of direct heating methods. In practise this means that any product containing moisture (be it food, paper, textiles etc), can be rapidly dehydrated using the GP-15 machine. Individual applications will of course require different techniques, and QMTI utilise their wealth of experience when designing and commissioning their GP series machines in order to gain optimum performance.

During the QMTI GP series machine RF heating process, the product passes through the electrode system acts as the dielectric, which is then subjected to a powerful RF voltage at a frequency of 27.12 Mhz. The molecules of an ionic (or dipolar) substance such as water, readily re-align themselves with this high frequency electromagnetic field and become heated very rapidly. Non-ionic (or non-polar) substances on the other hand are not affected by this process.

The use of RF energy as a means of drying is highly efficient as only the water molecules become excited during the process, and energy is not dissipated throughout the mass of the product thereby causing unnecessary heating (or power consumption). In addition, the heating is uniform, causing the water molecules to become excited simultaneously. The use of RF energy as a drying medium also means that the inner working surfaces of the GP-15 (i.e., those coming into contact with the product) do not themselves require heating as a condition of the overall drying process.

Furthermore, the drying process is self-regulating in that the amount of moisture removed from a product is in direct proportion to the amount of RF power applied to it.

Conventional heaters give off energy in the form of heat whereby the outer surfaces of the device itself become heated during the process. This is not the case with RF heater hence external surfaces of RF generators however remain relatively cooler, thus providing a more comfortable working environment for operators. The operation and maintenance of GP series machines is a relatively simple task. The number of controls are minimised thereby enabling rapid training of operators.

When calculating the energy required to raise the temperature of a specific product, the following formula may be employed as a general guide : -

$$Energy = \frac{Mass \times (Final\ T - Initial\ T) \times Specific\ Heat}{863}$$

Energy = kW h

Mass = Kg

Temp. = °C

Specific heat = Kcal/Kg / °C

Incase of phase change we will need to add "Latent heat in Kcal/863" to the above

An allowance of 10-20 % should also be made for unavoidable heat loss.

During the GP-15 machine heating process, the product moves along a conveyor belt between two electrodes which act as capacitive plates, where they are exposed to RF power at a frequency of approximately 27.12MHz. The molecules of a dipolar substance (such as water), readily re-align themselves with this high frequency electromagnetic field and become heated very rapidly. Non-polar substances (such as starch etc) are not affected by this process and remain unchanged during the process.

The use of RF energy as a means of drying is highly efficient as only the water molecules become excited simultaneously during the process and energy is not dissipated throughout the mass of the product thereby causing unnecessary heating (or power consumption). The result is that since heat is developed in direct relation to the quantity of bi-polar molecules, or moisture content of the product, automatic profiling of moisture occurs. This moisture profiling effect is both accurate and repeatable and can be maintained, in some cases, within upto ±0.25%, with the added advantage of adjustment of the mean moisture level if so desired.

STRUCTURE OF THE MACHINE

The GP machine is built with metal structure. It consists of a generator, oven, conveyor, and environment management unit. The rectangle box shaped oven is the main unit where products are passed through for drying process. The conveyor located below the oven passes the products through the oven.

The generator unit, located on this model, behind the oven, supplies the required RF Power to the oven. The environment management unit has an air-cooling system. Its air extraction unit sucks the hot air out of the machine, avoiding condensation and building of water droplets inside the machine.

The air-cooling system is used to cool the internal parts.

The control panel interface is a separate unit and all operations of the GP machine are

performed using the control panel interface only.

HOW IT WORKS

The conveyor passes the products requiring RF heating through the oven. RF heat is applied to the product via a movable electrode assembly. The distance set between the two electrode assemblies determines the amount of RF energy applied to the product.

The speed of the conveyor is closely monitored by the PLC to retain the product inside the oven, for the required process time.

With the help of preset conveyor speed and RF Power settings, the products achieve the desired moisture levels when they come out of the oven.

This is a fully automatic operation using Preset/Recipe settings set in the control panel interface.

This is a simple explanation the working of the machine. Refer to chapter, Operating the Machine *on page 52*, for detailed and step by step operation.

CHAPTER 2. Safety

ALL personnel should be fully conversant with the information contained in this manual prior to operating or maintaining the machine.

The use of any radio frequency equipment having relatively high output power, involves the use of potentially lethal voltages

Operating the equipment with removed covers, open doors or damaged door seals will expose the operator to high levels of non-ionizing radiation.

Due to the very high voltages involved with the operation of the triode valve only a QMTI GP machine trained competent person is permitted to conduct live fault finding within the GP machine

Ensure only a competent person is allowed to perform service or maintenance tasks on the GP-15

Do NOT



The following are prohibitive and will lead to electrical hazard if not complied with.

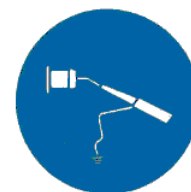
- Attempt to change components or make any internal adjustments with mains power applied to the GP-15.
- Use interlock or cut-out switches to isolate the power when accessing the GP-15 compartments
- Tamper with any interlocks - never attempt to defeat an interlock by short circuiting, removal, or other means. An interlock should not be relied upon to protect personnel from shock
- Look inside the R.F. duct aperture (tunnel) while the GP-15 is in operation
- Insert tools or objects into the R.F. duct aperture (tunnel) while the GP-15 is in operation
- Use excessive power for the product. The product will overheat and combust.



- Pressure wash, or use copious quantities of liquid to clean the GP-15
- Attempt to modify or make any internal adjustments to the R.F. circuits.

ALWAYS

- Fully isolate the GP machine prior to maintaining, servicing, or accessing compartments within the machine
- Ensure that all processing parameters are properly maintained within the GP machine design specification. For further information refer to the Appendix B Technical Description of the Machine
 - on page 131.
- Isolate ALL units (in case of multi-zone units) before servicing when multiple GP machines are installed.
- **Ensure that the supply cannot be reconnected except by authorised personnel.** (i.e., by removing fuse links or by removing the master key from the interlock). It is possible that certain components (i.e., capacitors), may retain sufficient charge even **after** the power has been removed as to be hazardous to health.
- Components that hold a charge after switching off are labelled adjacent to their location. Discharge with a suitable grounding probe before performing maintenance or servicing
- Ensure the product is suitable for R.F. heating before processing through the GP-15.
- Keep away from 'live' circuits - Do not attempt fault-finding on LIVE high voltage circuits.



GENERAL SAFETY

- This equipment incorporates automatic recycling which restarts the triode valve H.T. supply following a generator trip-out. **DISABLE BY SWITCHING OFF & ISOLATING BEFORE TESTING**
- It is important to both establish and observe, routine operating and maintenance procedures, to protect personnel and safeguard other plant. This manual is intended as a guide in the preparation and implementation of the customer's in-house safety, operating and maintenance procedures.



- As part of the in-house procedures, all operating, and maintenance personnel should read and fully understand the contents of this manual before working with the GP machine.
- Customers are strongly advised to develop their own operating, safety, and maintenance procedures for use with the GP machine. Before operating the equipment, always ensure that all national and/or local authority requirements pertaining to the country in question are fully observed.

SAFETY DURING INSTALLATION

The following requirements must be met, to satisfy the installation safety code.

- The equipment must be properly earthed.
- All electrical supply wiring must conform to local and national codes.
- The GP machine must be installed in an area large enough to freely accommodate operating and maintenance personnel.
- The chosen area must have adequate lighting and ventilation.
- Suitable lifting aids should be available during the installation of heavy components i.e., H.T. transformer, etc.



SAFETY INSTRUCTION CHECKLIST

Do not allow personnel to operate or maintain the equipment until they have familiarised themselves fully with the information provided in this manual.

Only properly trained, competent, and authorised personnel should be permitted to operate, maintain or service the equipment.

Do not operate the equipment unless all covers, guards, doors, and electrical systems are present and in full working order.

Before performing any maintenance, always isolate the machine fully.

Follow all national and local regulations, together with any environmental regulations that may apply to the operation of this equipment.

Should there be a need to work above the machine adequate fall protection must be provided. This can be in the form of a restraint harness, enclosed catwalk with railings, or an elevated safety platform.

QMTI representatives are available to answer any customer queries regarding the safe operation and/or maintenance of the GP machine. (Contact your local QMTI representative should you require assistance).

SAFETY OF OPERATORS AND SERVICE PERSONNEL

The highest priority for any individual performing his/her job is to safeguard their personal safety. This should not only be a subject for discussion at safety meetings but should be always borne in mind throughout the working day. A single moment's loss of concentration or careless act can have serious consequences not only for the individual concerned, but for others closely associated with that person.

The following recommendations should therefore be regarded as the minimum requirement when working with such equipment:

- Avoid loose fitting clothing or the wearing of ties, hair bands, rings or watches when operating machinery. Tie back long hair or wear suitable head protection.
- Always use the safety equipment provided, i.e. wear safety glasses in an environment where airborne particles could be injurious to the eyes. Use safety goggles and/or shields where they may be a danger from splashed chemicals. Special clothing i.e., coveralls, hard hats, protective gloves, safety boots, ear protection may be required in some areas. Always ask your immediate supervisor what safety equipment is required for the job in hand.
- Know your work area. Keep your work area clean and free from possible hazards that may cause injury through a fall etc. Always wipe up any oil and grease spills that may occur; do not leave them for someone else to do.
- Do not alter the mechanical or electrical components of the equipment in your charge unless you are specifically authorised (and qualified) to carry out the modification.
- Always lock-out and tag any equipment before adjusting or performing routine maintenance. Do not rely on interlock switches for protection.
- Customers should develop and maintain a specific safety programme for the GP-15.
- Anyone wearing a heart pacemaker should be warned that HF (RF) equipment is present.
- Pregnant women should also be warned about the presence of non-ionising radiation.
- Report any hazards that come to your attention immediately to your supervisor. If in doubt always ASK.



- Know the location of any safety equipment i.e., fire extinguishers, emergency showers, eyewash station, first aid kit etc.

SPECIFIC SAFETY PRECAUTIONS

Hazards which may be encountered while operating or maintaining the GP machine are summarised below:-

- Health hazards arising from the use of cleaning agents or chemicals.
- The product become overheated to a point where combustion results
The GP-15 must be fully isolated before removing overheated product from the applicator area. Only CO2 fire extinguishers can be used. It is essential that overheated product does not remain inside the machine with the conveyor stationary.
- Danger of burning, severe shock, or fatal injury when service personnel gain access to a live compartment or fail to discharge high voltage components before handling. Always ensure the machine is fully isolated before entering any compartment within the GP machine.
- Open only those compartments necessary for the task in hand, and before work commences discharge any high voltage points, components, terminals etc using a suitable grounding probe. Wear suitable eye protection to avoid the possibility of 'flash' blindness.



- Scalding or burning during operation OR maintenance, due to contact with:

Hot product

Exhaust duct

Oven Doors

- Components inside the GP machine - Hot water remaining in the cooling system.



- Exposure to high noise levels caused by air-moving fans, etc. When operating (or testing) the GP machine, wear suitable ear protection
- Heavy oven doors. Ensure doors are supported fully when opening. Do not allow to drop.

H.T. DC GROUND POINT

The high-tension DC return (0 Volt supply) is permanently bonded to the machine earth or chassis by means of a stud bolt located inside the main contactor compartment.



WARNING: DO NOT UNDER ANY CIRCUMSTANCES DISCONNECT THE SAFETY EARTH!

TRAINING

Thorough and adequate training of all operating and maintenance staff is vital to the safe use of this equipment.

Do not allow any personnel to operate or carry out maintenance on the GP machine, unless they can demonstrate a full understanding of the equipment together with its safe operation. Personnel should receive regular training and should be tested in order to maintain a high level of proficiency and effectiveness.

Operator training should cover the following basic topics: -

- Basic understanding of GP machine operation
- Functions of control and safety devices
- Understanding fault finding
- Operator awareness of potentially hazardous operations

CHAPTER 3. Technical Information

This chapter illustrates the technical details of the RF Machine using diagrams and tables. The various parts of the machine are also illustrated, and their function explained.

The representative image of GP-15 machine machine is illustrated next, with its components marked.



NOTE: The parts and image shown is representative and the actual machine components may vary based on your order customisation.

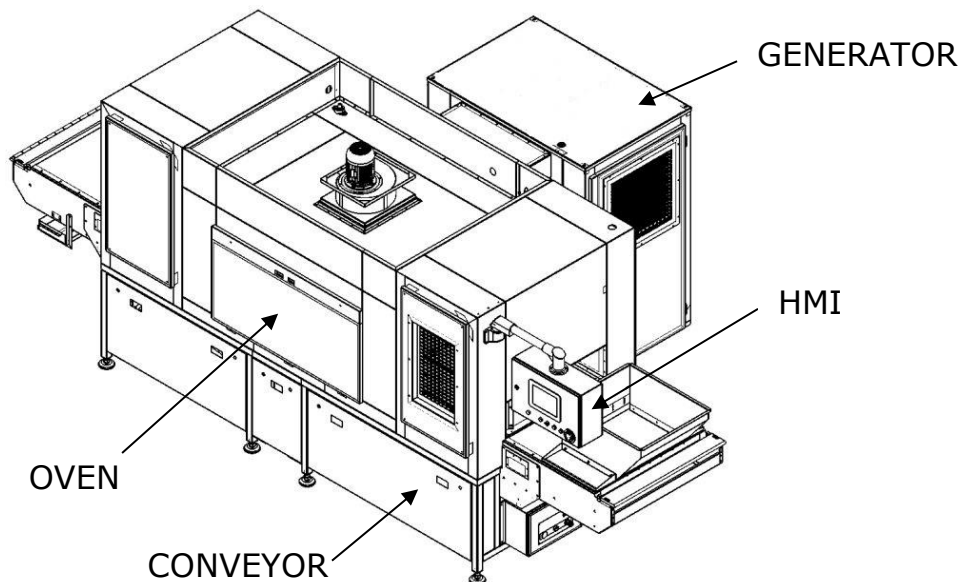


Illustration 1: GP-15 machine – Representative Image

PHYSICAL DESCRIPTION OF PARTS

The parts of the GP-15 machine are:

Generator

Oven

Conveyor

Control Console (HMI Interface)

Environment Management Unit

GENERATOR

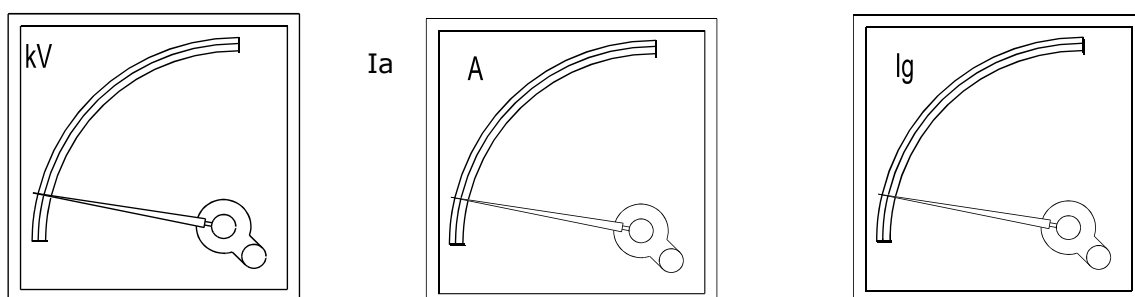
Generator refers to the RF oscillator and the ancillary electrical control systems. The RF oscillator within the GP machine provides a source of electromagnetic energy oscillating at 27.12MHz. The generator usually has two decks with one deck housing the resonator (oscillator circuit), oscillator valve, and capacitors.

The other deck consists of H.T. transformer, blower, oscillator filament transformer, rectifiers, and contactors. Incoming electrical mains supply is also connected here.

Analogue meters are fixed outside the generator unit to show grid and anode current. The generator can be divided based on three basic functions:

Analogue Meters

The machine is provided with three analogue meters which are located on the door of the main transformer compartment. These meters provide an indication of the status of the equipment by displaying the three fundamental operating parameters; H.T. (high tension) voltage, anode current, and grid current.



kV: This meter indicates the High-Tension DC supply voltage applied to the Anode of the triode valve. The kV level is set during installation and only requires alteration during installation & commissioning in another location, or when 'burning-in' a new triode valve.

Ia: This meter indicates the Anode current (DC) flowing through the Anode of the triode valve in Amperes. It is directly proportional to the KW of RF energy applied to the product.

Ig: This meter indicates the Grid current (DC) flowing through the Grid of the triode valve in Amperes. It gives information on the operation and condition of the triode valve. The grid current is set to suit a specific valve.



WARNING: Alteration of the grid current value must be carried out only by QMTI service personnel.

The anode high tension supply (V_a) is nominally 9.5kV DC which is supplied by the H.T. transformer. The secondary output of the transformer is full wave rectified on all three phases by a rectifier stack of three solid state avalanche diode assemblies. A low-pass spike filter network is incorporated between the rectifiers and the oscillator as a safety circuit.

OSCILLATOR

This GP-15 machine has an oscillator circuit, made of Aluminium and high conductivity copper parts. The 'tank circuit' essentially forms an LC resonator which couples with the load circuit in the oven through transformer action. The moving electrode assembly in the oven works as a capacitive tuner to couple the load into the Oscillator circuit

Safety:

There are high proportions of aluminium components within the cavity area which are part of the 'active' circuitry and are therefore electrically 'live'.

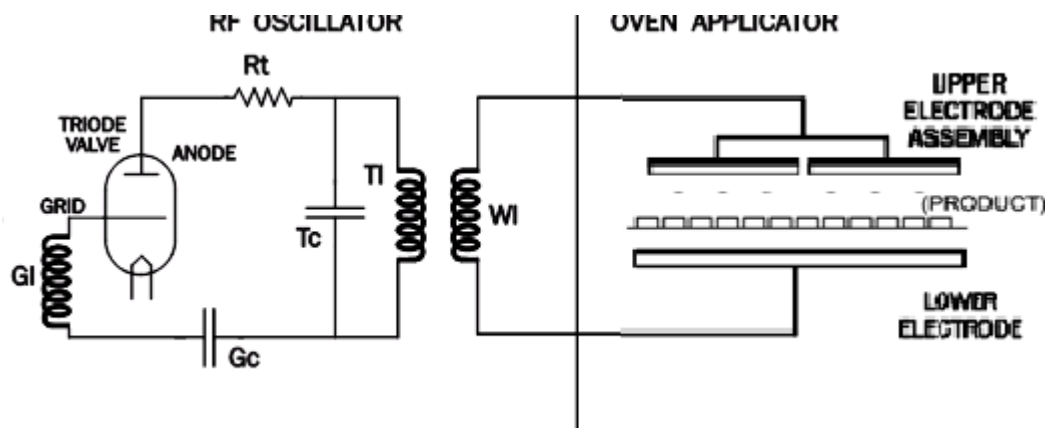


Illustration 2: Oscillator Electrical Circuit

OVEN

Oven is the area where the RF energy is applied to the product. Therefore, it houses the applicator electrodes and called the Applicator. The conveyor belt passes through this area, moving the product.

RF POWER CONTROL

The actual RF power required to process a particular product may be determined with knowledge of the products moisture content and then in practice by 'fine tuning' the power output of the GP-15 machine.

Alteration of the RF power generated by the GP-15 machine is determined by altering the distance between the applicator electrodes. As the distance (gap) between the fixed lower electrode and the moving upper electrode increases, so the power applied to the product decreases.

Vertical movement of the electrode plates is controlled by means of a closed loop circuit. The electrode position is obtained via an optical encoder mounted directly to the output of the gearbox which mechanically moves the plate. Protection against faulty alignment or faulty components in the system is provided by two independent upper and lower limit switches linked to the electrode support frame by means of a control arm.

When mains power is applied to the machine an electrode homing sequence is initiated. This process is described in **Homing the Electrode on page 56**.

The operator has control over the electrode position using the touch-screen display, either directly or from one of the products Preset/Recipes where the electrode travels to the indicated pre-set position when the RF power is switched on.

Using the formula and graph in on page 44 Section, the operator should be able to determine an approximate power level necessary for the product. Having established the power level, the GP-15 machine is switched on and the electrode height lowered until the calculated power level is reached. The product is then sampled, and the power level adjusted according to the requirements of the customer. The conveyor band speed should be adjusted always to present the maximum possible amount of product under the electrode – whether in monolayer or preferably as a bed of product if at all possible. In this way the voltage across the product for a given power level (and therefore the chance of arcing or flashing) is minimized.

Safety:

The door interlock system in the machine ensures that the oven doors cannot be opened without switching off the unit. Personnel should always wear safety protection for hand and eyes.



CONVEYOR

The conveyor is taken as being the whole base assembly which provides a framework upon which the applicator is located. The input to the machine is termed the in-feed and the output is referred to as the out-feed. In addition, of course, the conveyor provides guidance and support for the conveyor belt itself.

Attenuation Ducts are located next to the conveyor. These enclosed areas are created at the in-feed and out-feed surrounding the conveyor belt, which are carefully designed to prevent RF leaving the machine.

CONTROL CONSOLE UNIT

The Control Console Unit houses the Control Console/Human Machine Interface (HMI). This touch screen panel is located as shown in the outline drawing in the Engineering Drawings Appendix in the Engineering Drawings Appendix of the *Manual*

A touch-screen Operator Display (HMI) provides the interface with the Programmable Logic Controller (PLC). The display allows the operator to input data for GP-15 machine settings and control. It is here that alarms are displayed and logged to provide a history of faults. (Refer to Troubleshooting in the Manual)

ENVIRONMENT MANAGEMENT UNIT (EMU)

The EMU is responsible for maintaining the temperature of the various internal parts as well as in the oven chamber. This is effectively done by cooling the internal parts using Air, as well as removing the moisture inside the oven chamber due to RF heating using Air circulation. There are Two systems that provide these functions.

- Air cooling
- Product Air Circulation

These are collectively termed as the Environment Management Unit. They are located adjacent to the Generator, on either side.

AIR COOLING

Forced air circulation reduces the internal temperature within the generator cubicle. Air is circulated by means of a centrifugal fan fitted on the top of generator and connected to the RF cavity in the oscillator compartment.

PRODUCT AIR CIRCULATION

To remove the moisture removed from the product, and to compensate for a wide variety of operating parameters, it is necessary to provide air circulation around the product.



NOTE: This is separate from the cool air required to remove excess heat generated by the working areas within the GP-15 machine described in the previous section.

Under normal operating conditions, the area within the GP-15 machine (including the conveyor belt and the product), is always subject to warm air circulation even when RF heating is not applied. Maintaining the ambient temperature within this area avoids water condensation in the Oven chamber

Ambient air is drawn into the GP-15 machine through washable filters. It passes over a series of heating elements and forced into the main processing area containing the RF electrode assembly by a blower fan. The air is then drawn over the product surface and exhausted out via an extraction fan mounted at the top of the oven of the GP-15 machine. The illustration below shows movement of air in the GP-15 machine.

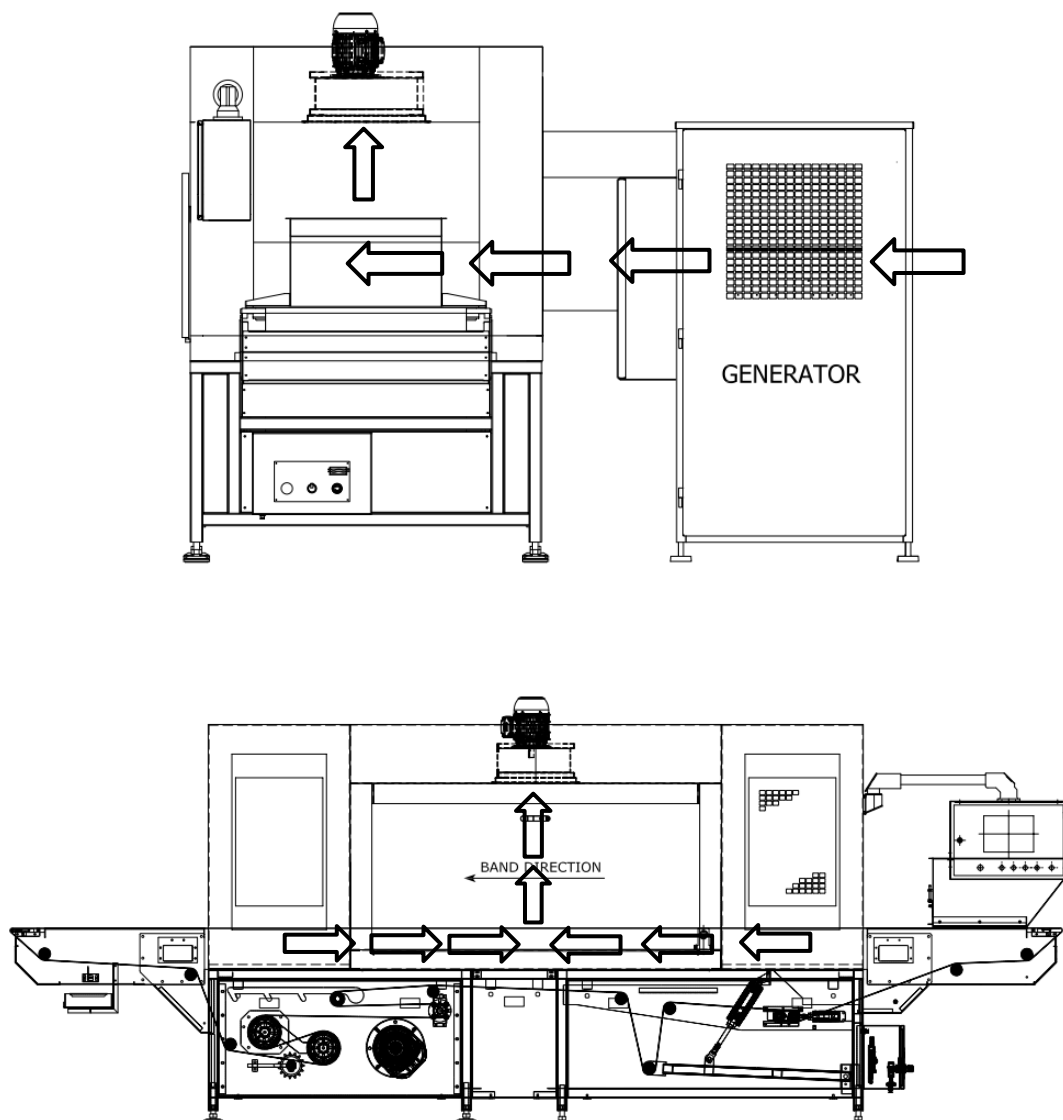


Illustration 3: Representative Image of Air Circulation on Product

CHAPTER 4. Installation

The next step is to install the GP-15 machine and perform a trial run. This chapter explains the steps required from unpacking to installation and running a trial.

Safety

Refer to Safety during Installation on page 16 before proceeding with this chapter.



UNPACKING

The GP-15 machine must be unpacked and installed as per instructions given in this chapter. Proper set-up and installation of the machine will ensure its efficient working and support from GP-15 machine.

The GP-15 machine is shipped mounted on MS channel frames. Certain loose parts like capacitors, coils etc. are packed in corrugated boxes and placed inside the structure of the machine. The exact location of the boxes and the contents depend upon the type of machine on order. Please refer the packing note sent along with the machine for further details.

Depending on the size of the machine, the two frames could be shipped in one container or in two separate containers. Machines made of more than one zone would have a slightly different shipping configuration. In general, the machine is disassembled and shipped as follows

Shipping Frame A:

- Generator
- Hopper

Shipping Frame B:

- oven
- conveyor stands

Shipping Frame C:

- Conveyor Extensions
- Wooden Box of Transformer

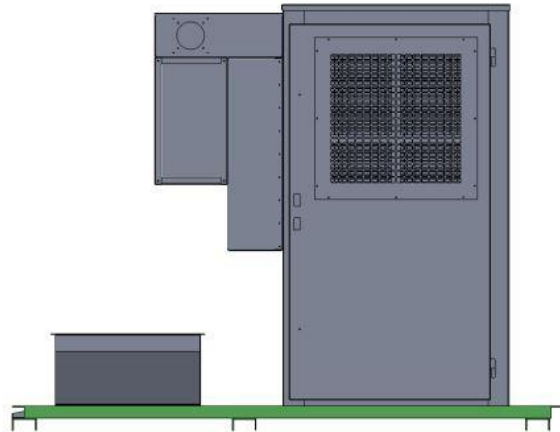


Illustration 4: Shipping Frame A

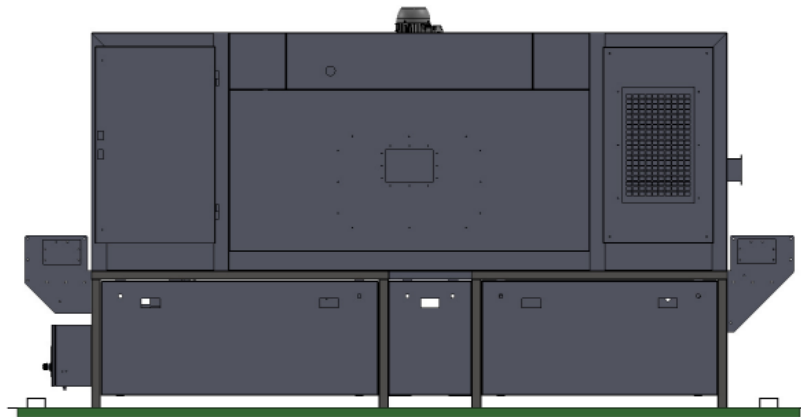


Illustration 5: Shipping Frame B

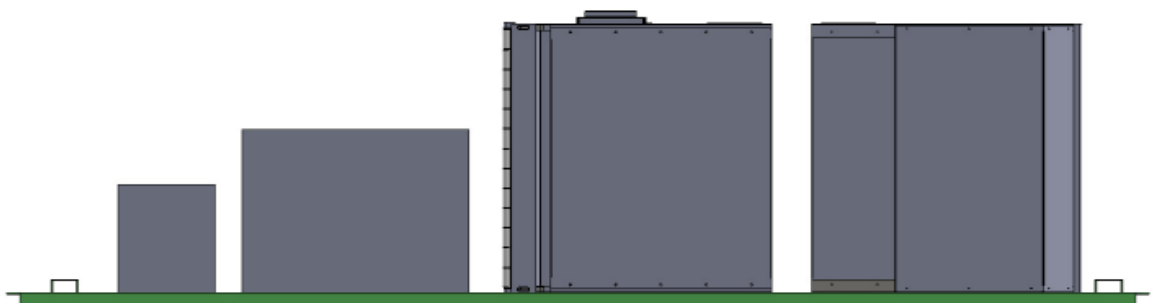


Illustration 6: Shipping Frame C

LIFTING AND HANDLING

All safety and warning instructions must be followed before unpacking and handling.



WARNING: The assembled GP-15 machine weighs approximately 1500kg per zone. If using a crane, ensure that suitable lifting methods are employed as shown in the figures below.

Safety

NEVER LIFT THE ASSEMBLED RF GENERATOR AND GP-15 machine BY MEANS OF THE LIFTING EYES

Raise and lower the assembled machine carefully. Follow all safety procedures and guidelines



EQUIPMENT REQUIRED

- An overhead crane suitable for a 5000 kg load
- A forklift suitable for a 5000 kg load
- Polyester (high tenacity) of 5000 kg SWL capacity for 2.5% elongation and 6-meter length
- Protective sleeves for the slings or Anti Abrasion padding where the slings contact the machine's structure to avoid damage to the surface.

PLACING THE UNIT

- The unit must be placed in position and the electrical-mains supply cables connected before installing the Oscillator Triode valve.
- Space must be provided to open the doors
- The support base must be flat and level
- The atmosphere must not be damp or excessively oily
- Steam pipes or other sources of heat must not pass close to the RF Generator
- Excessive vibration must be avoided
- Ambient Temperature must not exceed 43°C



NOTE: Customers must supply and fit their own mains isolator between the ac supply and the GP-15 machine

Safety

BEFORE SERVICING OPERATIONS COMMENCE, THIS ISOLATOR
MUST BE LOCKED IN OFF POSITION



All cables and electrical components (i.e., the customer supplied isolator) must be capable of carrying the full load current of the RF generator as well as any auxiliary equipment supplied.

LIFTING THE UNIT

Suitable methods must be employed when lifting an assembled unit, i.e., lifting straps, forklift truck, chain block, etc.

Safety



When lifting the unit using flat slings, ensure that the:

- Angle between diagonally opposed legs is $\leq 60^\circ$
- Working Load Limit (WLL) of the lifting equipment is not exceeded.
- Load is divided equally between each leg.

PLAN VIEWS OF ROOF SHOWING
LIFTING EYE ORIENTATION

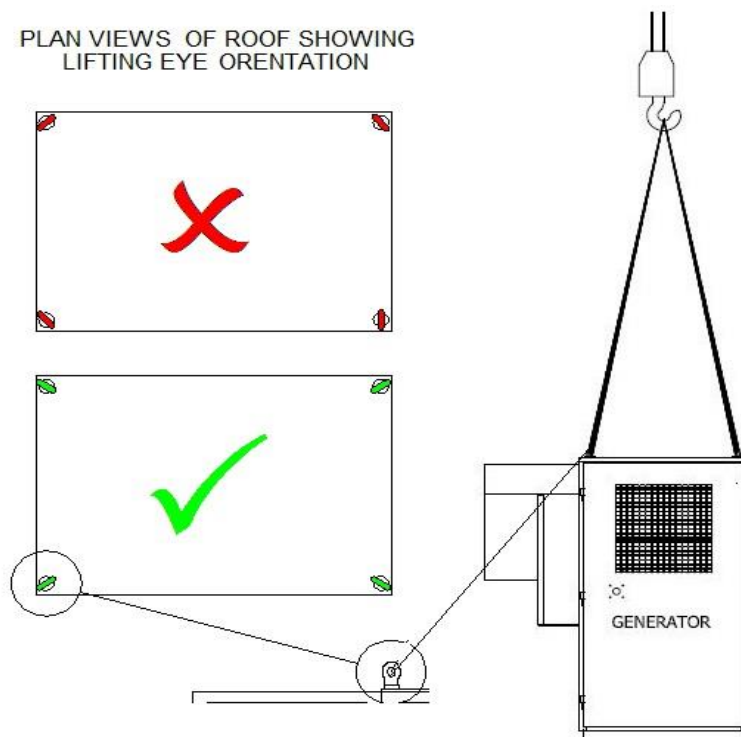


Illustration 7: Lifting the Generator Unit using Eyebolts

LIFTING THE CONVEYOR AND GP-15 UNIT

When using slings ensure that

- The unit is lifted from underneath
- A spreader bar and suitable supports are placed under the unit.
- Packing must be placed at the sides to avoid damage being caused by the slings.
- First place the conveyor assembly on the floor and place the GP-15 on the conveyor.

This is illustrated next.

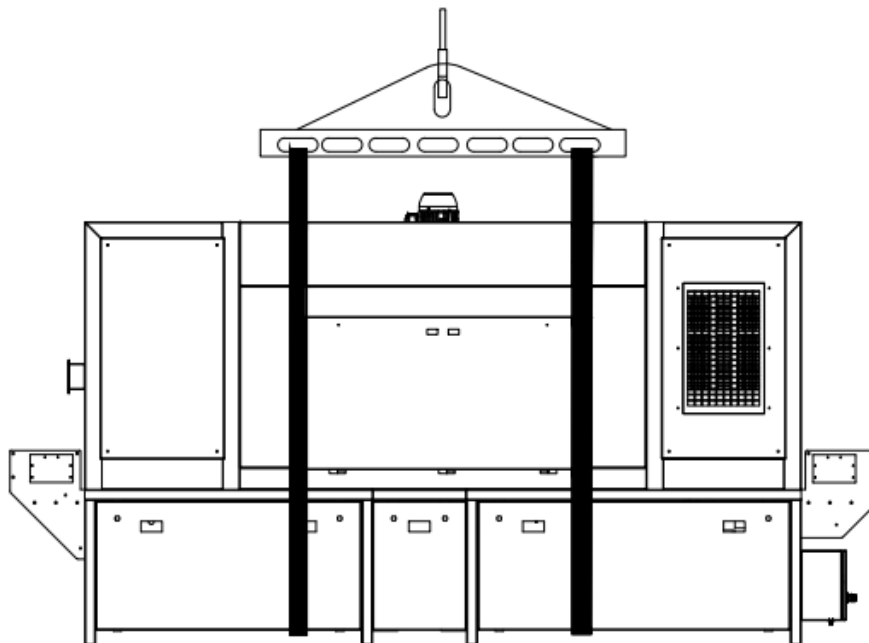


Illustration 8: Lifting the Conveyor and GP-15 Unit

PRE-INSTALLATION REQUIREMENTS

AREA

Well ventilated space, preferably a separate room, free of mechanical vibrations, dust, fumes, suspended particles like lint, fiber, fluff etc. The size of the actual machine will be approx. 5.5 x 2.9 x 2.2 Mtrs. (L X W X H). At least 2 meters of space should be kept free around the periphery of the machine to allow opening of all doors and for easy access during maintenance



NOTE: Please refer the Layout Drawing associated with this machine model

ACCESS CLEARANCE

Please provide at least 3.0 meters height and 3.0 meters of width at the doorway to get the equipment into the required location

ENVIRONMENT

The machine should be installed in a clean chemical and dust free environment with ambient temperature not exceeding 40C (except occasionally)

VIBRATIONS/FOUNDATION

No foundation is required for the machine, but levelled concrete flooring is a must. The total weight of equipment will be about 2550 Kgs. It is advisable to have anti vibration pads between foundation bolts and floor if the equipment is not installed on the ground floor or the location is prone to excessive vibration, the pads can be supplied at cost by QMTI.

STABLE VOLTAGE

The equipment must be provided with an electrical connected of AC 3 Phase, 60 Hz, 600Volts through a dedicated servo voltage stabilizer as per the following specification.

CAPACITY	40 kVA, 3 Phase, 60 Hz
TYPE	Unbalanced
INPUT VOLTAGE	560V to 640V
OUTPUT VOLTAGE	600 V with variation of +/- 1%
CORRECTION RATE	20 V/sec
COOLING	Oil Cooled
INDICATORS	Analog type Voltmeter and Ammeter for Input and Output
ADJUSTMENTS	Manual and Auto for Voltage



No microprocessor controls and instantaneous or magnetic trips

Input voltage trips should not be provided, however audible alarms for undervoltage may be provided if so desired by the user

MAINS ISOLATER

An isolator rated for 60 A or more, of a reputed make should be installed

EARTHING

Atleast 2 separate and dedicated earthing connections thru dedicated earth pits to be connected to the machine with 25 X 3 mm copper strips. The earth pit should be of less than 1 ohm resistance to be connected at length of the machine).

POWER CABLE

A suitable 3 ½ core 20 sq. mm cable of Copper conductor will be required for completing the electrical power connections

ASSEMBLING THE GP-15 MACHINE

Remove all packing material and loose parts stored in the oscillator compartment of the generator and inside the oven

Run interconnecting cables using suitable conduit. Secure cable terminations to the required locations being careful to observe the numerical identification provided on both cables and termination points. A blank gland plate is provided so that the customer may fit suitable cable glands during the installation. Refer to the engineering drawings provided in the Engineering Drawings Appendix of the manual.

Make all of the necessary cable connections between the electrical panels, PLC etc., and any remote connections required.

INSTALLING THE HT TRANSFORMER

Install the H.T. transformer in the compartment marked as HT Transformer compartment in *Illustration 25: General Arrangement showing compartment references* with the tappings facing the door opening. Make certain that the transformer primary taps are set to the local supply voltage.



WARNING: The H.T. transformer weighs 200Kg

Safety

Always exercise proper care and handling precautions when installing this component. The use of a fork-lift truck or similar should be considered



INSTALLING THE CONVEYOR

Align and level the input end of the conveyor.

Assemble and bolt together the remaining support frames, except the exit (delivery) end. Ensure that each section is leveled following placement.

Lift Generator/GP-15 machine assembly into position on the assembled conveyor.

Manoeuvre the exit (delivery) end unit into position against the rest of the conveyor assembly, making sure that the complete conveyor is now aligned and level.

Proceed with the installation of all services, i.e. mains electricity, compressed air etc.

Continue with the remaining installation of the conveyor belt and associated parts, fitting the conveyor belt in accordance with the belt manufacturer's own manual provided.



NOTE: The belt has a top and bottom surface; top to carry the product. Routing should be as shown in Illustration 13: Belt Path

CONVEYOR BELT ASSEMBLY

FOLLOW THE PROCEDURE OUTLINED BELOW WHILE ACCESSING THE GRAVITY ROLLER FOR BELT INSTALLATION. IMPACT FROM THE ROLLER CAN CAUSE PERMANENT INJURY

1. Compressed Air Supply has been provided to the Gravity Roller Pneumatic Assembly. The required pressure is 4 bar
2. Use the pneumatic switch shown below to activate the cylinders and lift the gravity roller upwards
3. Insert a locking lever (a wooden block) shown in the picture below to hold the roller arm in place.
4. Turn off the pneumatic switch to allow the roller to move downwards till it rests on the locking lever as shown in the picture.
5. After installing the Belt as outlined in the next sub heading is complete, turn on the pneumatic switch to lift the roller & remove out the locking lever.

6. Release the pneumatic switch to allow the gravity roller to move downwards and rest on the belt.

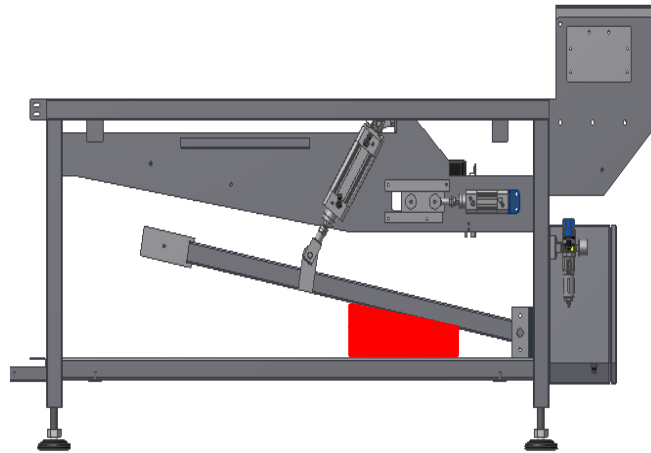


Illustration 9: Gravity Roller rested on Locking Lever

INSTALLING THE CONVEYOR BELT

The conveyor belt is constructed from a plain reinforced Teflon material, providing stability in use and extended life. The belt can be joined by means of inserting a wire like component between the interlocked spiral ends from both open ends of the belt.

Incase the belts are not fixed with the spiral ends then the conveyor belt that is received along with the equipment, which is longer than the required working length, needs to be wrapped thru the belt path of the machine as shown above. Since the belt is longer than the working length the two ends will overlap each other. Mark the length of the overlap and then accurately cut this length maintaining right angles to the longitudinal edge of the belt. The spiral ends will then need to be machine stitched with a strong polyester thread, on both the free ends of the belt as shown in the Picture below. **Error! No sequence specified.** The stitch pattern is very important as shown in the Picture. and must be done on a professional sewing machine.

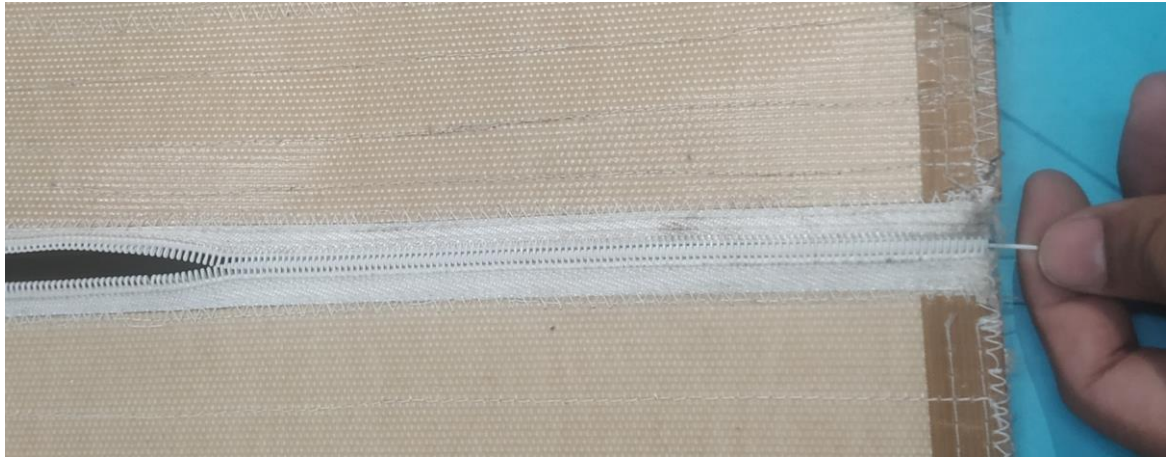


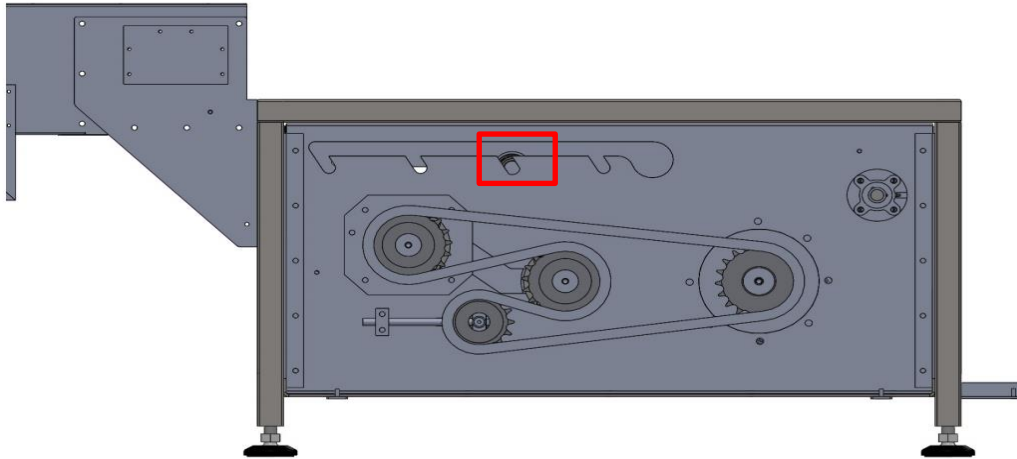
Illustration 10: Belt joining

Prior to cutting and joining of the new belt, it must be installed and routed around the conveyor path (i.e., round all rollers, guides, tensions, etc.). Before routing the belt ensure the following:

The belt tension pressure is released by operating the lever switch shown in Select position “Released”.



The slotted lost belt rack roller is positioned such that the roller sits in the second slot from the center of the conveyor (shown next), before the joint has been completed and the belt tension is re-applied. This will allow for adjustment either way - if the belt were to shrink or stretch slightly.



During fitting the working surfaces of the belt must be kept free from contamination. Fit the belt as shown in the diagram below

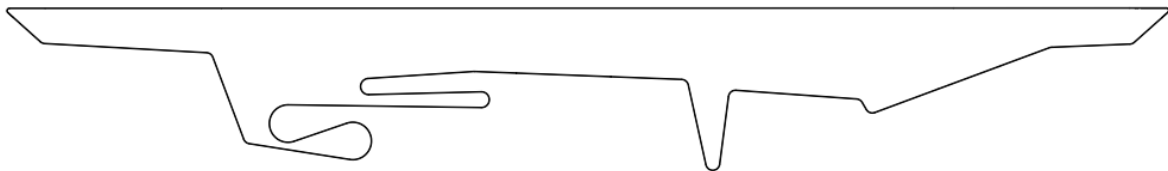


Illustration 11: Belt Path

STEPS TO FIX BELT PATH EFFECTIVELY

Ensure that the belt follows the indicated path exactly, without missing any of the rollers.

The working surfaces of the belt must be kept free from contamination.

With the un-cut belt in position, apply suitable tension either manually or using an improvised wooden board. The tension must be sufficient to lift weighted rollers present in the system. Tension on the belt may be maintained whilst measuring and marking.



WARNING: Before proceeding further, visually inspect the entire belt run, ensuring that it is correctly positioned and routed. Any alteration in path length will necessitate re-making of the joint at a later date.

CONVEYOR BELT TRACKING SYSTEM

The belt tracker assembly provides automatic correction to the moving conveyor belt when lateral displacement is detected during normal operation. This is achieved by means of a two roller assembly over which the conveyor is routed during installation.

The tracking system is incorporated to compensate for slight discrepancies and occasional uneven belt loading. It cannot correct badly installed conveyors or poor quality belts.

Whilst the belt remains perpendicular to the conveyor framework, (i.e. running centrally around the guide rollers), no action is taken by the tracker assembly. If for any reason the belt should move off-center, the edge of the belt will come into contact with a sensor mounted at either side of the framework. Activation of either sensor triggers lateral movement of the actuators supporting the tracking roller. Operation of the unit is such that movement of the actuators creates a 'skewing' effect in the opposite direction to the current position of the belt. Once the belt is running straight and true again, the belt tracker becomes inactive once more.

BELT TRACKING ASSEMBLY

Installing The Belt Tracking System

All rollers must be installed square to the conveyor framework, and the tracker unit must be mounted so that the rollers are also square both vertically and horizontally.

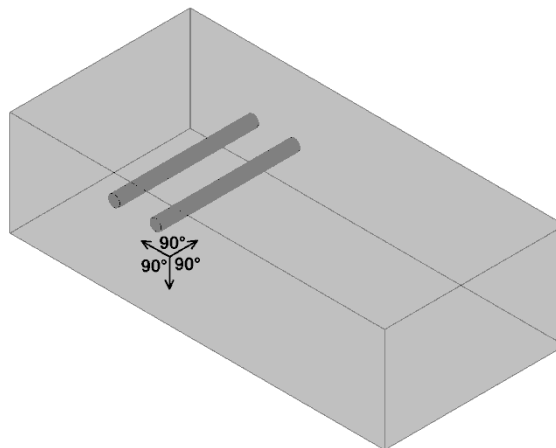


Illustration 12: Tracker Roller Alignment



NOTE: A clean dry compressed air supply will extend the working life of the tracker, and will reduce the possibility of breakdown.

The belt tracking system requires a compressed air supply capable of providing 0.4 MPa (4 bar or 60 psi) working pressure.

Install the tracker unit and check that all rollers are true to the conveyor framework. Adjust the belt tension pressure between 40 to 60 psi based on the type of belt by adjusting the belt tension regulator. Ensure that the manual belt tension release is set to "APPLIED" as shown next.



Illustration 13: Belt Tension control Panel

Testing The Belt Tracking System

With the belt centrally positioned on the conveyor, the brackets containing the optical sensor should be approximately set - such that the optical sensor beam is 10 to 15 mm away from the belt edges. Correction will therefore be initiated upon (10 to 15mm) sideways drift of the belt.

Movement of each piston should only be apparent when the adjacent sensor is activated. At all other times the pistons should not move.

Operation of the sensors should be tested keeping the lateral displacement of the tracking roller assembly in the both directions as shown in the figures below.

Belt Moving To The Left

Conveyor belt moving to the left makes contact with the sensor beam (1) causing the pneumatic cylinder to move the rollers (2) so the entire roller assembly is 'skewed' in the opposite direction, thus restoring the desired belt position (3).

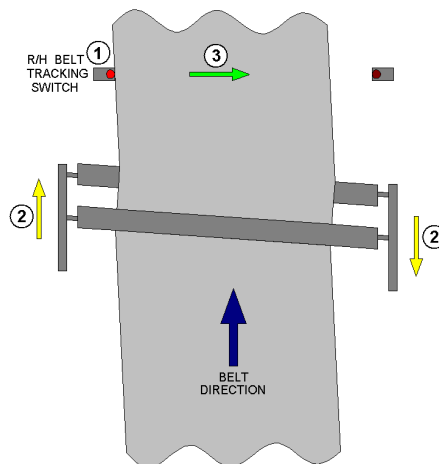


Illustration 14: Left Sensor Activated

Belt Moving To The Right

Conveyor belt moving to the right makes contact with the sensor beam (1) causing the pneumatic cylinder to move the rollers (2) so the entire roller assembly is 'skewed' in the opposite direction thus restoring the desired belt position. (3)

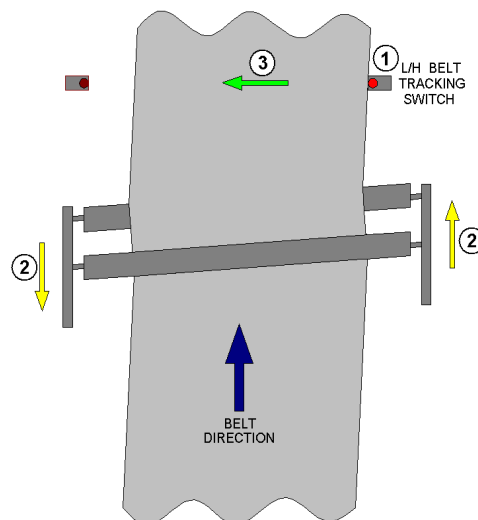


Illustration 15: Right Sensor Activated

If after installation as described above the actuators fail to move in response to activation of either sensor assembly, please check the electrical schematics to ensure that correct electrical connections to the sensor and the actuators have been made.

EXHAUST DUCT

The exhaust duct is an integral part of the Environment Management Unit. It is via this duct that the excess heat in the oven chamber is vented out of the GP-15 machine. This activity is a MUST for safe and reliable operation of the GP-15.

Drying chamber (Oven) is fitted with a centrifugal fan. The fan is capable of discharging a volume of air up to 31.1 meters³/min (1100 ft³/min).

Specifications of Exhaust Duct

A minimum duct diameter of 250mm (10 inches) is required for each extraction fan.

The air duct must be fitted with a condensation trap close to the fan in order to prevent water condensate from running back into the machine. Refer to the General Arrangement drawing for details on fan outlets and condensation trap.

It is important to ensure that all bends in the duct-work are swept bends, and also to avoid any unnecessary bends. These will only unnecessarily increase the backpressure on the fans, reducing considerably the fan performance.

The total pressure drop in the exhaust duct must not exceed 250Pa (1" WG)

Duct work inside the building should be suitably insulated if required, in order to minimize condensation.



NOTE: Please refer the exhaust duct drawing in the appendix

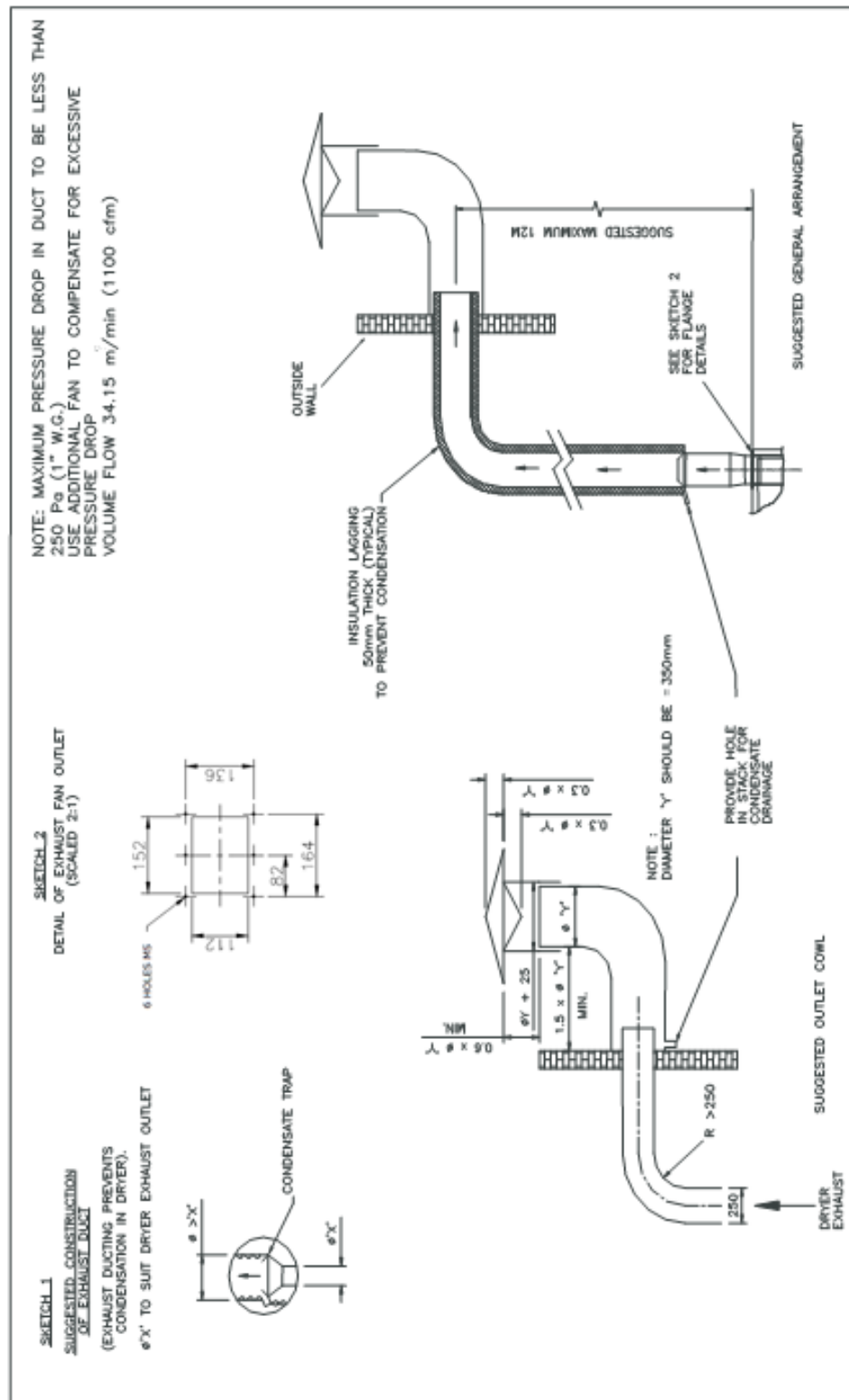


Illustration 16: Typical Exhaust Duct Construction

External Exhaust Stack

The excess heat vented out of the GP-15 machine has to be vented out of the premises/workshop also.

The “stack head” or vertical design is preferred to the weather (rain) cap one to prevent fumes or odours being deflected downwards and re-entering the building. The vertical design will require supports against side winds and adequate drain against rain when the machine is not in use. Other recommended designs are shown in Illustration 17: Alternate Recommendations for Exhaust Stack Designs.

If the fumes discharged are undesirable, the stack discharge point should be from 1.3 to 2 times the height of the building to prevent odour or exhausted steam and gasses from being entrenched into the building. (This is primarily a summer time issue if the roof has vents, which are open.)

The condensation from the stack should not be allowed to drip onto the ground in order to prevent it from entering the ground water system. It is therefore recommended that the condensate is piped back to a collection point where it can be pumped or drained as appropriate.

Safety

A bird mesh must be fitted to the duct discharge outlet.



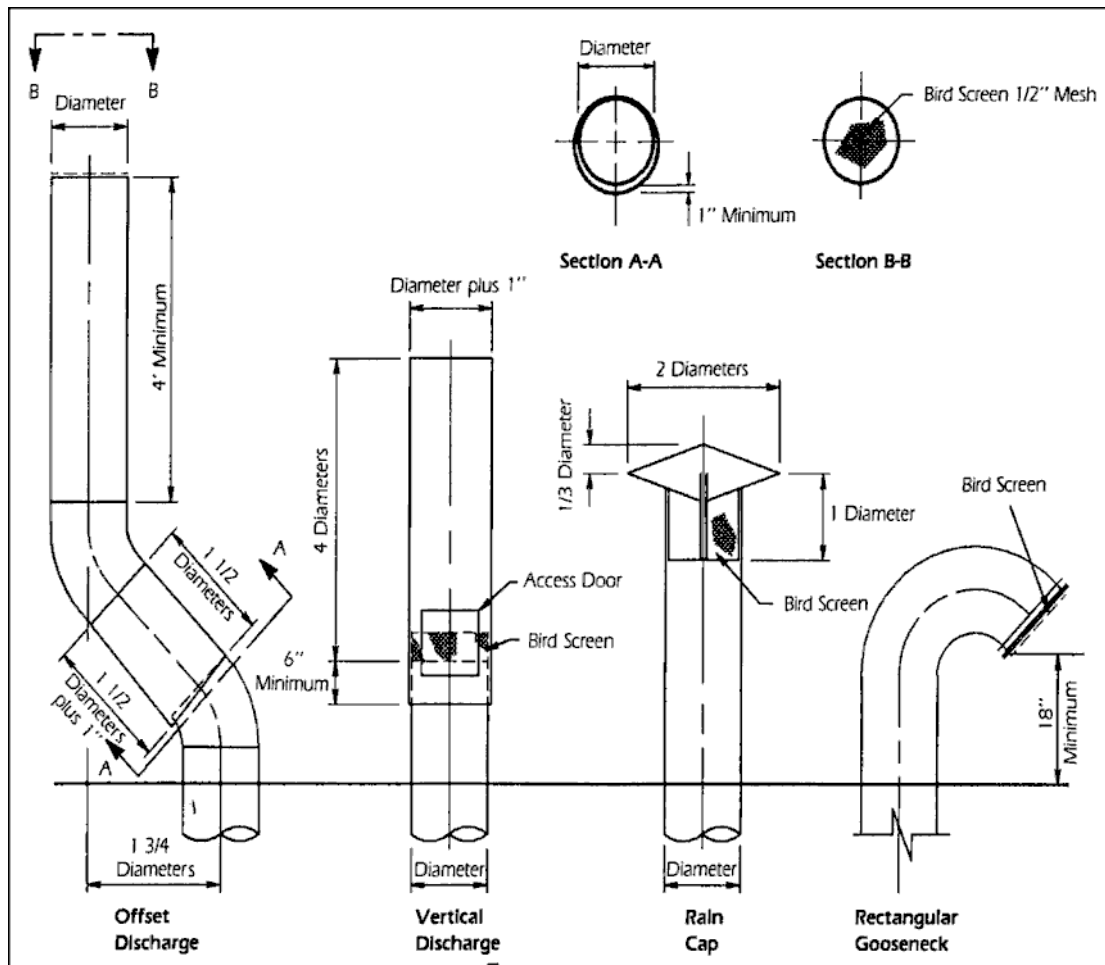


Illustration 17: Alternate Recommendations for Exhaust Stack Designs

UNINSTALLATION AND DISASSEMBLY (IF REQUIRED)

The procedure for dis assembling and uninstalling the machine requires specific instructions based upon the nature of the installation and are not part of this user manual. It is recommended that such tasks be carried out by a trained QMTI machine technician. Please contact QMTI's service department for further details.

CHAPTER 5. Settings & Starting

After successful installation of the various components of the GP-15, it is essential to set the initial Power, Temperature, Water, and other settings and controls required for operating the machine. This is explained in this chapter.

SETTING THE POWER LEVEL

To set the power level, understanding of the power required to do a production job. This is especially true when new products are run on the GP-15. This is explained by way of example/calculations below.

CALCULATION OF REQUIRED POWER SETTING

When processing new products, it is necessary to calculate the amount of RF power required to remove the desired amount of moisture. Having established the RF power setting in kilowatts, this figure will be used when setting the electrode height position during GP-15 operation.

Example: Let us assume we wish to process a total weight of 600 kg of product per hour with a final moisture content of just 3%.

- 1 To calculate the weight of the entire product less the moisture:

$$\frac{600}{1.03} = 582\text{Kgs (0\% moisture)}$$

- 2 If we know (or assume) a moisture content when entering the GP of 4%, we can calculate the total product weight including moisture:

$$582 \text{ Kgs} \times 1.04 = 606 \text{ Kgs Total Weight}$$

- 3 In order to meet our target of 600kg per hour processed product we must therefore remove 6kgs of moisture every hour.

As a rule of thumb, we should get 1Kg of water for every 1KWH of RF energy applied for products that have a very large surface to volume ratio and the base product that is to be dried is generally non susceptible to RF power.

For Products that have low surface to volume ratio or products that cannot be presented to the RF oven in a format that can take full advantage of the large surface to volume ratio or products that have base product that is susceptible to RF one should get about 0.6 Kgs of water for every 1KWH of RF energy applied.

For calculation sake let us consider 0.6 as a factor.

$$\frac{6}{0.6} = 10 \text{ kW of RF power}$$

- 4 Allowing an additional 10% for unavoidable losses, we require a total process power of 11kW for this application.

CALCULATION OF ANODE CURRENT

From the theoretical power level derived (above), the required anode current is calculated using the formula below.

$$\frac{2.6A}{15kW} \times 10kW = 1.9Amps$$

Where the Anode Current is at the full rated power of 15kW = 2.6 Amps (see test report)

The electrode height is adjusted until the Anode Current meter indicates 1.9 Amps. (This value may need slight adjustment after initial testing but should provide a starting point for the required settings).

Illustration 18: Moisture Removal

PHASING CHECK

Verification of the correct 'phasing' of the incoming electrical supply can be made by observation of the direction of rotation of the cooling system blower motors.



WARNING: All motors within the GP™ machine are correctly phased during manufacture. If the direction of rotation proves incorrect, alter the phasing of the incoming supply, DO NOT attempt to change the individual phasing of the motors.

If the rotation of the blower motors is reversed, then change over TWO of the incoming 3 phase supply lines connected to the main disconnect switch.

SWITCHING ON

- Ensure all emergency stop buttons are released
- Switch on Control Panel circuit breaker.

- Ensure the HMI has booted up and is displaying the Main Menu Screen
- Ensure there are no active alarms
- Ensure the extraction fans and oscillator cooling fans are running. The extraction fans run at low speed to allow the oven to reach operating temperature
- Switch on the H.T. circuit breaker.
- Select the production screen.
- Ensure the oven status display shows Ready For Production



NOTE: The above procedure must be followed in sequence

CONVEYOR– FREE RUN SWITCH(OPTIONAL)

Belt Free Run allows the user to quickly switch over the belt speed to the max speed allowed by the geared motor conveyor. This feature has been provided to allow the user to quickly remove any material within the oven during maintenance work or during an emergency. A switch is provided on the Infeed end as shown below to set the machine in belt free run mode



Illustration 19: Conveyor Free Run Switch



Note: The belt free run feature can only be used when the RF power is turned off.

The belt free run feature can also be used after any of the E-Stop buttons have been pressed.

CHAPTER 6. Operations

The GP-15 machine can be operated in two modes, viz., Manual and Preset/Recipe. This chapter explains the step-by-step operation of the machine in Preset/Recipe and manual modes.

Safety

Before operating the machine, the operator must read, understand and be fully conversant with the operation of the touch-screen display



EMERGENCY STOP SYSTEM

Apart from the various settings and initiations, the operator must know the Emergency Stop system in the machine before operating the machine. The emergency stop system consists of multiple ways to stop the machine in an emergency. These are explained next.

EMERGENCY STOP PUSH BUTTONS

These red circular push buttons bring the machine operation to a halt when pressed.

Totally there are four red circular Emergency stop, push buttons in the machine. Two are located on each end of the conveyor, third on the Generator door and the fourth on the HMI panel.

When the emergency stop push button is pressed, it De-energies all the motors, fans and transformers (except the control transformer). The safety relay turns off the RF contactors as soon as any of the emergency stop buttons is pressed. When they are operated the H.T. supply is removed from the generator & RF cycle is turned OFF. For additional safety the HT Circuit Breaker is tripped using an UV coil.

To get the machine back into operation, these buttons have to be reset (Pulled out) first and reset by pressing the **"RF OFF"** push button located on the HMI panel.



NOTE: The safety relay monitors all contactors, if any contactor fails to de-energise the safety relay cannot be reset.

CONTROL PANEL ACCESS

Operation of the GP-15 machine and access to certain high voltage compartments is controlled by the use of Safety Switches. The Safety Switches system uses **Solenoid operated safety switches, Trojan safety switches and a Door Unlock Switch** . For normal operation, this switch should be in OFF position.

SOLENOID DOOR LOCK

Access to certain high voltage compartments is controlled by the use of solenoid locks. A single **Door Unlock Switch** as shown below is provided on the Infeed end.



Illustration 20: Door Unlock Switch

Activating the Door Unlock Switch unlocks the solenoid switch doors and automatically trips the HT, and brings the operation to a halt.



Illustration 21: Solenoid Operated Safety Switch

The safety circuit of the machine requires that all doors are closed and the door unlock switch is turned off to resume normal operation.



If an attempt is made to keep the protected doors open and turn off the Door Unlock Switch, the alarm will not deactivate.

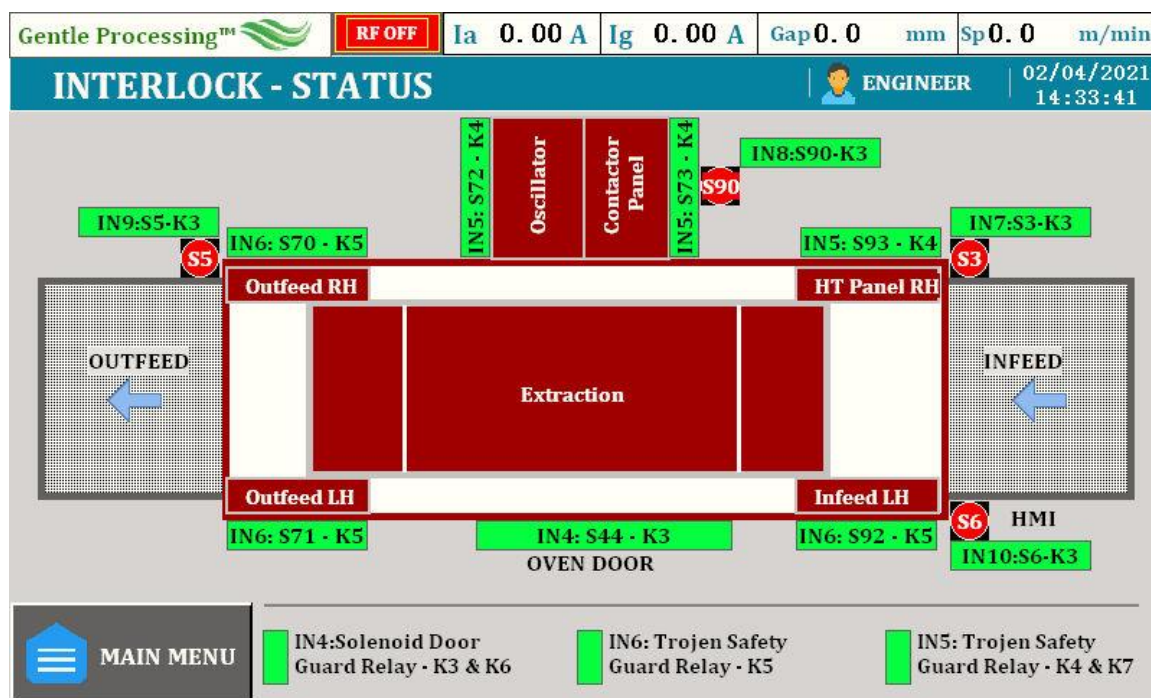


Illustration 22: HMI screen showing status of emergency stop and machine doors

Safety



It is mandatory for the operator to know these Emergency Stop methods and use it correctly in need.

To turn on the machine after an E-stop condition or Door interlocks opened



1. Ensure all the Doors are closed and master key interlock is closed, Check that all E-Stop push buttons are reset
2. Reset and acknowledge all faults using the RF OFF push Button on the HMI Panel.
3. Navigate to product Menu screen.
4. Ensure the oven status display shows Ready Condition Press "RF ON PB" For RF ON.

BASICS OF HMI PANEL

The HMI (touch screen) panel on the control console is driven by menus. The understanding of the various buttons, menus, and their purposes is required to run the machine in Preset/Recipe or manual mode. The HMI panel is used for the following:

- Preset/Recipe and Manual operation of the machine
- Monitoring production operation
- Setting product output settings and saving them as Preset/Recipes for reuse in Preset/Recipe mode
- Service and Troubleshooting
- Conveyor operation and monitoring
- Fault/Errors message viewing (Alarm causes)

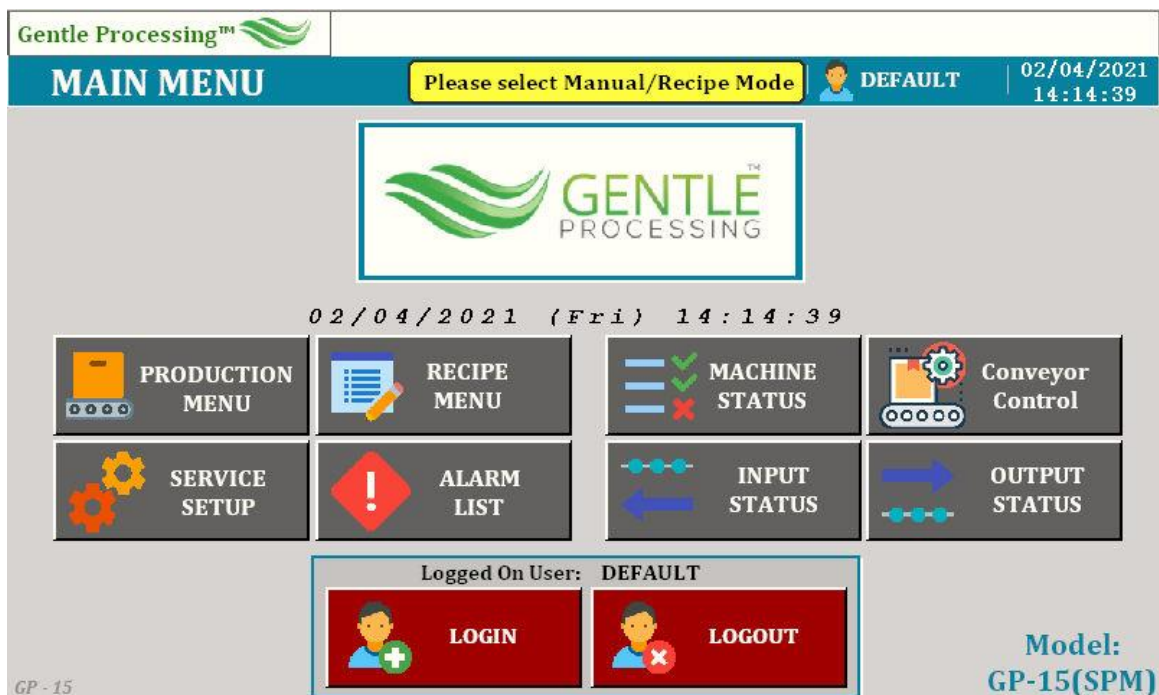


Illustration 23 : Home / Main Menu Screen – HMI Panel

The Home screen/Main menu is the first screen. All other screens are accessed from this page.

Production Menu: Displays the production screen. This is the main screen for running the machine in production

Recipe Menu: Displays the recipe menu screen. The product recipe functions (load, edit, copy) are accessed from this screen.

Service Setup:	Displays the service setting screen. Limit settings are accessed from this screen
Alarm List:	Displays Active Alarms and Historic Alarms, showing last 100 faults
Conveyor Control:	Displays the conveyor setting screen. Belt Limits / tracking can be accessed from this screen
Machine Status:	Displays the Status of machine Emergency Stop push buttons and Doors
In/Out Status:	Displays the current status of PLC inputs and outputs
Login:	Allows users to log in to the HMI to access menus for preset settings, alarm list, and service settings.
Logout:	Exits the current login user



Note: If a screen is idle for 5 minutes, then the User will be logged out automatically

PERMISSION LEVEL

Access to all the screens is restricted based on user roles and permissions. Based on their role, they need to log on the HMI and can access specific screens. There are:

User Role	Login	Screen Access
Operator (DEFAULT)	No	Access to Product Menu and Status Screens <u>Warning: No Manual operation access</u>
SUPERVISOR [S]	Yes	Same menus as Operator. Full access to Pre-sets / Recipe menu and Alarms. Manual operation access
ENGINEER [E]	Yes	Full access to all menus. Access to Service Menu, except some Trip Settings (view only access)
Gentle Processing	Yes	Full Access without any restriction including all menus.

The default main screen will display User menus. Additional Menus will be displayed based on permission level. From the Home screen, the Users can press any menu button to view and work with each of them.

OPERATING THE MACHINE

Step-by-step instruction to operate the machine using HMI screens in Preset (Recipe) and Manual mode is given next. These instructions are valid after the machine is switched ON, and all power supplies are in order.

RECIPE/PRESET MODE

The GP-15 machine is operated from the control console panel that has a touch-screen display. The Preset/Recipe operations using the control console panel is menu driven. The operating parameters required for the product are preset as named Recipes in the RECIPE MENU and accessed from the RECIPE MENU.

Preset / Recipe mode of operation requires 1 operator to monitor the in-feed of the product into the oven via the conveyor, as well as its out-feed on the other end of the conveyor. To run the GP-15 machine for a specified product, set its specific parameters as a preset and follow the steps below.

- Ensure that the product is readily available.
- Ensure there are no active alarms and the red lamp is illuminated steady red and "Ready Condition" message is displayed and the R.F. START button starts flashing green.
- Select a preset / recipe (setting for a product) from the Recipe selection screen using the +/- Sign button to select the Recipe number. Refer to Recipe Select Menu on page 64.
- The mode indicator on Product menu screen(page60) will turn from 'Manual Data in Use' to 'Recipe in Use' to indicate the selected mode.
- Ensure that the product name displayed is correct.
- Press the green START push button. The R.F. START button illuminates steady green.
- At the end of the production run allow product to 'run-out' of the oven, then press the GP-15 OFF button to stop processing

Safety

Even after the GP-15 machine OFF button is pressed, the oven chambers and other parts may be very hot. So, personnel should observe caution.



MANUAL MODE

Manual operation is recommended when processing new or unknown products, as it is not possible to determine the required parameters without a trial run.

In manual control, all product specific settings are entered individually prior to starting the GP-15 machines. Adjustments are made to the settings when the GP-15 machine is on to obtain the desired results. Once the settings have been ascertained, on stopping the GP-15 machine, the settings can be saved as presets with a name/identifier.

During manual operation, the operator adjusts the electrode position using the buttons on the production screen to achieve the desired power levels. *Refer to on page 44.*

The steps are given below.

- Ensure that product is available.
- Select the product Menu screen from the Main screen (Illustration 23 : Home / Main Menu Screen – HMI Panel on 50).
- Ensure there are no active alarms and the red lamp is illuminated steady red and “Ready Condition” message is displayed and the R.F. START button starts flashing green.
- Ensure manual control is selected by using the +/- Sign button to select the pre-set number to zero.
- Use the key adjacent to the conveyor setpoint display to adjust the setpoint until the desired setpoint is displayed.
- Adjust the electrode gap setpoint to the required value.

Safety

The amount of power (determined by the electrode height) applied to any product should be limited to known safe values to reduce the risk of overheating and fire hazards. **If in doubt, contact the QMTI sales and service department for assistance.**



- Ensure that the conveyor infeed is fully loaded with product.
- Press the green GP-15 machine start push button. The R.F. START button illuminates steady green.
- The electrode height is adjusted to obtain the power level that gives the best results in moisture removal and/or profiling.



NOTE: The actual electrode position is not altered immediately the setpoint is altered. The electrode is not moved to its new position until

both buttons have been released for 0.5s.

- The anode current, grid current, electrode height and conveyor speed are monitored from the production screen
- At the end of the production run allow product to 'run-out' to ensure there is no product in the GP-15, then press the GP-15 OFF button to stop processing.

RECYCLING

Recycling is the automatic restarting of the GP-15 heating process in case of a temporary overload or fault condition in the machine that stops the heating process. Example: A lump of scrap dough entering the GP-15 machine. The faults trigger an alarm, and the alarm banner is displayed. The alarms can be viewed on the alarm list screen. Refer to Alarm List on page 84.

The recycling system is initiated by the occurrence of one of the following trips

- Anode current trip (software set)
- Anode current trip (Over Current Relay)
- Grid undercurrent trip
- Grid over current trip
- Arc suppression

When running the GP-15 in production mode these are collectively known as a recycle fault

Recycling is limited in the number of attempts at re-starting the process within a certain time. The recycle count reset time interval is automatically calculated and is dependent on the conveyor speed, the maximum number of restarts and the restart interval. After this time, the restart counter is reset.

In the event that the trip condition is permanent the GP-15 machine is stopped after the permitted number of restarts.

The message RECYCLE LOCKOUT is displayed on the production screen

Recycling settings are accessed via the Oven Limits and Trips screens. The settings are

Parameter	Description	Typical Value
Permitted Restarts	The maximum number of restarts permitted before the Oven is shutdown	4
Restart Delay	The time between the recycle fault occurring and the RF restarting	2s

RECYCLE OPERATION

The steps below explain how the recycle function works in the GP-15 machine. There are two methods:

RECYCLE OPERATION WITHIN THE SET RESET ATTEMPT

- RF power is applied, and the restart count reset timer is started.
- A flash over occurs in the applicator area causing a trip. The RF power and count reset timer is stopped. The restart interval timer is stopped. The electrode drives up.
- The Restart Interval timer times out, and the restarts counter is incremented by one. RF power is restored, and the electrode drives down to its setpoint. The count reset timer is restarted.
- A second flash over occurs before the count reset timer has timed out. Steps 2-3 are repeated. The restart count is now 2.
- No further flash overs occur, and the count reset timer times out. The restarts counter is set to zero.

RECYCLE OPERATION BEYOND THE SET RESET ATTEMPTS

- 1 Steps 1-4 are repeated until the restarts counter value has reached the max restarts value. (5 in this example.)
- 2 Further restarts are inhibited and a Recycle lockout message is displayed in the Oven status display. An alarm is generated.



NOTE: Recycle lockout occurs after 4 attempts have been made at automatic power restoration, i.e. following the occurrence of the fifth flash over.

These steps are illustrated in the next image.

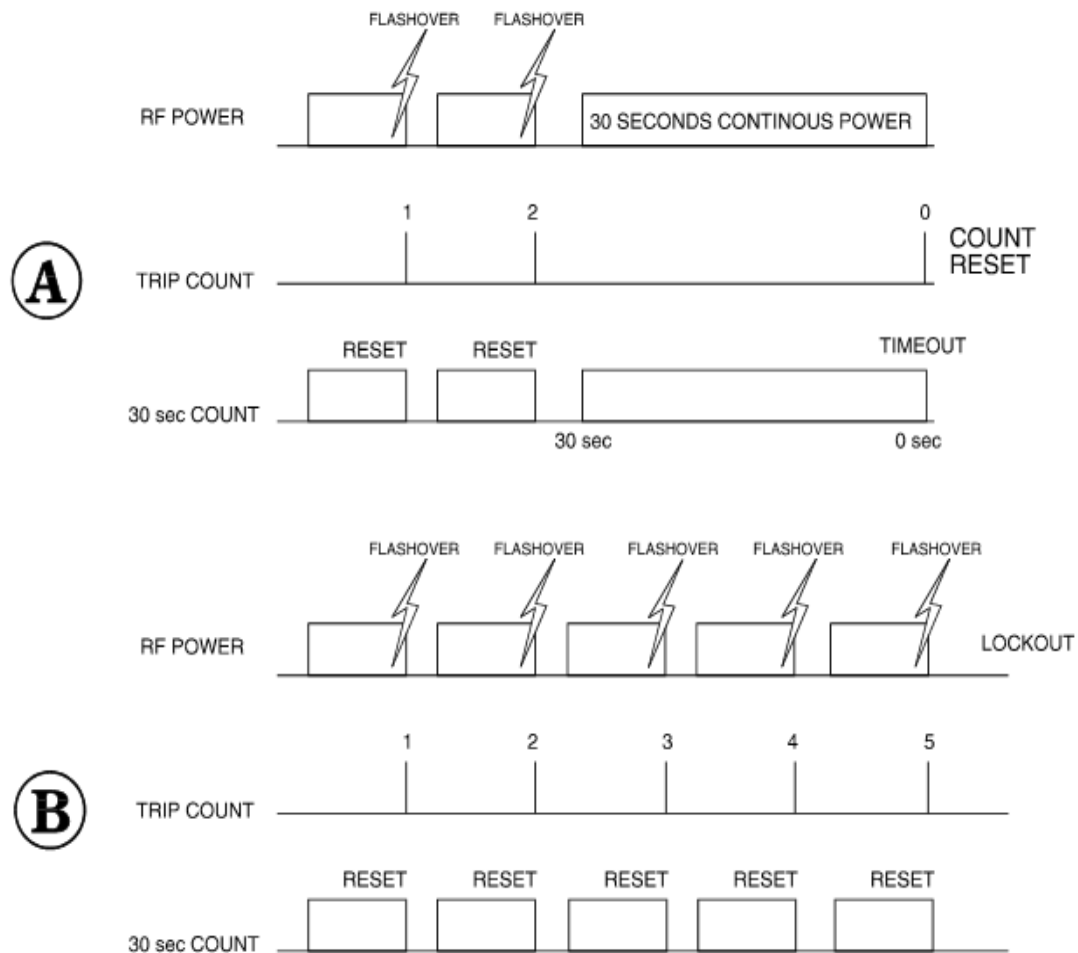


Illustration 24: Recycle Operation



WARNING: Always investigate the cause of repeated arcing and recycle lockouts. Ensure the problem is rectified before continuing production. Failure to investigate can cause irreparable damage to the machine.

HOMING THE ELECTRODE



On applying mains power to the machine, an electrode homing sequence is initiated. During homing, the electrode is driven up to the upper limit switch. The electrode gap counter is reset. Then, the electrode is driven down to the max gap limit position. This ensures proper electrode positioning in case the electrode drive system had been disturbed.

- The electrode is driven up to the upper limit switch
- The electrode gap counter is reset to the index switch height value.
- The electrode is driven down to the max gap limit value.

During normal operation, the operation is as follows.

- The electrode max gap setpoint is set to a value between the index gap point and the min gap limit
- The operator panel RF ON button is pressed.
- The GP-15 machine is switched ON
- The electrode is driven down from the max gap point
- The electrode is driven down to the electrode gap setpoint

Since the operator has control over the electrode position using the touch-screen display, he can determine an approximate power level necessary for the product. With this approximate power level, the GP-15 machine is switched on and the electrode height lowered until the calculated power level is reached. The product is then sampled, and the power level adjusted according to the requirements of the customer. The conveyor band speed should be adjusted always to present the maximum possible amount of product

INDICATING LAMPS

Control of the RF power is provided by means of two push buttons with indicating lamps to indicate the operational state of the machine. Green and Red lamps illuminate/flash to indicate smooth operation or fault in the GP-15 machine. Indications for the lamps are given next. Refer to the alarm list screen also on page 84.

Push Button	Meaning
Green Solid	The GP-15 is in operation
Green Flashing	The GP-15 is ready for operation and no faults.
Red Solid	The GP-15 machine is off; no faults present and The GP-15 is ready for operation waiting for start command
Red Flashing	The GP-15 is in fault. An emergency stop conditions present.



NOTE: In case of an active fault, an audible siren starts on. It is silenced by pressing the RF OFF button on the HMI console.

OPTIMUM GP-15 MACHINE USE

- 1 The longer the GP-15 machine runs empty, the less energy efficient the system becomes. At the same time the service life of the triode valve life is being reduced.
- 2 Dirty air filters or heat exchangers reduce the overall efficiency of the machine and may lead to down-time due to thermal tripping causing possible damage and premature failure of RF components.
- 3 Hardened or damaged door seals will reduce the efficiency of the internal cooling system and causes the RF seals to be less effective, which may cause radio interference to others. All panel fixings must be replaced following maintenance or servicing as they fulfil a vital role in the reduction of interference.
- 4 Conveyor belt transport systems must be kept clean and well maintained. Correct belt tensioning and tracking is essential. A faulty conveyor will cause extensive loss of production.
- 5 Dirty or contaminated electrodes can lower the efficiency of the GP-15 machine. Avoid build-up of product inside the GP-15 machine as it will constitute a potential fire hazard and an obstruction to moving parts.
- 6 Reduced air extraction from the GP-15 machine due to dirty impellers, incorrect air flow settings or similar faults will lower heating efficiency. This in turn can lead to excess condensation build-up within the GP-15 machine leading to RF arcing, electrical breakdown and extensive damage to the RF circuitry.
- 7 Poorly maintained cooling systems in general will shorten the working life of the triode valve - an expensive item to replace!

SOUND LEVELS

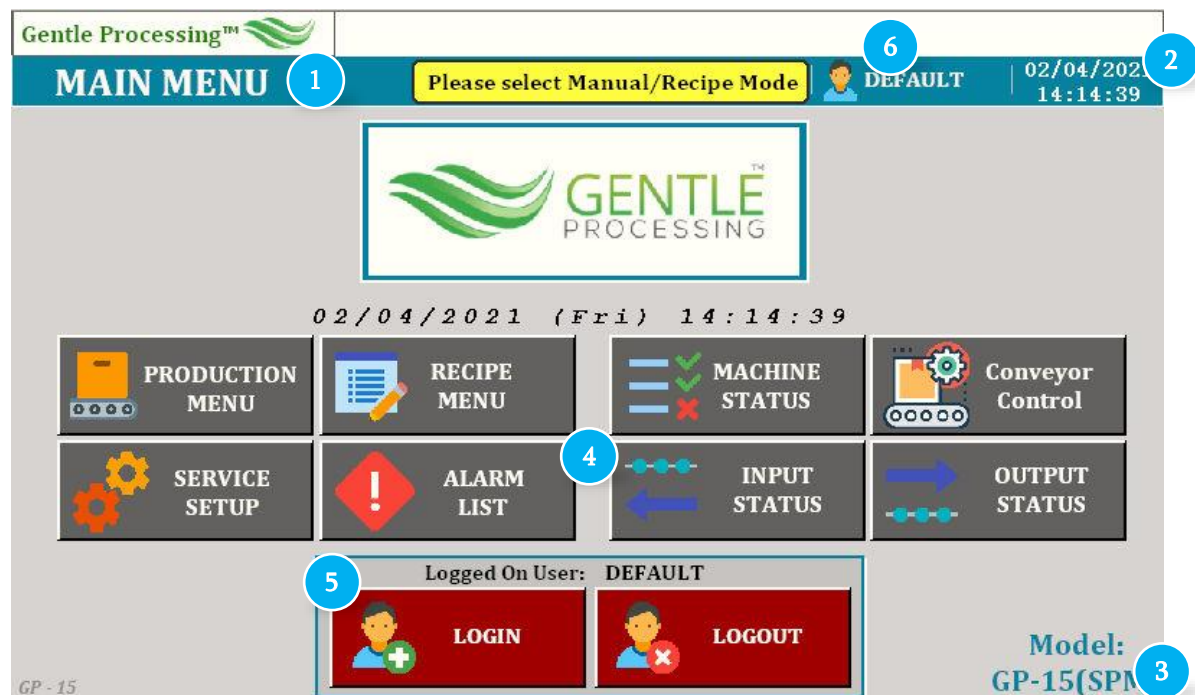
Suitable extraction/exhaust ducts need to be installed prior to the operation of the GP-15 machine. It is unlikely that sound levels will go beyond 75 dB when the machine is completely installed. However, in case of exposure to high noise levels during testing of the air exhaust system of the GP-15 machine, wear suitable ear protection



CHAPTER 7. HMI Screens

The various screens used to set product, Preset/Recipe, and conveyor settings are explained next.

MAIN MENU



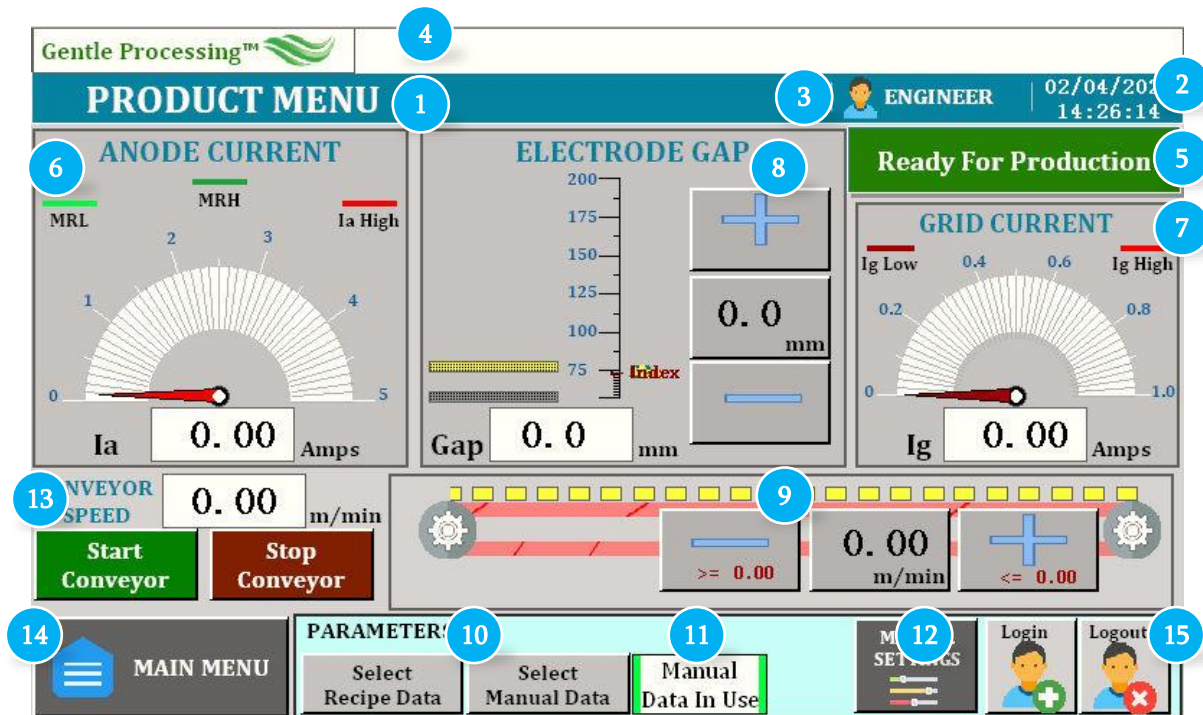
Screen 1: Main Menu

The Main Menu allows the operator to move to any of the operating screens.

1. Menu Name
2. System Date and Time
3. Machine Model
4. Allows navigation between screens as explained in Basics of HMI Panel on page 50
5. Login and Logout Buttons.
6. Displays Logged in user's name.

PRODUCTION MENU

The production screen displays the current product preset / recipe settings, and machine parameters. This screen has to be selected before RF Drying is started. The various sections in this screen are explained next.



Screen 2: Production Menu

1. Menu Name
2. Date and Time.
3. Displays Logged in user's name.
4. Active Alarm Banner shows up when fault/warning is present.
5. Operational state of the GP™ machine. Shows actual running condition or actions required to be taken.
6. Anode current shown numerically, and on dial graph. Alarm set points can also be seen.
7. Grid current shown numerically, and on dial graph. Alarm set points can also be seen.
8. Electrode height adjusts numerical entry and +/- Sign (Active in Manual control mode). Actual Electrode Height shown numerically.
9. Conveyor Speed adjusts numerical entry and +/- Sign (Active in Manual control mode).

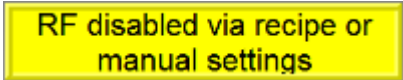
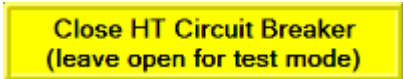


Note: Red graphic indicates stopped; Green indicates running.

10. Preset or Manual Select buttons. The display box shows the selected Preset number and Zero represents manual mode selection.
11. Current state. Shows Manual/Recipe.
12. Displays pop up for manual settings of Anode current – Meter Relay High (MRH) and Meter Relay Low (MRL) and is visible when manual mode is selected with engineer or Gentle Processing login.
13. Conveyor Start/Stop button
14. Exits current screen and goes back to the Main Menu screen
15. Login/Logout buttons.

OPERATIONAL STATE INDICATION MESSAGES

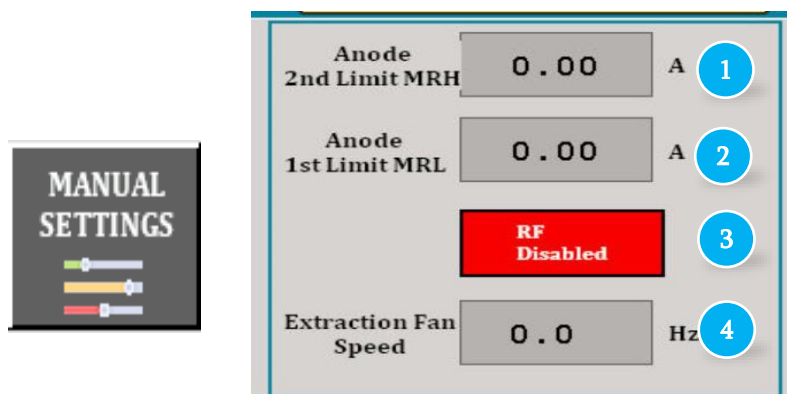
The various possible operational states of the GP™ machine is illustrated next.

Message	Meaning
	The emergency stop relay in the console is not made. To reset press the “acknowledge and reset all alarms button” on the alarm list screen.
	Indicates that the Main Circuit Breaker is OFF.
	The RF Power is disabled. To enable the RF change the recipe setting or change via the manual settings button
	The conveyor is not running or is below the minimum permitted speed (see zone limits and trips for settings). The RF power will not start with conveyor off.
	The triode valve filament is allowed time to reach operational temperature before the RF is allowed to start.
	The H.T. circuit breaker is not closed. The GP-15 operates in this condition, but the H.T. contactors do not close. The electrode functions normally

Not Ready See Alarm List For Details	A condition is preventing the GP-15 from reaching the ready state. Look at the Alarm list for active alarms
Ready For Production	The GP-15 is ready with no faults
In Production	The GP-15 is running
Anode Current Control In Operation	A change in the product moisture content has caused the anode current to increase until it reaches the limit Set Point.
Recycle Lockout number of restarts has exceeded setting	The recycling function has exceeded the permitted number of restarts in the time period (see zone limits and trips for settings)

MANUAL SETTINGS

The manual settings button is part of the parameter settings in the Production menu. This button displays the Manual Settings screen and is visible when manual mode is selected with Supervisor, Engineer or Gentle Processing login.



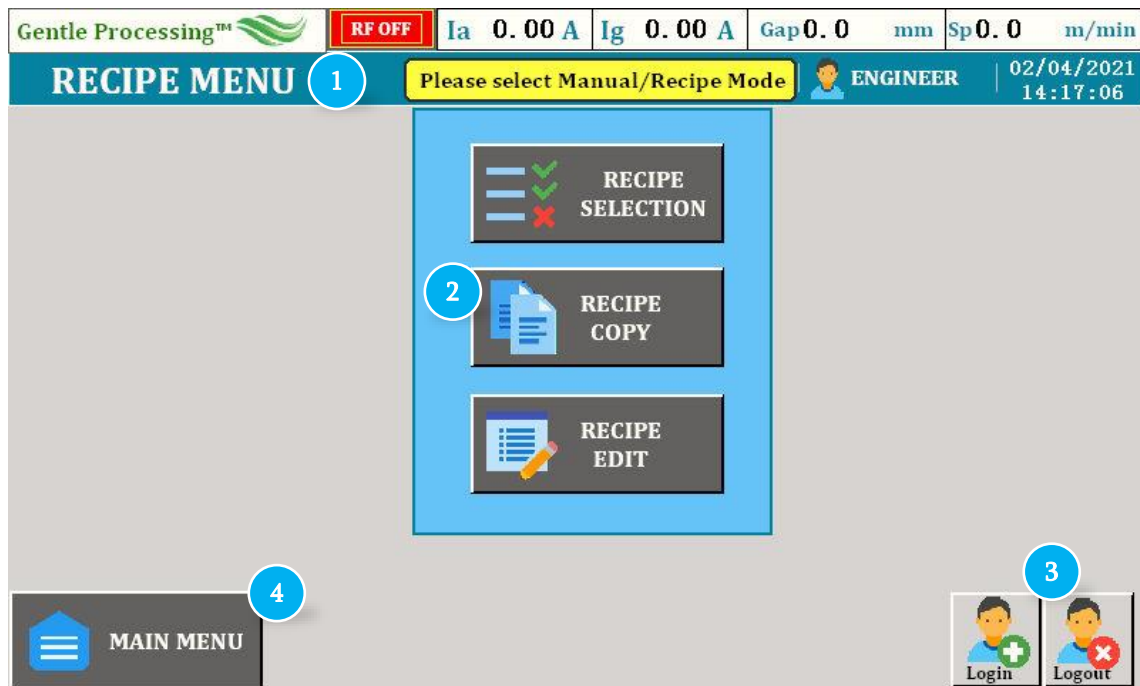
Screen 3: Manual Setting Screen

1. **MRH:** Sets the maximum permissible anode current. The electrode gap is automatically increased when the anode current exceeds this value.
2. **MRL:** Sets the anode current level at which the electrode drive is stopped above this limit.
3. Toggle button for enabling/disabling the RF Power.
4. Sets the extraction fan speed (Hz). 5-60 Hz. This determines the rate of water vapour removal and prevents condensation inside the GP™ machine.

RECIPE MENU

The recipe menu is where settings are saved for the Product after initial trial and error applied using manual control and settings screens. Access to all the recipe function screens is limited. The recipe editing and recipe copy screens are visible to Supervisor, Engineer, and Gentle Processing User roles only.

All users have access to the recipe selection screen.



Screen 4: Recipe Menu Screen

- 1 Menu name.
- 2 Recipe function screens to select the recipe, edit existing recipe, or copy a recipe to save as a new one.
- 3 Log in and Log out buttons. The name of the logged on User is also displayed above these buttons.
- 4 Main Menu button

The recipe function screens are explained next.

RECIPE SELECTION SCREEN

The Recipe Selection screen displays the available recipes and the current recipe in use. Recipes can be selected and downloaded from here.

Recipe Parameter	Preview	Current	Compare
Product Name			×
Conveyor Speed	0.00 mtr/min	0.00 mtr/min	×
RF Enable ??	RF Disabled	RF Disabled	×
Extraction Fan Speed	0.0 Hz	0.0 Hz	×
Electrode Gap	0.0 mm	0.0 mm	×
Anode Current Control 2nd Limit	0.00 Amps	0.00 Amps	×
Anode Current Control 1st Limit	0.00 Amps	0.00 Amps	×

Buttons: Download, [Down Arrow] 0 [Up Arrow], RECIPE MENU

Screen 5: Recipe Selection Screen

- 1 Menu name.
- 2 Parameter table of the selected recipe (**Preview**) vs recipe in use currently (**Current Recipe**). The parameters for the selected and current recipes are compared and its result is displayed in the **Compare** column.



Note: The recipe in the Current column is the recipe which will be run when the product is run on recipe mode. In other words, that is the downloaded recipe currently in use.

- 3 Buttons to select the recipe, and download the same. It also indicates the name of the recipe selected for Download. When a recipe is downloaded, it overwrites the current recipe settings with the one displayed/selected.



Note: Recipes have to be selected and downloaded for recipe operation mode.

- 4 Exit button to return to previous menu.

RECIPE COPY SCREEN

This screen is not visible to the Operator User, and therefore access is denied. The Supervisor, Engineer, and Gentle Processing users can copy a recipe from one record to another. This recipe can then be altered via the Recipe Edit Screen. This allows recipes for products with similar settings to be created easily.

The screenshot shows the 'RECIPE COPY' screen. At the top, there's a status bar with 'Gentle Processing™', 'RF OFF', and various parameters: 'Ia 0.00 A', 'Ig 0.00 A', 'Gap 0.0 mm', and 'Sp 0.0 m/min'. Below this is a header bar with 'RECIPE COPY' (callout 1), a user icon labeled 'ENGINEER', and a timestamp '02/04/2021 14:27:32'. The main area is a table with four columns: 'Recipe Parameter', 'Preview', 'Current', and 'Compare'. The table lists several parameters like 'Product Name', 'Conveyor Speed' (callout 2), 'RF Enable ??', 'Extraction Fan Speed', 'Electrode Gap', 'Anode Current Control 2nd Limit', and 'Anode Current Control 1st Limit'. Below the table is a 'Select Recipe' section with up and down arrow buttons and a display showing '0' (callout 3). At the bottom, there are three buttons: 'Copy' (callout 4), 'Rename Copy' (callout 5), and a 'RECIPE MENU' button (callout 6).

Recipe Parameter	Preview	Current	Compare
Product Name			✗
Conveyor Speed	0.00 mtr/min	0.00 mtr/min	✗
RF Enable ??	RF Disabled	RF Disabled	✗
Extraction Fan Speed	0.0 Hz	0.0 Hz	✗
Electrode Gap	0.0 mm	0.0 mm	✗
Anode Current Control 2nd Limit	0.00 Amps	0.00 Amps	✗
Anode Current Control 1st Limit	0.00 Amps	0.00 Amps	✗

Screen 6: Recipe Copy Screen

- 1 Menu name.
- 2 Selection table enables recipe copying by listing the parameters on the left most column. The next two columns are to copy 'From' and 'To' records/recipes. The last column compares parameters in the 'From' and 'To' columns and indicates if they match or not.
- 3 The **Select Recipe** arrow keys allow increase or decrease of the record number to be copied from Column From to Column To.



Note: To save a recipe record for a product via manual settings, scroll the arrows to the top or last until a record number does not display. This displays the manual settings record and the name MANUAL appears in the product name field) A recipe for a product is obtained thus using manual control, to be saved to a record.

- 4 Once the copy **From** and **To** recipes are selected and their comparison column is satisfactory, the recipe is copied by pressing the **copy** button.
- 5 Since the **copy** button does an identical copy, the **Product Name** in the

first row also gets copied as is. The **rename copy** button allows changing name of the product so that duplicate names and resultant errors are avoided.

- 6 Exit button to return to Recipe menu.

RECIPE EDIT SCREEN

This screen is not visible to the Operator User, and therefore access is denied. The Supervisor, Engineer, and Gentle Processing users can create a new recipe, edit an existing recipe, or upload the current settings using this screen.

This screen is divided into three sections, besides some buttons. The top section is used to enter Product name, Conveyor settings, and the middle section is used to enter the GP™ machine settings. The last section is used to select a recipe number. These are explained next.

Steps to Edit a Recipe

- Select the required recipe using the Recipe Selection Screen
- Exit and come back to Recipe Menu Screen
- Select the recipe editing option, to display the following screen.



Note: The current values are displayed in the while text boxes in the Recipe Editing screen. To alter/edit these values, press the buttons with the setting name. The keypad will be displayed to facilitate entering of new values.

The screenshot shows the 'RECIPE EDIT' screen. At the top, there's a status bar with 'Gentle Processing™', 'RF OFF', and various parameters: 'Ia 0.00 A', 'Ig 0.00 A', 'Gap 0.0 mm', and 'Sp 0.0 m/min'. Below this is a header bar with 'RECIPE EDIT' (1), a user icon labeled 'ENGINEER', and a timestamp '02/04/2021 14:28:51'. The main area contains a 'Recipe Parameter' section (2) with 'Conveyor Speed' set to '0.00 m/min'. Below that is a control panel (3) with 'RF Disabled' (3) and an 'Enable' button. It also includes 'Extraction Speed' (4) at '0.0 Hz', 'Electrode Gap' at '0.0 mm', 'Anode Control 2nd Limit (MRH)' at '0.00 A', and 'Anode Control 1st Limit (MRL)' at '0.00 A'. At the bottom, there's a 'SELECT RECIPE TO EDIT' section (5) with a keypad showing '0', a 'RECIPE FUNCTION' section (6) with 'Cancel', 'Upload', and 'Save' buttons, and a 'RECIPE MENU' button (7) with a home icon.

Screen 7: Recipe Edit Screen

- 1 Menu name
- 2 The name of the selected product, and its conveyor setting, are displayed, which can be altered.



Note: Press the conveyor speed button to make changes to conveyor settings other than its speed. The Conveyor menu will appear. Refer to Conveyor Menu on page 67.

- 3 The button is a toggle button to enable or disable RF power. The adjacent green indicator indicates the state of RF power.



Note: When RF power is enabled, the settings are displayed and can be edited. When RF power is disabled, its settings are not visible for alterations.

- 4 The various RF settings that are displayed when the RF power is enabled, and can be altered.
- 5 The up and down arrow buttons to assign a recipe number (displayed between them). Each recipe is identified with a number internally.



Note: A maximum of 30 recipes can be stored.

- 6 These buttons are used for alteration management. Changes are discarded when cancel button is pressed, and they are saved to the current recipe when save button is pressed. The upload button displays the current settings in use.



Warning: When the recipe number is changed before edits made to a recipe are saved, the new recipe settings are displayed, and the previous edits are lost.

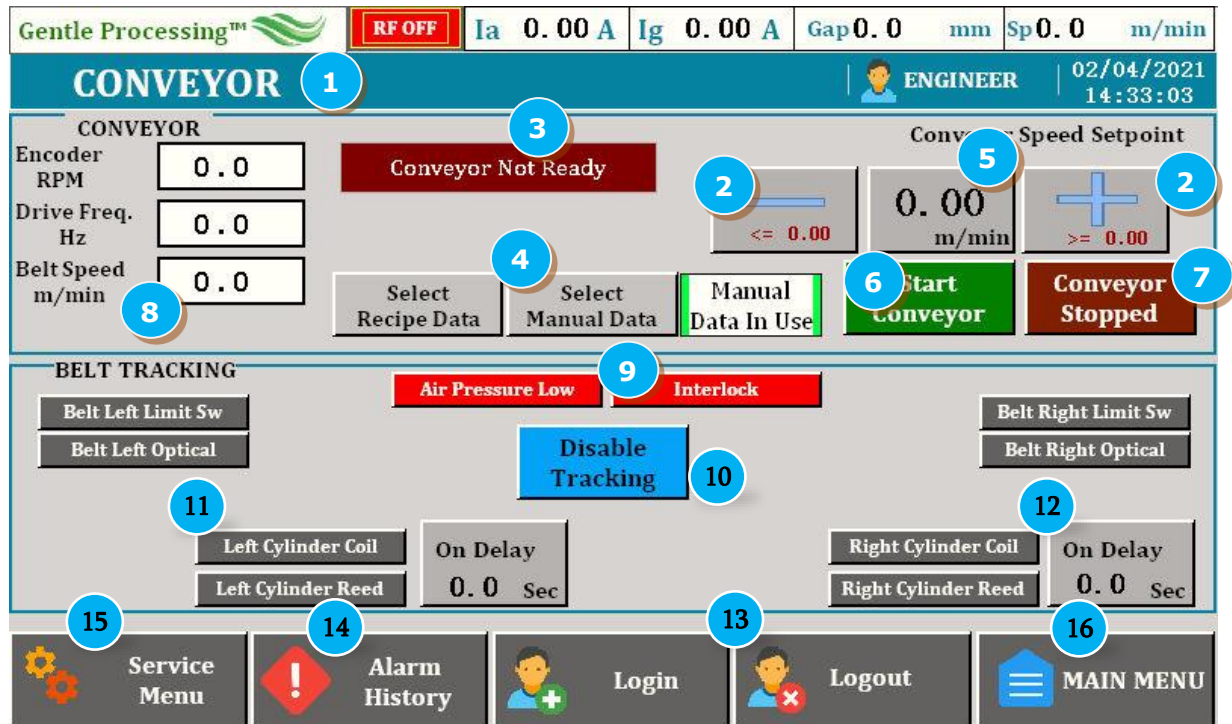
- 7 Exit button to return to previous menu.

CONVEYOR MENU

The conveyor screen provides controls to start, stop, and set conveyor speed. Controls are also provided to set up the belt speed feedback encoder. This menu is not applicable for the Operator user. Only the Supervisor, Engineer, and Gentle Processing users can work on this menu and set the conveyor options.

The screen is divided into two sections, the first section is for conveyor control, and the second section is for VSD/Encoder scaling settings.

These are explained next.



Screen 8: Conveyor Menu Screen

- 1 Menu name.
- 2 Displays, and allows setting of belt speeds and motor drive frequency. The +/- keys are used to adjust the speeds and drive frequency during Manual operation. The numbers in the +/- keys indicate high and low limits set for conveyor speed.
- 3 Displays the status of the conveyor, Conveyor Not Ready or Conveyor Running or Conveyor Ready.
- 4 Toggle button to switch between Manual and Recipe selection.



Note: Once the conveyor is started, the speed setpoint will not change even if the mode is changed from Manual to Recipe or vice versa.

- 5 Allows Manually Start and Stop of the conveyor at speed set via Manual data mode or Recipe data mode
- 6 Start Conveyor: Press the Start Conveyor button to start the conveyor. The conveyor drive starts and the start button text changes to Conveyor Running
- 7 Stop conveyor: Press the Stop Conveyor button to stop the conveyor. The conveyor drive stops and the button text changes to Conveyor Stopped

- 8 Displays the conveyor belt speed from the conveyor control section. The other values and settings are the counted pulses from the conveyor belt speed encoder, which indicates encoder current count rate.
- 9 Displays the status of the conveyor tracking Interlocks, air pressure etc.
- 10 This toggle switch allows enabling or disabling the automatic belt tracking.
- 11 Tracking controls Left side: Optical sensor/ limit switch, tracking coil / cylinder status & delay timers.
- 12 Tracking controls Right side: Optical sensor/ limit switch, tracking coil / cylinder status & delay timers.

Belt Tracking control block

Belt Left / Right Limit Sw

Indicates the status of end Limit Switch. If the Belt limit switch is hit a warning message is displayed.

If this Limit Switch is activated, it indicates that the belt tracking is out of control and requires operator attention.

Belt Left / Right Optical

Indicates the status of Optical sensor used to track belt position.

Left / Right Cylinder Coil

Indicates the status of PLC output for the respective cylinder.

Left / Right Cylinder Reed

Indicates the status (ON/OFF) of the reed switch mounted on respective cylinders. Reed switch helps to check proper operation and position of the cylinders.

On Delay (Sec)

Set this On Delay timer in sec for the time the cylinder should be in extracted state after the belt is tracked. Once the timer is completed the extracted cylinder retracts.

Belt Tracking can be disabled only for maintenance purpose when the RF cycle is in OFF state to check the operation of individual tracking cylinders. Once the Tracking is disabled, will show up the Toggle Switches to operate the Tracking Cylinders manually.

- 13 Login/Log out buttons. Also shows current logged in User's name.
- 14 Button to display Alarm list.
- 15 Exit button to return to Service menu.
- 16 Button to display Main Menu screen.

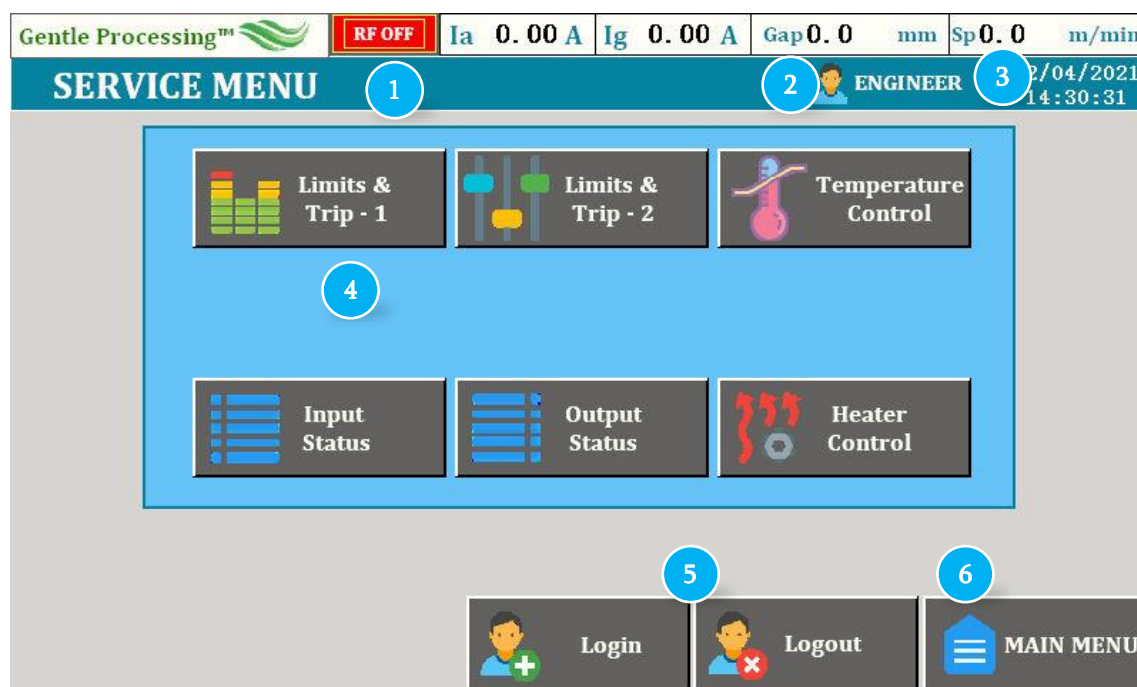
OPERATIONAL STATE INDICATION MESSAGE

Message	Meaning
	An Interlock is preventing the conveyor from reaching Ready condition. Look at the Alarm list for active alarms.
	The Conveyor is Ready and can be Started by Pressing the "Start Conveyor" button.
	Indicates the Conveyor Start button is pressed and waiting for minimum threshold speed to achieve.
	Indicates the Conveyor is running.

SERVICE MENU

The Service setup permits to set the critical levels at which the RF power is disabled following the detection of flash over or other fault conditions and various parameters which affect the running of the machine

Access to all the Service function screens is limited to Supervisor, Engineer, and Gentle Processing User roles only.



Screen 9: SERVICE MENU

1. Menu Name.
2. Displays Logged in user's name.
3. Date & Time
4. Service screens navigation buttons
5. Login /Logout Buttons
6. Exits current screen and goes back to the Main Menu screen

OVEN LIMITS AND TRIPS - I

This screen is used during the commissioning and testing of the machine. There are multiple sections in this screen. Gentle Processing and Engineer users have permissions to work on this screen. Refer to the Test Report in the *Maintenance Manual* for actual values.



Warning: The anode current trip, and grid under current trip setting adjust buttons are visible to, and can *only* be operated by **Gentle Processing** user.

Screen 10: Oven Limits and Trips - I Screen

1. Menu name

2. Anode Current Trip (Ia Trip) set to 0.0

3. Grid Current Trip (Over Trip) set to 0.0

4. Electrode Gap settings (Index Switch Height, Max Gap Limit, Min Gap Limit, Position Dead Band, RF Switch Point, Slow Speed @ mm, Slow Speed @ Ia, Slow Speed, Fast Speed)

5. Meter Scaling (Anode Current, Grid Current) set to 0.0

6. Electrode Control (Fast Speed, Slow Speed, Actual Gap, Setpoint) set to 0.0

7. Test Function (RF OFF, OPERATOR PANEL, ENABLED, Control Button Selection, Grid Undercurrent Trip)

8. Login/Logout buttons

9. Alarm History button

10. Service Menu button

11. MAIN MENU button

Screen 10: Oven Limits and Trips - I Screen

1. Menu name
2. Sets the anode current level at which the GP™ machine heating is switched off and a trip alarm generated. The trip prevents the triode valve from sustaining damage due to an overload condition beyond its maximum ratings.

Ia Trip

Sets the anode current level at which the GP™ machine heating is switched off and a trip alarm generated. The trip prevents the triode valve from sustaining damage due to an overload condition beyond its maximum ratings.

Min M-R Diff.

Sets the difference between Anode 1st limit and 2nd limit value. The actual difference between the two set-points is typically 1.0 Amps (if the settings are too close, the electrode may hunt – (i.e. travel upwards and downwards rapidly)

3. The maximum and minimum permissible grid current values are set. Beyond these values, a trip will occur and prevents the triode valve from sustaining.

Over Trip

The maximum permissible grid current values are set. Beyond these values, a trip will occur and prevents the triode valve from sustaining.

Under Trip

The minimum permissible grid current values are set. Below these values, a trip will occur and prevents the triode valve from sustaining.

4. The various values of the electrodes such as their distances, upper and lower limits, gap, and other related information is added.

Index Switch Height

This is the distance between the electrodes when the top electrode is driven up to, and stopped by, the upper limit switch. The electrode gap distance is preset with this value when the upper limit switch is operated during the home sequence. A home sequence must be carried out when this value is altered for it to take effect.

Max Gap

The operational software upper limit. This is the position the electrode drives to when the GP™ machine is off (no RF).

Min Gap

The operational software lower limit. This is set so that when the electrode is driven down it stops at this gap before it hits the lower limit switch.

Position Dead Band

The distance around the setpoint where no action occurs i.e. any change in setpoint has to be greater than the dead band/2 from the electrode gap for it to initiate a change in the electrode gap.

RF Switch Point

This is the position of the electrode drives to switch the GP™ machine on.

Slow Speed @ mm

Set the band where the electrode drive slow speed is activated wrt set electrode gap.

Slow Speed @ Ia

Set the band where the electrode drive slow speed is activated wrt set anode current.

Slow Speed

Set the electrode drive speed when the electrode is in the slow speed band.

Fast Speed

Set the electrode drive speed when out of slow speed band.

5. This parameter allows setting of scaling factor used to calibrate the digital readings of Anode and Grid Current. Using the Analog Meters as the master, the scaling factor can be adjusted so that the digital reading of the current on the HMI matches that of the analogue meters
6. The actual gap is displayed. The manual setting of the setpoint control is done using the Setpoint Control buttons and the arrows. The Home Electrode button is used to initiate homing. The button changes to Homing during the homing sequence. Refer to Homing the on page 56 for more information.
7. These buttons are for conducting tests without allowing a trip to occur when the RF heating is changed. The control button selection button allows starting and stopping of the GP™ machine using the push buttons mounted adjacent to the meters on the generator. The grid undercurrent trip button allows control and open circuit H.T. tests to be carried out without the GP™ machine tripping. On exiting the screen both the controls are restored back.
8. Login/Log out buttons. Also shows current logged in User's name.
9. The alarm list (refer to Alarm History) displays the active and inactive alarms.
10. Exit button to return to Service menu.
11. Button to display Main Menu screen



NOTE: Both the stop buttons are operational irrespective of the control mode.

OVEN LIMITS AND TRIPS – II

Gentle Processing and Engineer users have permissions to work on this screen. Refer to the Test Report in the Maintenance Manual for actual values.

Screen 11: Oven Limits and Trips - II Screen

1. Menu Name
2. Displays the conveyor belt speed from the conveyor control section.
3. The minimum speed threshold set the speed at which the GP™ machines can operate. When the conveyor speed equals or exceeds this value the GP™ machine heating is enabled.
4. The encoder speed scaling value is applied to the encoder counter value to convert it to metres/min. The value is obtained by running the conveyor and measuring the speed of the belt. The encoder speed scaling value is adjusted until the indicated speed (m/min) matches the measured speed.
5. The Mtr@50Hz setpoint scaling parameter is applied to conveyor speed setpoint. The value is obtained by running the conveyor at 50Hz drive frequency and entering the length travelled by the conveyor (m/min). During this test, VSD gain parameter should be kept at 1.0.
6. The VSD Gain setpoint is used to fine adjust the conveyor speed scaling.
7. The maximum number of restarts and the time between each restart (recycle) is

set. The Monitor Interval sets the time period in seconds within which to count a restart, initiated by the first trip.

8. Extraction Fan settings
9. The time for which the Triode valve is on.
10. The time for which the RF power is on
11. Disable MRL MRH Function.



Warning: MRL/MRH Disable setting buttons are visible to, and can *only* be operated by **Gentle Processing** user. This function is used to operate the equipment beyond recommended settings to check the OCR

12. Login/Log out buttons. Also shows current logged in User's name.
13. The alarm list (refer to Alarm History) displays the active and inactive alarms.
14. Exit button to return to Service menu.
15. Button to display Main Menu screen

INPUT STATUS

DIGITAL INPUT STATUS

This screen lists all the input of the PLC and displays their current status.

A green lamp indicates active or high state.

This screen helps the user analyse faults if any in the machine, and check the current operating condition of the machine.

The inputs with green colour text indicate that the respective input should be high or active (represented by "Green Colour" square) for healthy condition of the machine.

INPUT STATUS - 1

[RACK 0] PLC: 1766-L32 BXBA 20 DI

I:0/0 IN0	Electrode Encoder A 500 PPR - B3	I:0/6 IN6	Trojan Door Safety Relay - K6
I:0/1 IN1	Electrode Encoder B 500 PPR - B3	I:0/7 IN7	Emergency Stop PB RH Infeed - S2
I:0/2 IN2	Conveyor: Encoder-A 500 PPR - B4	I:0/8 IN8	Emergency Stop PB LH Outfeed - S4
I:0/3 IN3	Conveyor: Encoder-B 500 PPR - B4	I:0/9 IN9	Emergency Stop PB RH Outfeed - S5
I:0/4 IN4	Solenoid Door Safety Relay - K3	I:0/10 IN10	Emergency Stop PB LH Infeed HMI - S6
I:0/5 IN5	Trojan Safety Guard Relay - K4 & K7	I:0/11 IN11	RF ON PB - S16

MAIN MENU **OUTPUT STATUS** **NEXT**

Screen 12: Digital Inputs Status - I Screen

1. Menu name.
2. List of Inputs with status indicator.
3. Navigates to Output status screen.
4. Returns to the Main menu.
5. Navigates to the next Input status screen.

OUTPUT STATUS

DIGITAL OUTPUT STATUS

This screen lists all the output of the PLC and displays their current State.

An Output with green lamp represents active or high state.

This screen helps the user analyse faults if any in the machine and check the current operating condition of the machine.

The screenshot shows the 'OUTPUT STATUS - 1' screen. At the top, there's a status bar with 'Gentle Processing™', 'RF OFF', and various process parameters: 'Ia 0.00 A', 'Ig 0.00 A', 'Gap 0.0 mm', and 'Sp 0.0 m/min'. Below this, the title 'OUTPUT STATUS - 1' is displayed with a user icon and 'ENGINEER' and a timestamp '02/04/2021 14:34:44'. The main content area is titled '[RACK 0] PLC: 1766-L32 BXBA 12 DO' and contains a table of 12 digital outputs. The table is divided into two columns. The first column lists outputs 0:0/0 to 0:0/5, and the second column lists outputs 0:0/6 to 0:0/11. Each output has a green status indicator (a green square with a white '1') and a description of the output and its relay. At the bottom, there are three navigation buttons: 'MAIN MENU' (labeled 4), 'INPUT STATUS' (labeled 3), and 'NEXT' (labeled 5). The 'INPUT STATUS' button has a blue arrow pointing left, and the 'NEXT' button has a blue arrow pointing right.

Output Address	Output Description	Output Address	Output Description
0:0/0 OUT0	Right Solenoid Extract (Out) Relay - K39/S64	0:0/6 OUT6	Warning Lamp Relay - K47
0:0/1 OUT1	Left Solenoid Extract (Out) Relay - K38/S63	0:0/7 OUT7	Electrode MPCB Shunt Coil Relay - K37
0:0/2 OUT2	Emergency Stop Safety Reset Relay - K10/K4	0:0/8 OUT8	HEATER FAN 1 CONTACTOR RELAY
0:0/3 OUT3	Ready Lamp (HMI Station) Relay - K26/P6	0:0/9 OUT9	HEATER FAN 2 CONTACTOR RELAY
0:0/4 OUT4	Fault Lamp (HMI Station) Relay - K27/P7	0:0/10 OUT10	Heater #1 Relay - K33/Q105
0:0/5 OUT5	Hooter Relay - K28/P13	0:0/11 OUT11	Heater #2 Relay - K34/Q107

Screen 13: Digital Outputs Status Screen

1. Menu name.
2. List of Outputs with status indicator.
3. Navigation buttons to go to Input status screen.
4. Navigates to the Main Menu.
5. Navigates to the next Screen

TEMPERATURE CONTROL

Product moisture can be also controlled by reading product actual temperature by 6 optical temperature sensor mounted at the conveyor outfeed.

If the product actual temperature rises then the setpoint temperature, the electrode is moved up, resulting less power applied to the product. This process is repeated after every end of envelop time. Even after applying less power, the product temperature does not falls to setpoint temperature, the next correction factor is applied I.e increase in conveyor speed.

Similarly if the product actual temperature is low then the setpoint temperature, the electrode is moved down, in order to apply more power to the product. This process is repeated after every end of envelop time. Even after applying more power, the product temperature does not rises to setpoint temperature, the next correction factor is applied I.e decrease in conveyor speed.

The screenshot shows the 'TEMPERATURE CONTROL' screen. At the top, there's a status bar with 'Gentle Processing™', 'RF OFF', and various process parameters like 'Ia 0.00 A', 'Ig 0.00 A', 'Gap 0.0 mm', and 'Sp 0.0 m/min'. The main title 'TEMPERATURE CONTROL' is prominently displayed. Below it, there are several sections: 'TEMPERATURE' with 'Product' and 'Trigger' setpoints at 0.0 °C; a 'DISABLED' toggle with a 'PRESS TO ENABLE' button; 'Delay to reach product @ Outfeed' and 'Envelop Time' both at 0.0 Sec; 'TEMPERATURE LOW' and 'TEMPERATURE HIGH' correction factor sections with 'Ia' and 'Conv Speed' adjustments; a 'SELECTION' row with four 'OFF' buttons for different temperature sensors; 'OPTRIS' readings for each sensor; a 'SELECT COUNT' of 0.0; a 'PRODUCT TEMP @OUT-FEED' of 0.0; a green 'MINT 8AI OK' indicator; and a bottom navigation bar with icons for 'Service Menu', 'Alarm History', 'Login', 'Logout', and 'MAIN MENU'.

Screen 14: Temperature Control Screen

1. Menu Name.
2. Toggles to enable or disable the temperature control mode.
3. Required product temperature setpoint. Trigger value is the setpoint from where the temperature correction starts.



Note: *Product temperature setpoint should be always greater trigger setpoint.*

Trigger setpoint should be greater than ambient temperature.

4. Delay to reach the product at the outfeed is the initial time required for the product to reach at the conveyor outfeed, required to start the temperature control mode.

Envelop Time defines the time period between correction applied and next correction to apply.

5. Temperature Low

Correction factor to apply if the product actual temperature is below the required product temperature.

6. Temperature High

Correction factor to apply if the product actual temperature is above the required product temperature.

7. Temperature sensor selection switches. Depending upon the product requirement, number of temperature sensors can be selected or not selected. Average product temperature calculation will be made based on only the temperature sensors selected.

8. Actual temperature read out of individual sensors.

9. Average temperature readout of product.

10. Indicates the status of temperature communication module and average temperature range.

11. Exit button to return to service menu.

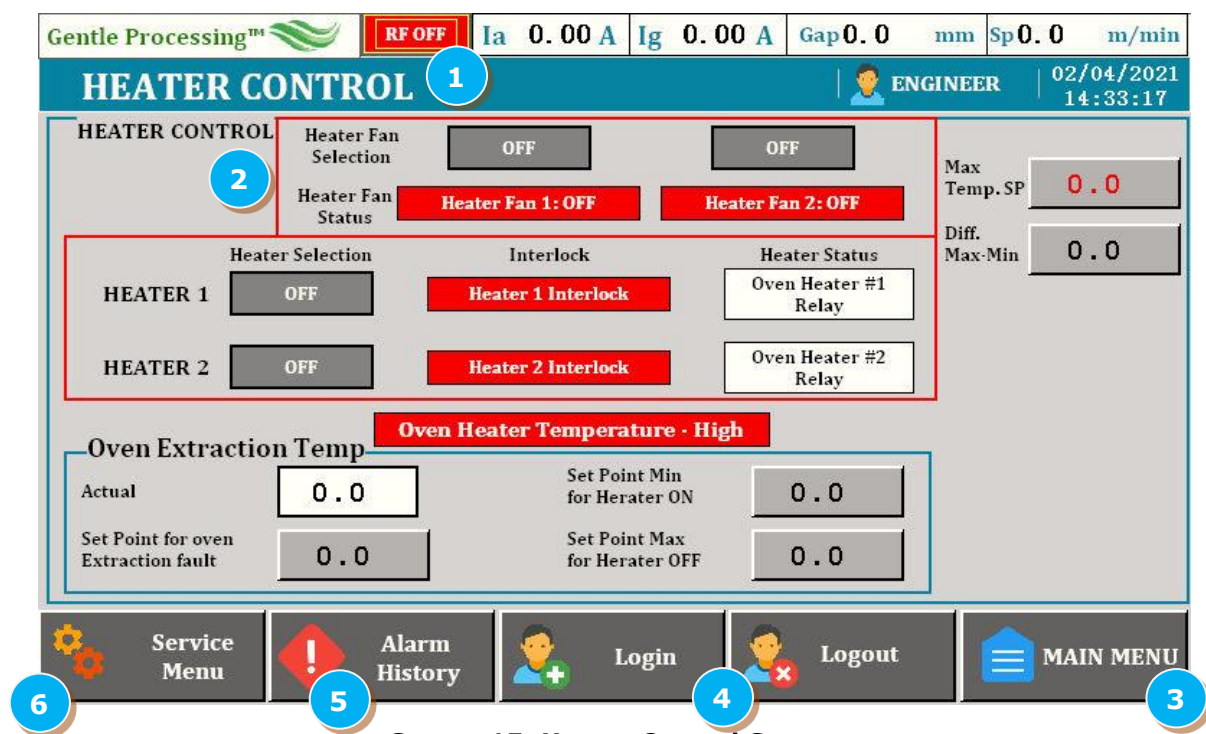
12. User Login/Logout button.

13. The alarm list (refer to Alarm List on page 140) displays the active and inactive alarms.

14. Exits and returns to Main menu screen.

HEATER CONTROL

This machine comes equipped with two electrical heater arrays. Air is forced to flow over the heaters and over the product via the base electrode. This air is eventually extracted from the machine via the extraction system. The heater screen allows the user to turn the heaters on or off.



Screen 15: Heater Control Screen

1. Menu Name
2. Heater and heater fan Control Block
Allows selection of the Heaters and Fans to be ON or OFF. Also can set the heater fan speed in Hz with actual fan speed indication.
3. Exits and returns to Main menu screen
4. User Login/Logout button.
5. The alarm list (refer to Alarm List on page 140) displays the active and inactive alarms.
6. Exit button to return to service menu.

NETWORK STATUS

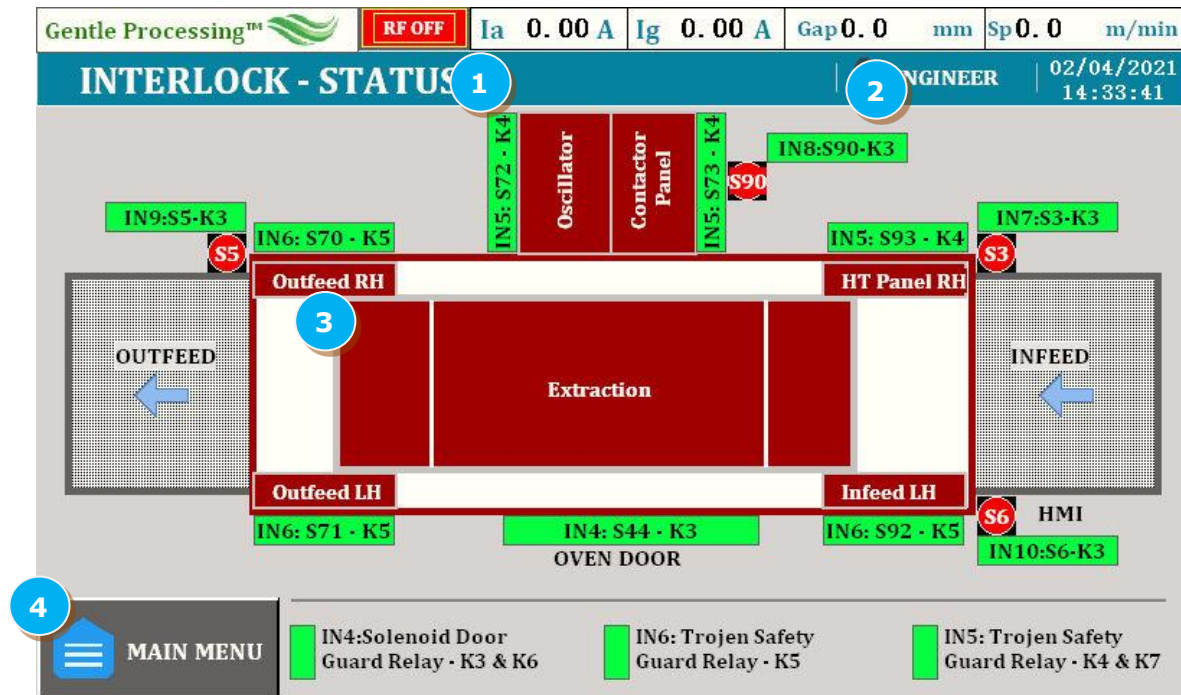
The communication protocol between the PLC and various devices is over Ethernet IP.

IP ADDRESS

S.NO	MODULES	IP Mask
PLC		
1	MicroLogix 1400 1766-L32BXXB PLC	192.168.002.160 255.255.255.000
HMI		
2	Panel View 800 HMI 10"	192.168.002.164 255.255.255.000
Drives		
3	Conveyor Drive PF525	192.168.002.165 255.255.255.000
4	Extraction Drive PF525	192.168.002.166 255.255.255.000
5	Electrode Drive PF525	192.168.002.167 255.255.255.000
EWON		
6	eWON Flexy 201	192.168.002.095 255.255.255.000
EWON		
RTD Module		
7	Mint 8AI	192.168.002.110 255.255.255.000

INTERLOCK STATUS SCREEN

This screen shows graphical status of the machine Doors and Emergency Stop Push buttons.



Screen 16: Machine Status Screen

1. Menu Name
2. Alarm Banner showing active alarms in the machine
3. Representation of the machine with all doors switches and emergency stop push buttons.

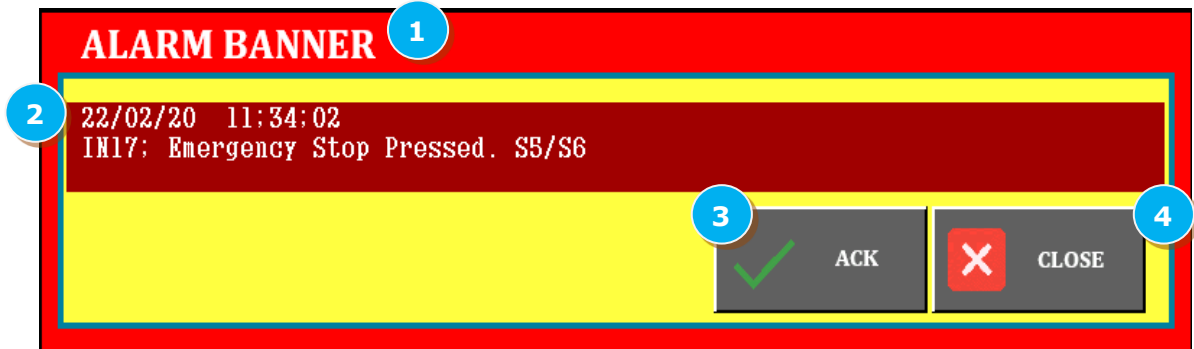


Red lamp indicates fault condition and green lamp indicates healthy condition

4. Exits and returns to Main menu screen

ALARM BANNER

The alarm banner is displayed when an alarm is triggered. The most recent alarm is displayed first.



Screen 17: Alarm Banner Screen

1. Displays the Menu Name.
2. Alarm banner displays the Alarm message.
3. Press to remove the alarm from the banner. The alarm is not removed from the alarm list. However, the next alarm is displayed.
4. Clicking this indicates that the user has acknowledged the alarm.

ALARM HISTORY

The alarm screen is visible for all users. However, only the Gentle Processing and Engineer users can operate the **clear history** buttons.

The alarm list screen displays a history of active and inactive alarms. The time and date the alarm was triggered is also displayed.



NOTE: A maximum of 100 alarms are displayed and stored.



Screen 18: Alarm List Screen

1. Menu name.
2. Alarm display area.
 - Red colour row indicates Active Alarms.
 - Green colour row indicates Inactive Alarms.
 - Selected alarms are indicated in pink background.
3. The arrow keys are used to scroll up and down to see the alarm messages.
4. Removes all alarms in the alarm list screen including active alarms.
5. Silence the Audio Alarm
6. Exits and returns to Main menu screen.

CHAPTER 8. Maintenance

Most failures in the GP-15 can be prevented with good housekeeping and maintenance.

This chapter covers the maintenance tasks to be followed for the long-term life and running of the GP™ machine. Maintenance tasks are divided in to general maintenance and periodic maintenance, as well as component based maintenance for ease of following them.



Warning: Due to the very high voltages involved, only a QMTI GP™ machine trained engineer is permitted to conduct live fault finding within the GP™ machine.

All maintenance and service personnel must refer to and follow all instructions in the *Safety Chapter on page 14*.

QMTI GP™ machine cannot assume responsibility of the safety of maintenance personnel unless directly deputed by QMTI.

The maintenance engineer should ensure that all guards, safety interlocks and earth circuits are correctly fitted, fully functional and the machine is in a safe condition to be operated after completion of the maintenance activity.



Safety

- Fully isolate the GP™ machine prior to maintaining or servicing the machine.
- Open and lock off the customer supplied primary isolator
- Open and lock off the Control Panel circuit breaker and H.T. circuit breaker.
- Each person working on the equipment must individually lock off the circuit breakers and attach a Lock-out safety tag or label as shown on the right.
- All maintenance personnel must understand that:



Servicing LIVE industrial control equipment is highly dangerous. Severe injury or death may result from electrical shock or burns received due to contact with live electrical components.

‘ ELECTRICAL COMPARTMENT ACCESS

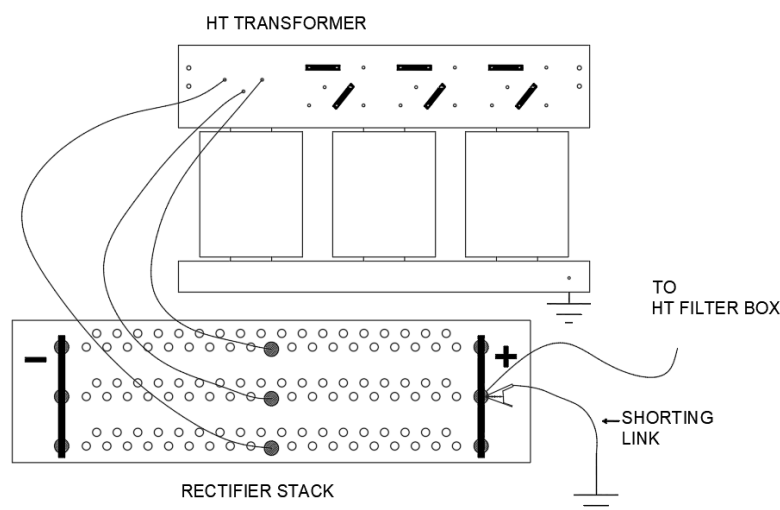
Switch off the customer supplied primary isolator. Lock off the Control Panel circuit breaker and the H.T. circuit breaker. Attach a suitable lock off tag.

Access to the Contactor Panel is via a interlock mechanism

To open certain cabinets, the door interlocks must be used. For details on its operation refer Solenoid Door Lock on *page 48*



Warning: Discharge the High Voltage Capacitors when accessing the H.T. and R.F. compartments. Components that hold a charge after switching off are labelled adjacent to their location. Discharge with a suitable grounding probe before performing maintenance or servicing





Warning: Always discharge the anode line of the triode valve using a grounding probe before connecting the shorting link.

A temporary safety earth shorting link **MUST** be connected between the HT lead through capacitor and suitable earth point as shown in the example above



NOTE: Remove this lead when the work is completed. **DO NOT** switch on the control circuit breaker until the link is removed

LOCATION AND EXPLODED VIEWS OF MAJOR ASSEMBLIES

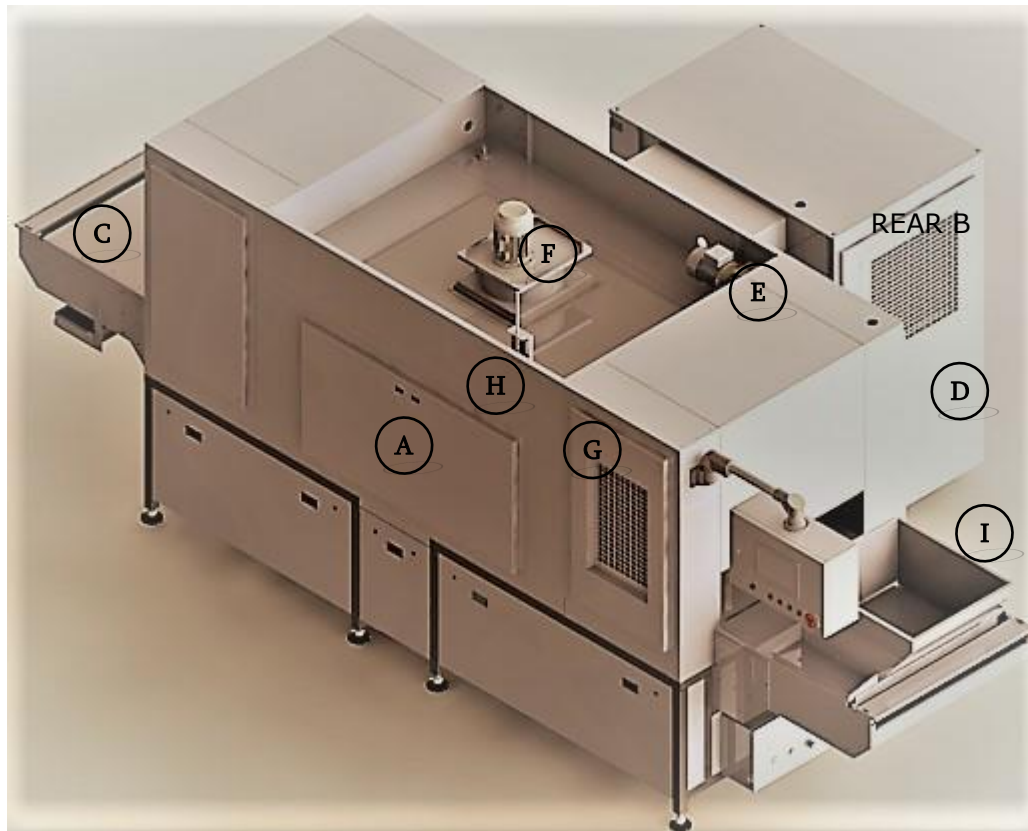
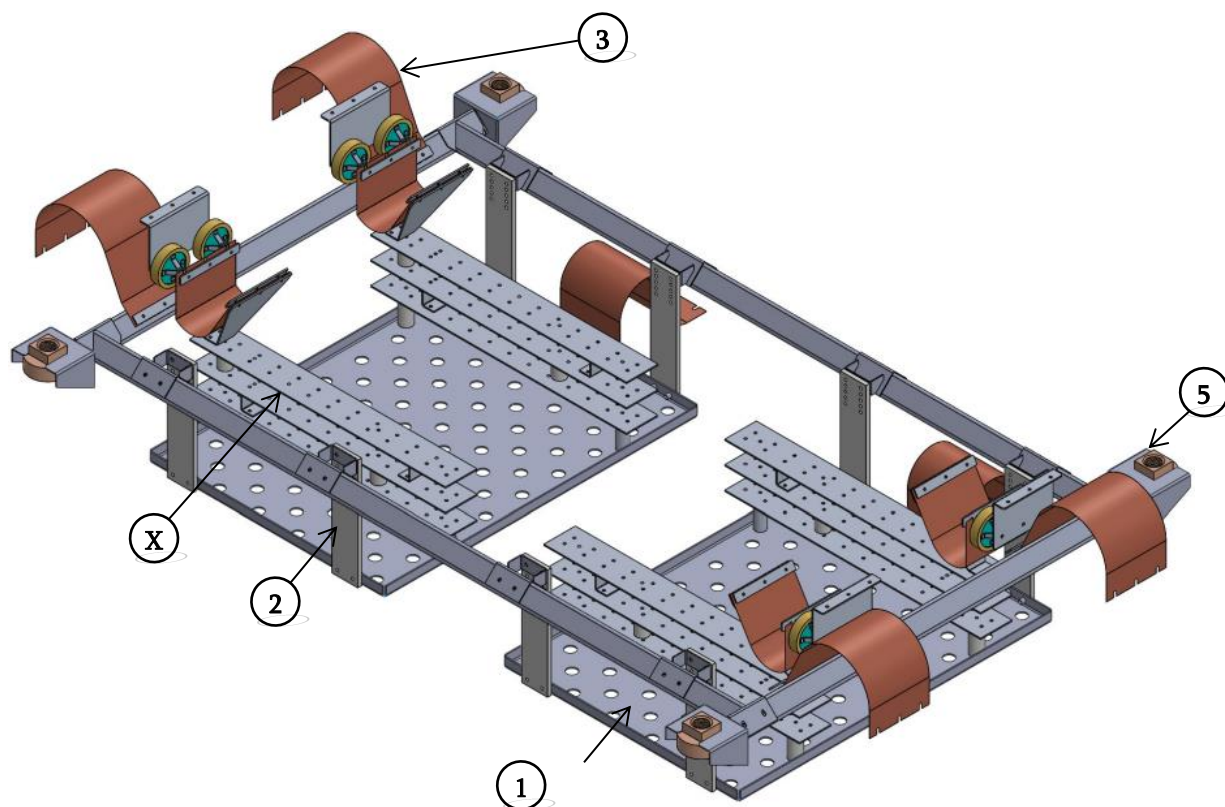


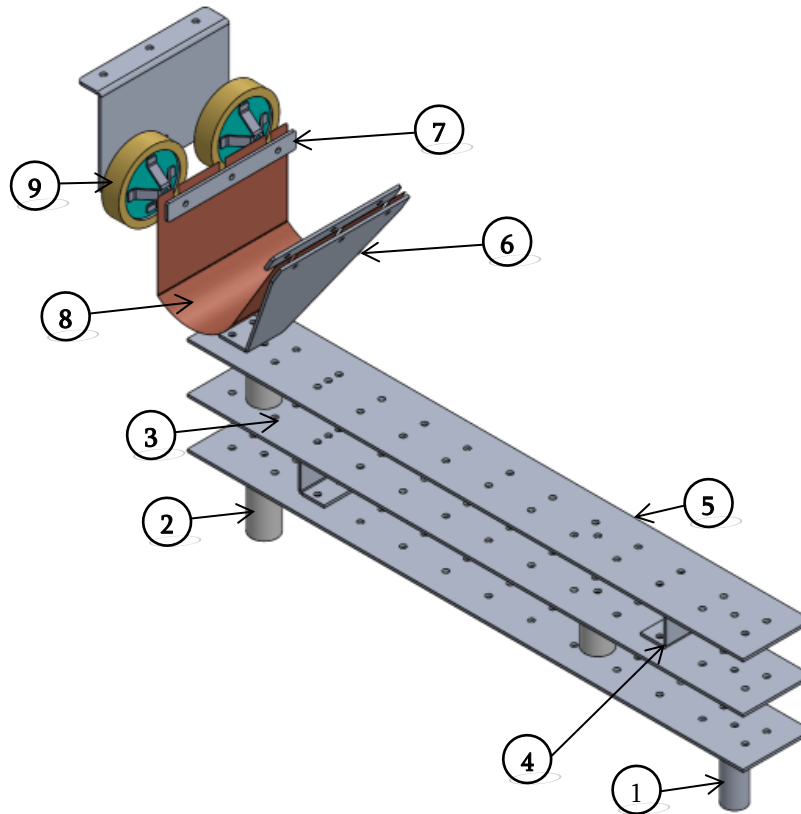
Illustration 25: General Arrangement showing compartment references

OVEN COMPARTMENT (DOOR A)



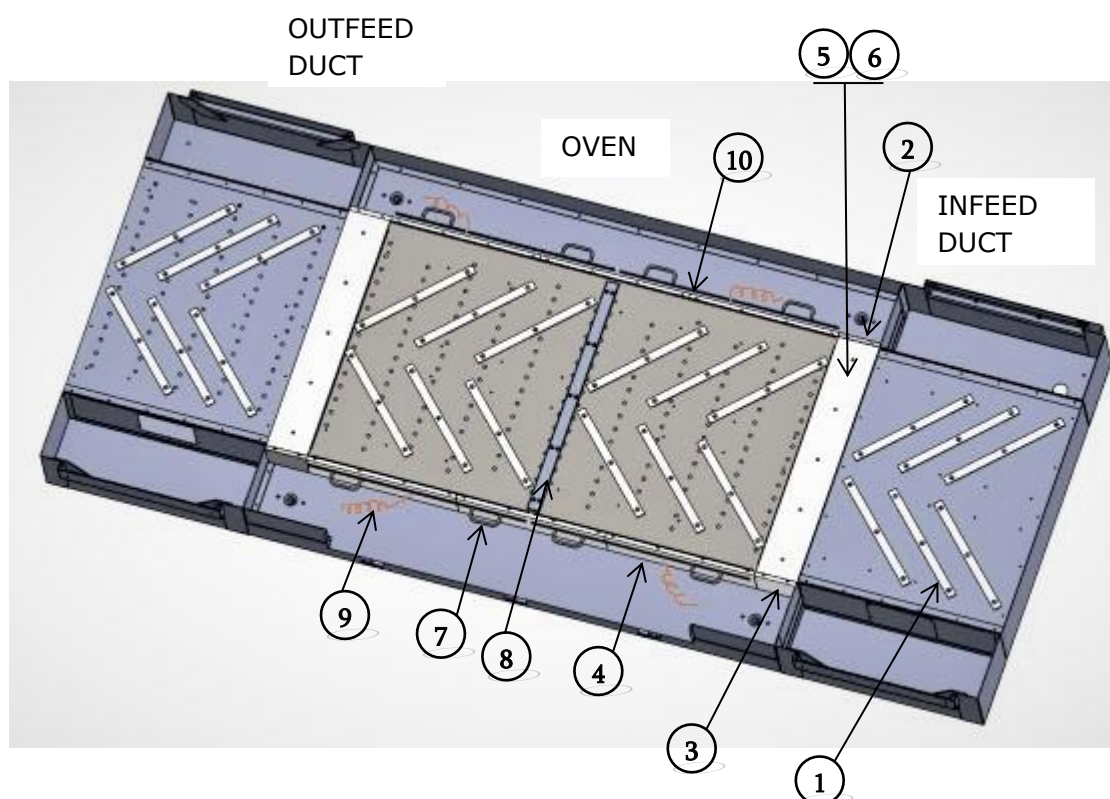
ITEM	QTY	PART NO	DESCRIPTION
1	2	11.05.433	TOP ELECTRODE
2	8	10.00.091	SILICON SUPPORT
3	6	10.07.874	FEED STRIP 18 ins
4	6	11.05.432	ELECTRODE FRAME LENGTH BEAM
5	4	10.07.201	ELCTRODE LEAD SCREW NUT

DETAILS OF X



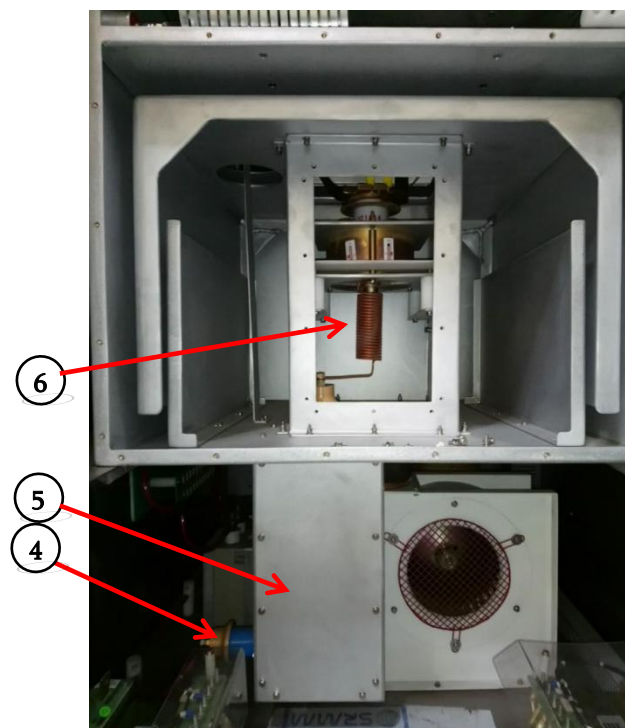
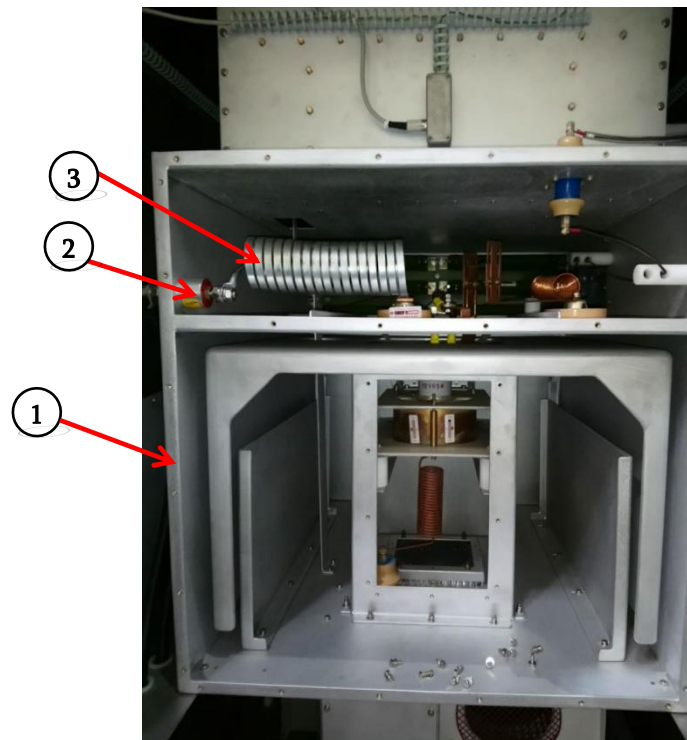
ITEM	QTY	PART NO	DESCRIPTION
	4	80.00.525	ALUMINIUM SPACER DIA 25X75
2	4	81.20.005	TEFLON SPACER DIA 25X75
3	8	81.20.006	TEFLON SPACER DIA 25X53
4	8	10.07.644	TUNING BRIDGE
5	12	10.07.643	TROMBOCONE
6	4	11.03.519	TUNING FEED PLATE
7	16	10.00.449	KEEPER PLATE EARTH RETURN
8	4	10.00.273	FEED STRIP 12ins
9	8	12-0010-0000	CAPACITOR DISC 1000Pf 20% 10KV

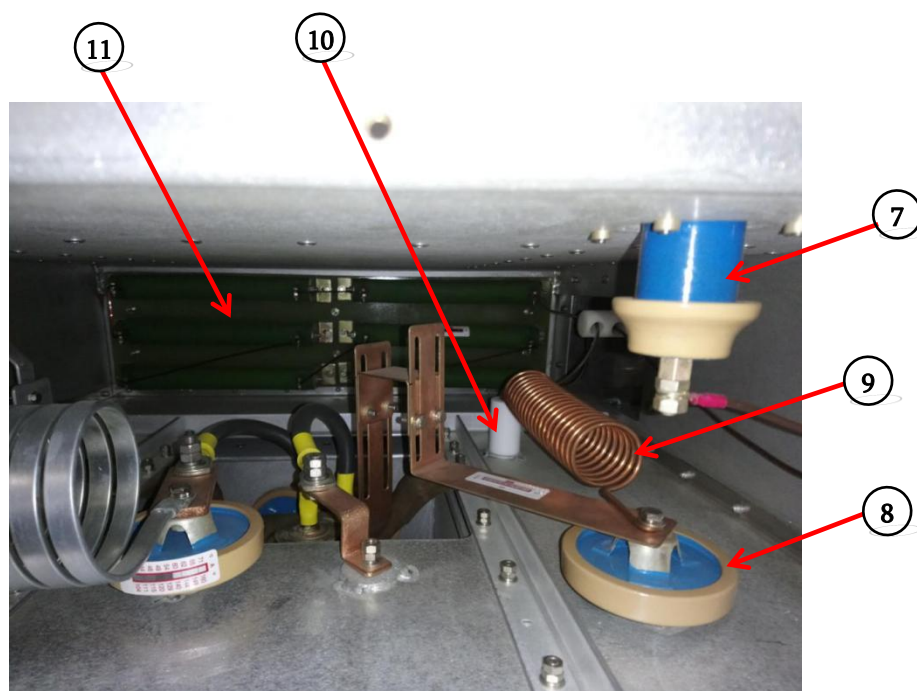
BOTTOM ELECTRODE ASSEMBLY



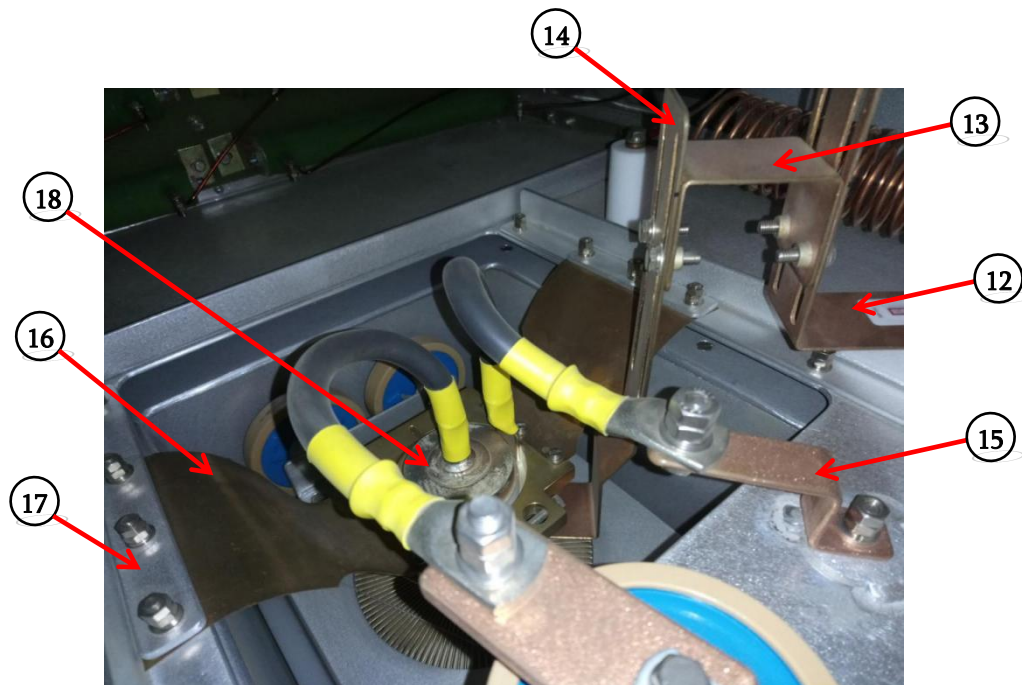
ITEM	QTY	PART NO	DESCRIPTION
1	24	41-0005-0692	WEAR STRIP TEFLON
2	2	41-0005-0693	SUPPORT 1 LH MATERIAL PET 25thk
3	2	41-0005-0694	SUPPORT 1 RH MATERIAL PET 25thk
4	6	41-0005-0695	SUPPORT 2 MATERIAL PET 25thk
5	2	41-0135-0005	TOP SHEET
6	2	41-0005-0696	SUPPORT FOR TOP SHEET
7	8	11-0126-0010	U HANDLE SS TYPE
8	1	41-0005-0697	JOINING PLATE
9	4	41-0296-0005	CHOKE 18TURN
10	2	21-0042-0180	BOTTOM ELECTRODE

OSCILLATOR COMPARTMENT (DOOR B)

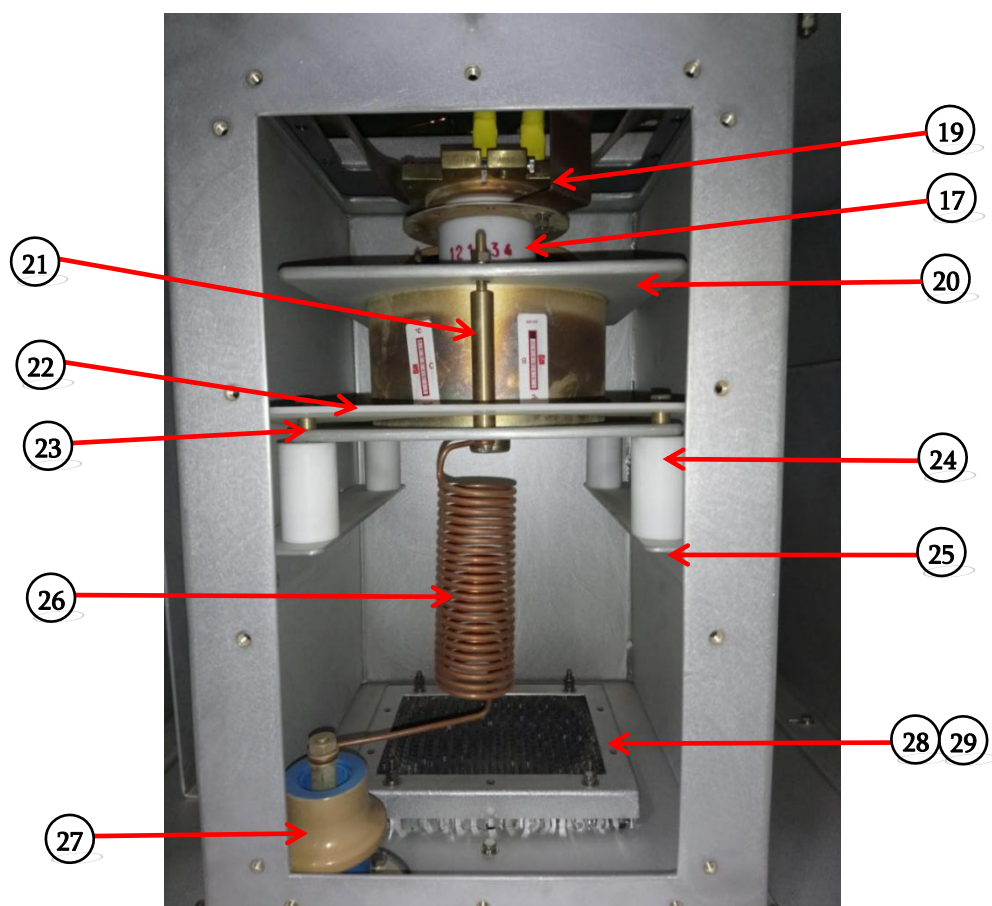




ITEM	QTY	PART NO	DESCRIPTION
1	1	11.04.660	CAVITY OUTER CASE ASSEMBLY
2	1	12-0010-0034	CAPACITOR FEED THROUGH 200A
3	1	11.01.693	CHOKE FOR CAVITY FLAT 9.5 X 3mm
4	2	12-0010-0017	CAPACITOR FEED THROUGH 1000 PF 14 KV
5	1	10.00.241	CHOKE 22 TURN
6	1	12.00.506	HT FILTER BOX
7	3	12-0010-0015	CAPACITOR 1000 pf 10KV/60KVA/30A Type-09 F/T
8	4	12-0010-0000	CAPACITOR DISC 1000 PF 13KV
9	1	10.00.243	CHOKE 19 TURN
10	5	81.20.006	TEFLON BUSH DIA 30 X 53 mm LENGTH
11	6	12-0009-0019	RESISTOR 200 ohms 200 W WIRE WOUND

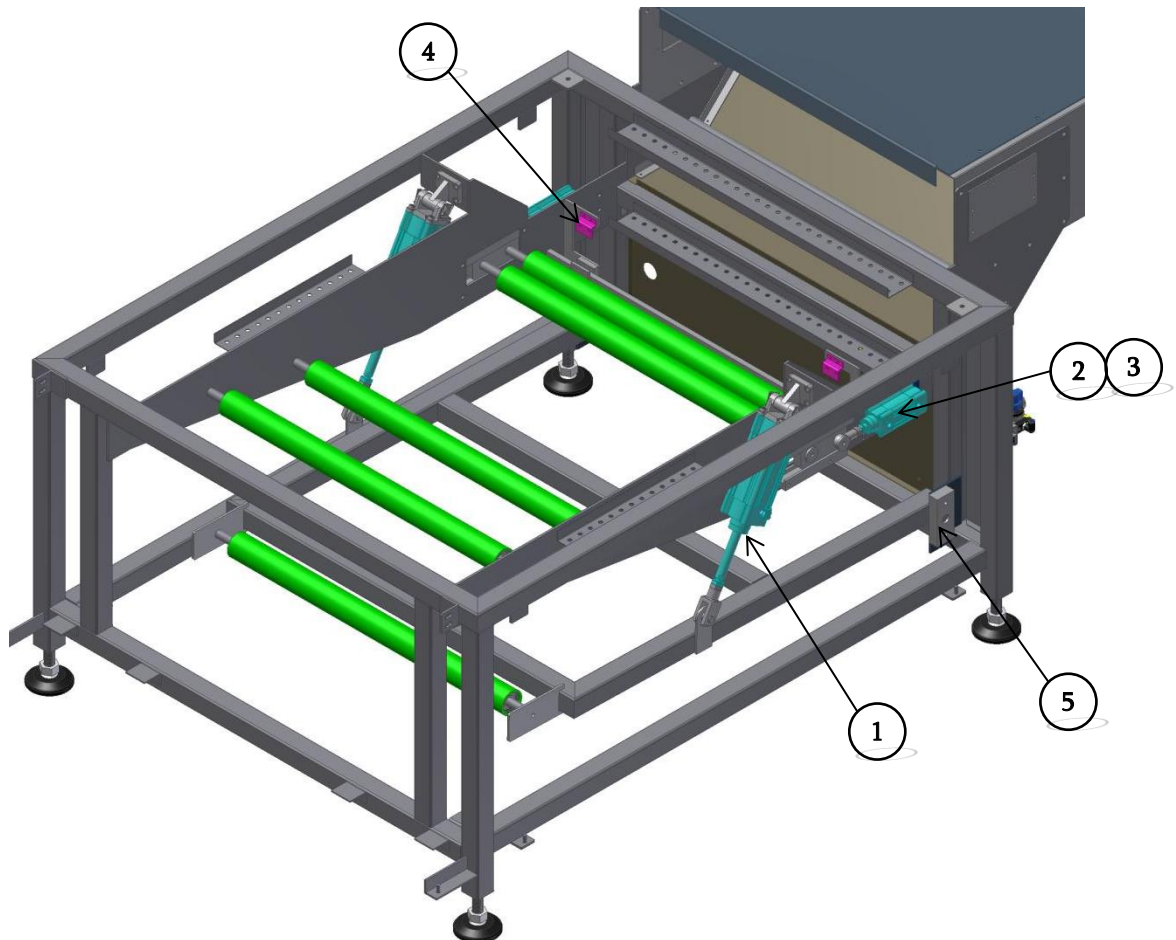


ITEM	QTY	PART NO	DESCRIPTION
12	1	10.05.281	CAPACITOR CONNECTION GRID LOOP
13	1	10.01.218	BRIDGE [GRID LOOP]
14	1	10.02.991	VALVE CONNECTION GRID LOOP
15	1	80.00.480	FILAMENT CONNECTION
16	2	10.02.992	CATHODE H F CONNECTION
17	2	10.01.209	CLAMP STRIP
18	1	12-0024-0013	Trode valve

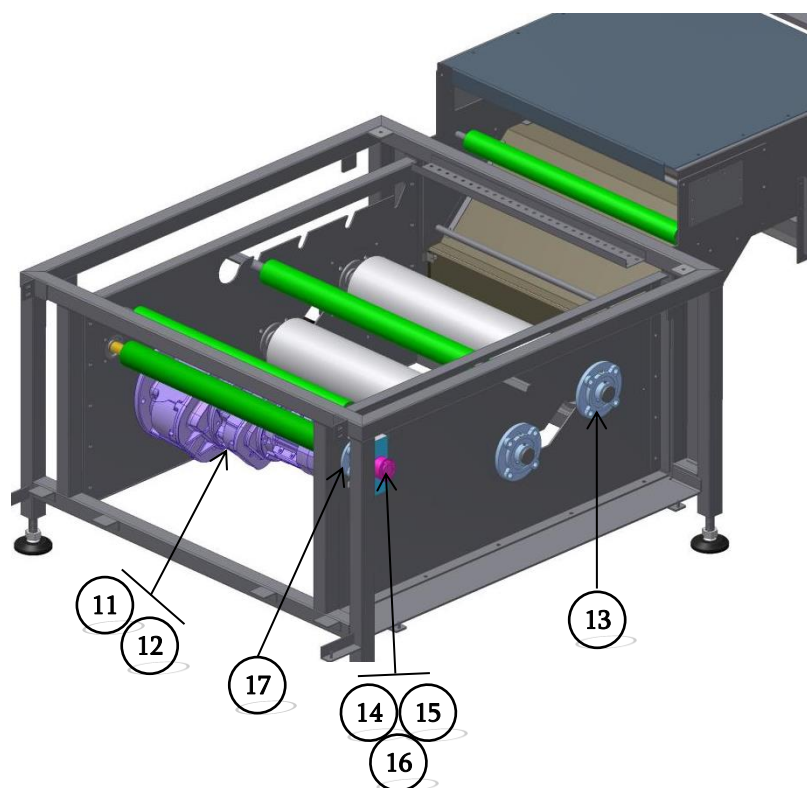
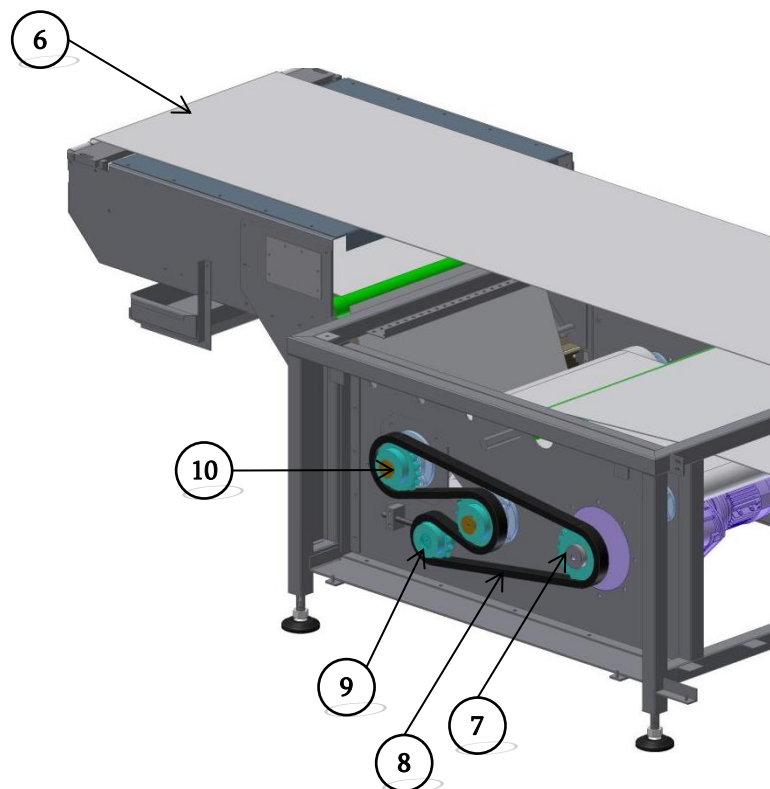


ITEM	QTY	PART NO	DESCRIPTION
19	1	10.02.990	VALVE CATHODE CLAMP
20	1	80.01.780	TOP SEATING PLATE
21	3	10.01.054	VALVE CLAMPING ROD
22	1	80.01.778	BOTTOM SEATING PLATE
23	4	10.00.048	BRASS SPACER
24	5	81.20.006	TEFLON BUSH DIA 30 X 53 mm LENGTH
25	2	10.01.113	VALVE MOUNTING ANGLE
26	1	10.00.241	CHOKE 22 TURN
27	2	12-0010-0015	CAPACITOR 1000 pf 10KV/60KVA/30A Type-09 F/T
28	1	10.01.206	FRAME HONEY COMB
29	1	10.01.149	HONEYCOMB

CONVEYOR (DETAIL C)

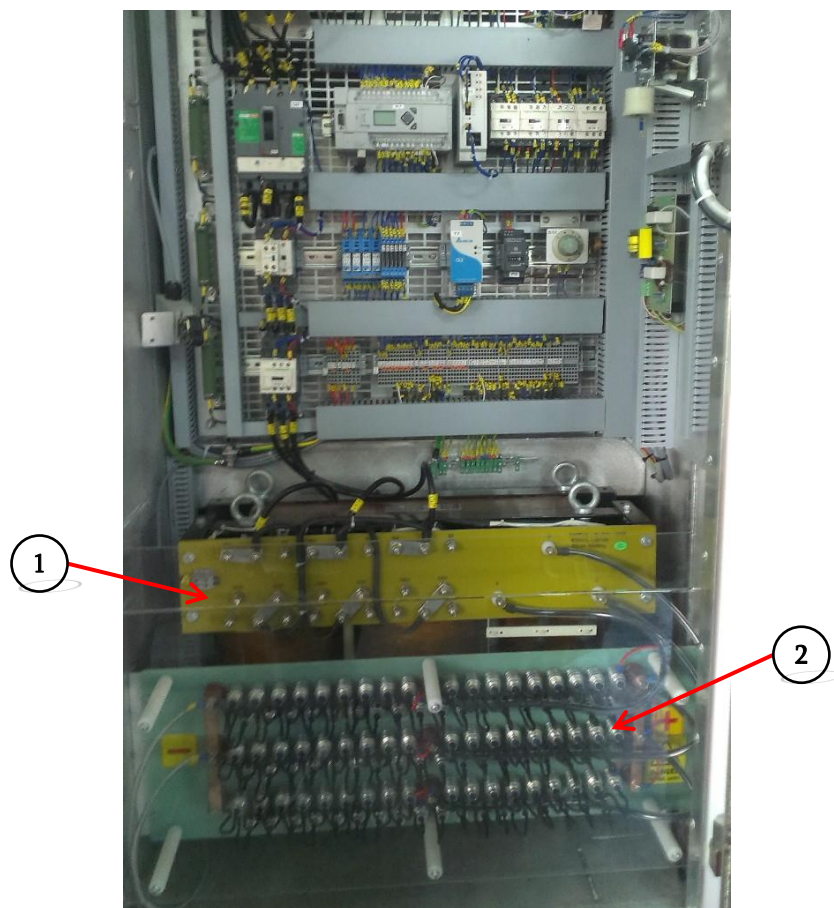


ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	11-0021-0033	TENSIONING PNEUMATIC CYLINDER DSBC-63-100
2	2	11-0021-0034	TRACKING CYLINDER PNEUMATIC DSBC-40-25
3	2	12-0031-0054	PROX SENSOR
4	2	1.46.203	PHOTOELECTRIC SENSOR
5	2	10.09.056	PP SUPPORT BLOCK



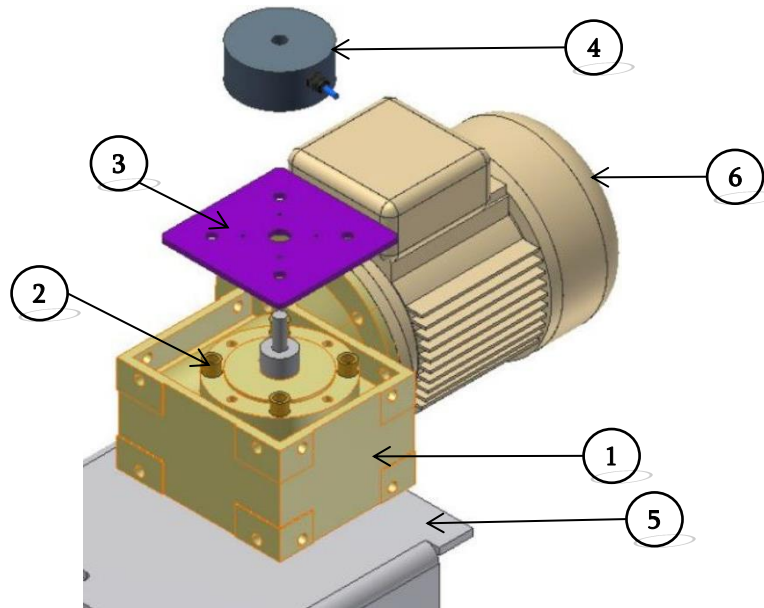
ITEM	QTY	PART NUMBER	DESCRIPTION
6	16 Mtr	11-0095-0040	BELT
7	1	41-0487-0029	DRIVE SPROCKET
8	10 FT	11-0147-0015	ROLLER CHAIN 16B-1
9	1	10.09.168/1	TENSIONING SPROCKET
10	2	10.09.167	DRIVEN SPROCKET
11	1	11-0043-0121	GEARBOX
12	1	12-0037-0082	MOTOR
13	4	11-0082-0023	BEARING UCFC 209
14	1	12-0042-0010	ENCODER 500 PPR
15	1	10.09.282	ENCODER MTG PLATE
16	2	41-0115-0027	ENCODER MTG SPACER
17	2	11-0082-0019	BEARING UCFC 205

HT TRANSFORMER COMPARTMENT (DOOR D)



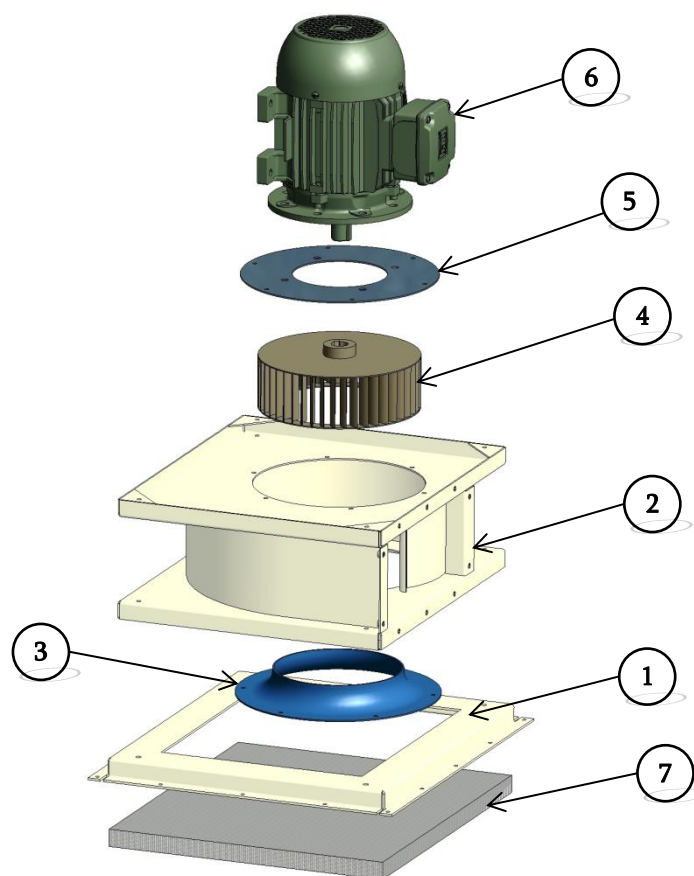
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	12-0001-0079	Transformer H.T.27 kVA, 410-610V/6700V,3PH
2	1	83.00.113	RECTIFIER ASSEMBLY

ELECTRODE GEARBOX (LOCATION E)



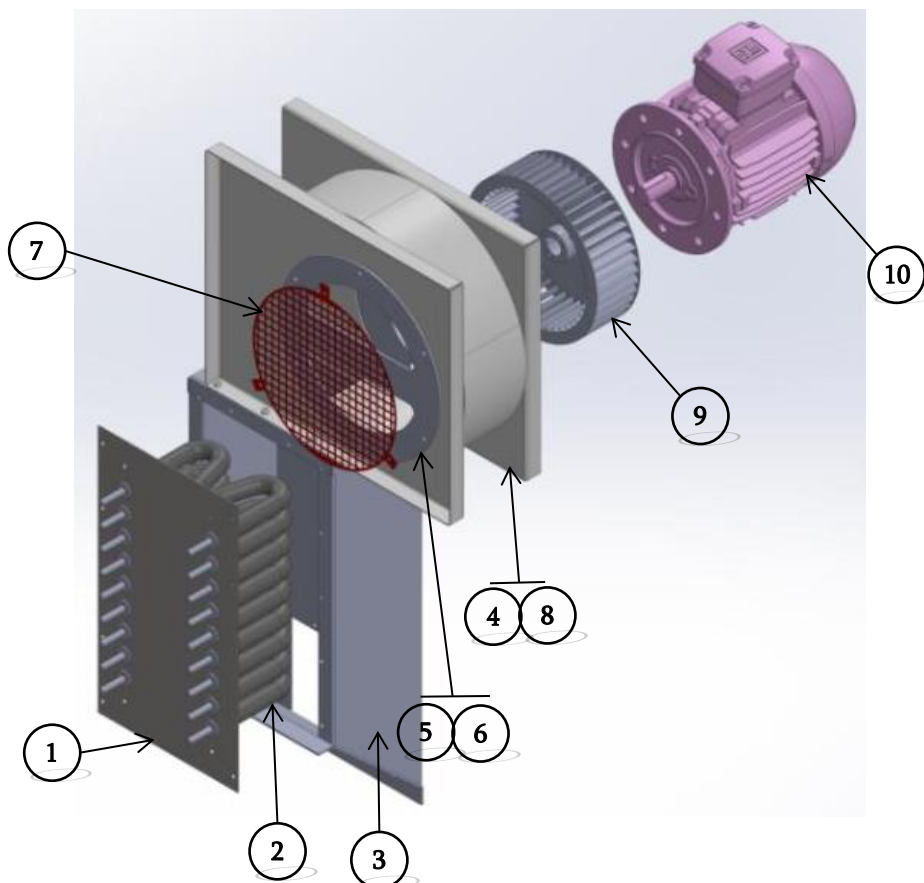
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	11-0043-0111	GEAR BOX
2	4	80.00.324	SPACER ENCODER BASE PLATE
3	1	41-0005-0477	ENCODER BASE PLATE
4	1	12-0042-0011	ENCODER
5	1	10.09.432	ENCODER GERAMOTOR MTG BRACKET
6	1	12-00037-0081	MOTOR

EXHAUST BLOWER (LOCATION F)



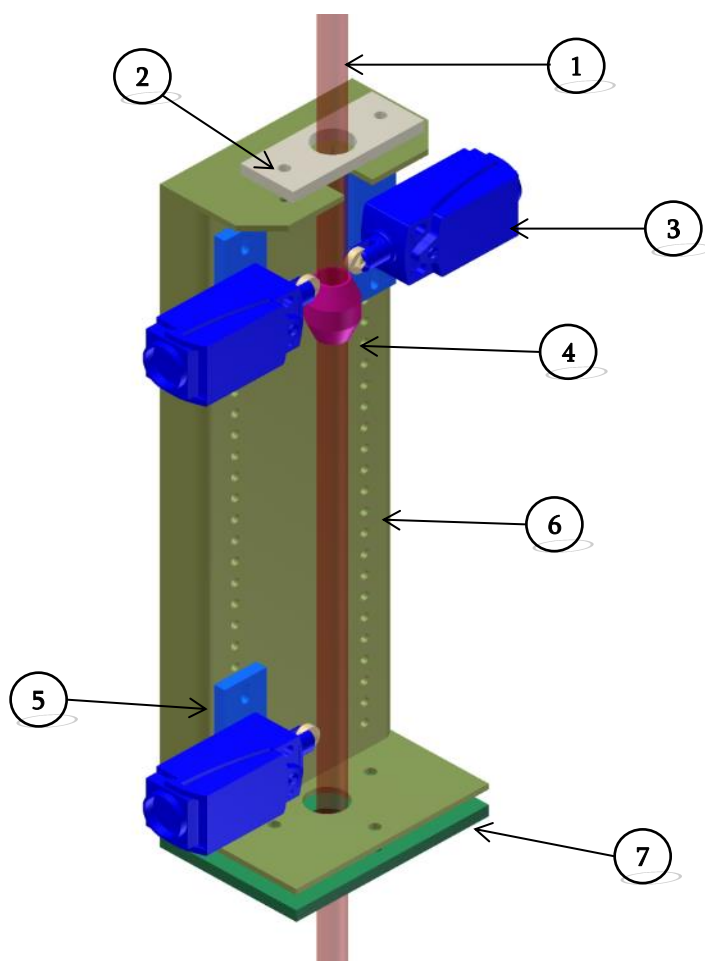
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	10.07.267	HONEYCOMB FRAME
2	1	80.00.222	FAN CASE – CC9
3	1	80.02.186	INLET CONE
4	1	41-0005-0417	IMPELLER
5	1	80.00.494	MOTOR MOUNTING PLATE
6	1	12-0037-0080	Motor flange mounted
7	1	80.00.920	HONEYCOMB

ATTN DUCT (LOCATION G)



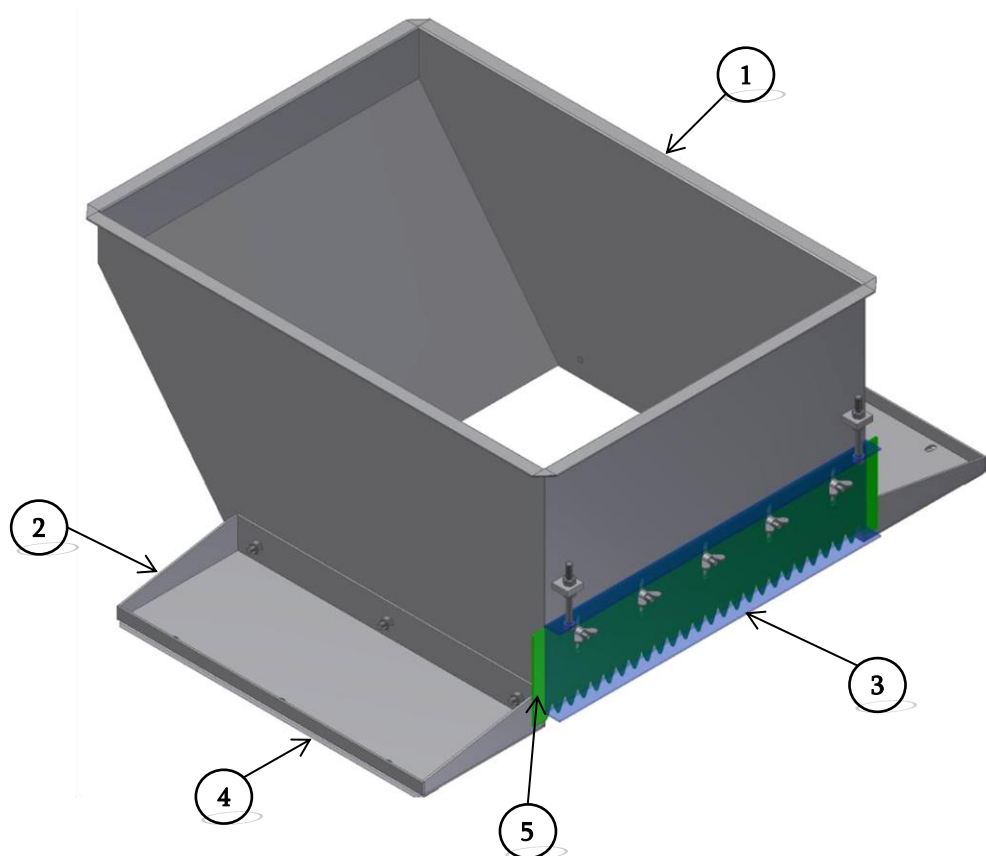
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	11.01.320	FRONT PANEL HEATER BOX
2	6	12-0044-0007	HEATING ELEMENT 1KW 119 OHM
3	1	80.00.007	HEATER BOX
4	1	80.00.221	FAN CASE – CC9
5	1	80.02.187	INLET CONE
6	1	10.06.158	AIR BAFFLE CA9
7	1	1-16-165	FAN FINGER GUARD
8	1	80.00.494	MOTOR MTG PLATE
9	1	41-0005-0417	IMPELLER
10	1	12-0037-0080	Motor flange mounted

LIMIT SWITCH ASSEMBLY (H)



ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	10.08.727	ACTUATOR ROD
2	1	10.07.714	KEEPER PLATE
3	3	1.05.230/1	LIMIT SWITCH
4	1	70.00.114	ALUMINIUM CAM CRASH INDEX
5	3	10.07.629	ADJUSTER
6	1	10.07.628	BRACKET CRASH/INDEX SWITCH
7	1	10.07.631	GUIDE PANEL

HOPPER (I)



ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	2-813-05	HOPPER INFEED
2	2	21-0042-0187	HOPPER INFEED MTG BKT
3	1	21-0042-0185	SIZING PLATE
4	1	41-0507-0001	SPACER
5	1	21-0042-0186	SIZING PLATE TEETH TYPE

COMPONENT ROUTINE MAINTENANCE

LOCATION	ASSEMBLY	CONTAINING	ROUTINE MAINTENANCE
A	OVEN AREA	Electrode Assembly Connecting Straps Lead screw Assemblies The GP™ machine Extraction Ducts Conveyor Belt	Inspection cleaning & maintenance
B	RF Generator Area	Oscillator Tank Circuit and Triode Tube (Valve) HT Filter	Regular inspection & Fault Finding
C	Conveyor	CONVEYOR DRIVE UNIT	Inspection & Cleaning
D	HT/Mains Panel	HT/Mains Panel, HT contators, Mains and HT Isolator, Filament Transformer, Control Transformer	Regular inspection & Fault Finding
D		Contactors Panel (Circuit Breakers, Contactors, PLC, safety relay, VFDs) Push Buttons & Indicator Lamps, Meters and OCR Panel Meters & Mains On Lamp on Door Electrode Limit Switch Assy Valve Thermal Arc Detection PCB	Regular inspection & Fault Finding
D	HT Area	HT Transformer Rectifier Stack	Regular inspection & Fault Finding

Spike Filter & HT Dropper

E		Electrode Drive, Encoder	Regular inspection & Fault Finding
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F	Extraction Area	Extraction Fan	Regular inspection & Fault Finding
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G	ATTN DUCT	HEATER ASSEMBLY	Regular inspection & Fault Finding
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MAINTENANCE SCHEDULES

The following tables provide information regarding the recommended daily and quarterly maintenance and lubrication schedules.

LOCATION DAILY MAINTENANCE

A	Clean all electrode plates of any foreign matter; pay particular attention to the lower plate.
	Check all fixer straps and fixings
	Clean all foreign matter from Lead screw assemblies

LUBRICATION CHART (HALF YEARLY)

LOCATION	SECTION	ITEM	RECOMMENDED LUBRICANT		
			SHELL	ESSO	MOBIL
C	CONVEYOR	ROLLER BEARINGS	Alvania RA or R2 or R3	Beacon 2 or 3	Mobilux EP No 2
		GEARBOX †	Omala 680 or Vitrea 680	Spartan EP680	Mobilgear 636
			or	or	
			Tivela WB	Glycolube 220	Glygoyle 30
C	DRIVE CHAINS	CONVEYOR	Retinax AM	Norva 275	Mobiltac No 81
			or	(IL 2880)	
			Retinax HDM	or	
				Esso Millgrease	
A	ELECTRODE DRIVE	ELECTRODE LEADSCREW**	Retinax AM or Retinax HDM	Esso Millgrease	Mobiltac No 81
		BEARINGS	Aeroshell Grease No 22	Unirex N3	Mobilgrease No 28
			or	or	
			Alvania RA	Beacon 2	Mobilux EP No 2
A	DRIVE MOTOR	GEARBOX †	Vitrea 220	Nuray 320	DTE AA



*NOTE: Items in **bold** text are SYNTHETIC lubricants. Items in normal text are MINERAL based lubricants.*

† Manufacturer recommends BP Energol GR-XP 680 or any of the listed mineral based lubricants.

If recommended lubricants are used, oil changes are due every:

10,000 operating hours (Mineral based) or 2 years

20,000 operating hours (Synthetic based) or 4 years,

...unless units are operated under extreme environmental conditions. In that case, oil changes should more frequent.

Should any of the above lubricants not be available in certain countries, it is permissible to use a locally available substitute with similar characteristics.



Warning: DO NOT mix lubricants of different brands or mix synthetic based lubricants either with other synthetic oils OR mineral based lubricants.

If the lead screws start to squeak or be noisy while operating, they must be lubricated immediately. DO NOT wait for the next maintenance.

Safety

When using lubricants always follow the safe-handling instructions provided by the manufacturer. Wash hands before eating, drinking or smoking - Avoid contact with the eyes.



PERIODIC MAINTENANCE

DAILY	Check the GP™ machine applicator areas for any accumulation of product and clean as necessary.
WEEKLY	Remove the lower electrode assemblies and clean both applicator areas fully. Clean all air filters and air intakes.
MONTHLY	Check all metallised door seals. Where there are visible signs of burning or carbonisation renew the seals ensuring that with the new seals fitted the door achieves a fully sealed closure. Frequent replacement of door seals may be due to poor alignment of the door catches, or failure to close the catches fully during operation. Alternatively the door panels themselves may have become permanently distorted causing uneven pressure on certain parts of the door seal. Hardened or damaged door seals will also reduce the efficiency of the cabinet cooling system, and may also cause radio interference to be radiated by the generator causing problems for radio systems generally. Remove accumulated dust etc from within the generator compartments. Grease the electrode lead screws as recommended in the lubrication chart on the previous page.
QUARTERLY	Check the entire air extraction system (fans, ducts etc,) for accumulated fat, condensate or grease and clean as necessary. Grease the chain sprockets of the electrode drive system located beneath the GP™ machine.
HALF YEARLY	Inspect the gearbox lubricant level on the conveyor belt drive, and top-up as required.
ADDITIONAL	Where an air line is employed as an aid to cleaning, ensure the air supply is dry and free from other contaminants. Remember that a poorly maintained cooling system can also shorten the working life of the triode valve - An expensive item to replace!

COMPONENT-WISE MAINTENANCE

ELECTRICAL CONTROL GEAR

PERIODIC INSPECTION

Industrial control equipment should be inspected periodically. The actual inspection intervals should be based on environmental and operating conditions, and kept under review, as a change in use will affect the intervals.



*NOTE: Magnetic starters, contactors and relays are designed to operate **WITHOUT** lubrication. NEVER apply lubrication to these components.*

CONTAMINATION

If inspection reveals a build-up of dust, dirt, moisture, or evidence of other contamination to the control equipment, the cause must be discovered and rectified. Possible faults may concern defective housings, failure to seal openings and doors (or conduits), or possible malfunction due to bad operating procedures. Always replace any damaged or brittle elastomer seals; repair/replace any other damaged or malfunctioning parts (e.g. hinges, fasteners, etc.). Dirty, wet or contaminated control devices must be replaced unless they can be cleaned effectively by vacuuming or wiping.

CONTACTS

Check all contacts for excessive wear or accumulation of dirt. Use a vacuum cleaner or simply wipe the contacts with a soft cloth to remove dirt.

The operation of contacts is not impaired by dis-colouration or *slight* pitting. NEVER file contacts as 'dressing' only shortens contact life. Aerosol contact cleaners *must not* be used, as they can leave residues which may cause sticking or cause high resistance contact closures.

Contacts should be replaced after the parent metal has become badly worn. Always replace contacts in complete sets to avoid misalignment and uneven contact pressure.

TERMINALS

Loose connections in power circuits cause overheating and equipment malfunction or failure. Loose grounding connections increase the hazard of electrical shock and also cause unwanted electro-magnetic interference (EMI). Ensure that all terminals are secure and that the connections are sound; securely tighten any loose connections.

Replace any parts or wiring damaged by overheating, also any broken wires or bonding-straps.

COILS

If a coil exhibits evidence of overheating (cracked, melted or burned insulation) it must be replaced. Should this type of damage occur, always check for (and rectify), a possible over/under voltage condition which is a possible cause of premature coil failure. Be sure to clean any residues of burnt or melted insulating material from other parts of the device, or replace where cleaning cannot remove the contamination.

ARC SHIELDS

These are protective devices fitted to the larger circuit breakers within the system. Periodically inspect for cracks, or deep erosion. Arc hoods and shields should be repaired or replaced if found to be damaged.

SOLID STATE DEVICES

Solid state devices require a periodic visual inspection. Printed circuit boards should be inspected to ensure they are properly located within the edge connectors and are not subjected to any twisting or bending stress. Termination screws must be tight, and wires secure in the terminations. Replacement of any solid state device which may be suspect should only be attempted at a PCB or plug-in component level. Replacement of individual components is not recommended.

HIGH-VOLTAGE TESTING

High voltage insulation resistance (IR) and dielectric withstanding voltage (DWV) tests should be made to verify the condition of the solid state control equipment. Before performing tests for IR or DWV of electrical equipment such as transformers or motors, any solid-state devices associated with these components **MUST** be disconnected prior to testing.

LOCKING AND INTERLOCKING DEVICES

Check these devices for normal operation and ensure they are capable of performing their intended functions. Where necessary adjust, repair or replace any device found to be unsatisfactory.

FINAL CHECK-OUT

Following periods of maintenance or repair, always test the **entire** control system fully

for normal functioning, in order to minimize the possibility of a malfunction when returning the system to operational status.

EMU COMPARTMENT

COOLING SYSTEM BLOWER

Monitor the forced air cooling fan for normal operation by momentarily applying power and listening for any unusual sounds i.e. binding, rubbing or metal-to-metal contact. Should the unit prove unserviceable, check and replace faulty wiring, fuse/s, blowers or fan motors as appropriate. Ensure the fins of the heat exchanger/s are clean to maintain maximum cooling efficiency.

HIGH VOLTAGE TRANSFORMER COMPARTMENT

Periodically check all H.T. terminals for tightness. Loose connections in power circuits cause over-heating that leads to equipment failure. Do not over-tighten the terminals; always use two spanners to prevent turning the terminal posts on the transformer panel.



QMTI is not liable for any damage to the high voltage transformer caused by lack of, or incorrect maintenance.

Replace any parts or wiring which may have suffered damage due to over-heating. Replace any broken wires or bonding straps. Remove accumulated dust with a soft dry cloth.

RECTIFIER STACKS

Remove dust that may have accumulated using a soft cloth dry cloth. Check all terminals and wiring for loose connections and signs of over-heating. Tighten connections and replace broken or over-heated components as necessary.

OSCILLATOR COMPARTMENT

GENERAL



Use only a clean lint-free cloth when cleaning within the oscillator compartment

Severe damage to the oscillator may result if water or solvent based cleaners are employed.

Ensure all nuts, bolts and other connections are tight. Do not to over-tighten. Any fittings requiring replacement must be of copper, brass or a similar material.

Inspect all other parts, especially capacitors and ceramic components. Replace any component showing signs of damage due to flash-over or arcing. Clean all ceramic insulators with a soft lint-free cloth to remove dust.

Inspect the Silicon Resin Bonded Fibreglass (SRBF) mounting boards for evidence of arcing or other damage. Clean or replace if necessary with QMTI original spare parts. Wipe off any traces of dust, oil or dirt from the compartment floor.

TRIODE VALVES

Ensure all electrical connections and mounting bolts are tight. Under or over tightening the connections and fixings can damage the triode valve. Carefully wipe the ceramic area around the triode valve using a clean lint-free cloth to remove any marks or build up of dust particles. Do not subject the triode valve to sudden mechanical shock or vibration.

RF CONNECTIONS

Ensure all electrical and mechanical connections are tight. Loose connections in the RF circuits can cause over-heating and equipment malfunction or failure. Do not over-tighten.

Replace any parts damaged by overheating or arcing with QMTI original spare parts. Wipe all components with a soft dry cloth.

Damaged fasteners must be replaced with components of the same original specification.



Warning: Magnetic **ferrous fasteners MUST NOT be used within the oscillator compartment.**

INSULATORS

Clean all insulators with a soft lint-free cloth to remove traces of oil, dirt or dust. Do not under any circumstances use water or solvent based cleaners.

THE GP™ MACHINE APPLICATOR AREAS

GENERAL

Remove all product that has accumulated in the applicator areas. Failure to do so could impair the working efficiency of the GP™ machine and is a fire hazard.

Blow out any dust or debris using a dry air line.

Safety

Always wear eye protection and observe standard codes of practice for use of compressed air lines.

If it is not practical to use an air-line, clean the area with a dry clean cloth

Ensure all deposits are removed from inside the duct tunnel areas where the product enters and leaves the GP™ machine. Inspect the GP™ machine lining (including the insulating blocks and other components) for signs of damage due to arcing or burning. Replace any damaged items with QMTI original parts.



RF CONNECTIONS

Ensure all electrical and mechanical connections are tight. Loose connections in the RF circuits can cause over-heating and equipment malfunction or failure. Do not over-tighten.

Replace parts damaged by overheating or arcing with QMTI original spare parts. Wipe all components with a soft dry cloth.

Damaged fasteners must be replaced with components of the same original specification.



Warning: Magnetic **ferrous fasteners MUST NOT be used within the oscillator compartment.**

INSULATORS

Clean all insulators with a soft lint-free cloth to remove traces of oil, dirt or dust.

Heavy arc marks can be removed by scraping the surface. All traces of arcing must be removed.

Do not under any circumstances use water or solvent based cleaners during this operation.

BASE ELECTRODES	Remove the tray by lifting it clear from the locating blocks. Normally it is sufficient to wipe the tray with a soft dry cloth or soft (non-metallic) brush. Stubborn marks i.e. arc marks or any material which adhere to the electrodes should be removed with a plastic abrasive pad i.e., 'Scotchbrite' or similar. Steam cleaning of the tray is possible provided it is dried immediately afterwards. Ensure that the bottom tray is inserted correctly into its original position. Disconnecting the coil adjacent to the base tray will cause permanent damage to the machine.
TOP ELECTRODE PLATES	These are cleaned in situ. Wipe with a soft dry cloth or soft (non-metallic) brush. Stubborn marks i.e. arcs or any material which adheres to the electrodes, should be removed using 'Scotchbrite' pads or similar.

DOOR SEALS

The condition of the various door seals within the GP™ machine, are essential to the overall screening of the unit, both from a safety point of view and to the prevention of Electromagnetic Interference (EMI). Inspect the Radio Frequency Interface (RFI) seals fitted to each door and replace where signs of damage are visible.

Remove all traces of the old gasket plus any contamination caused by grease, arcing or dirt from the channel and mating flange with a plastic abrasive pad ('Scotchbrite' or similar) before inserting the new gaskets.



NOTE: New gaskets should be secured at 6" (150mm) intervals by the application of a small amount of adhesive or silicone sealant. E.g., hot melt glue or 'DOW Corning Sealastic'.

CHAPTER 9. Troubleshooting

When there is a fault in the machine, an alarm is flashed, a siren goes on, and the machine stops. This helps users and engineers to identify the cause of fault and rectify that. There is also a Recycle operation that restarts the machine in case of fault. Refer to the *Installation and Operation Manual*

Safety

– Fault finding should only be performed by competent maintenance personnel who have been fully trained by QMTI technical representatives.



– Repeated Alarms must be investigated and the cause established and rectified before continuing operation or permanent damage to the machine will result.

The majority of faults that occur will generate an alarm which is displayed and logged on the HMI. A flashing beacon also indicates a fault condition. Refer to the, Alarm List on page 135.



NOTE: The alarm message indicates the location of the fault. Further investigation is required to determine the exact cause.

ELECTRICAL CIRCUITS; GENERAL

The most common fault associated with the RF oscillator and the GP™ machine electrode assemblies, is electrical breakdown in one of several possible locations, which can lead to flash-overs.

- Dirt (contamination) around the oscillator area.
- Excessive condensation forming around the capacitors terminals.
- Overheating of the ceramic areas around the capacitors.
- Impending triode valve failure.
- Oscillator instability.
- flash over in the triode valve and oscillator areas will cause the magnetic trip to operate. A flash over places excessive current loading on the H.T. supply. flash overs of this magnitude especially if recurring, will leave visible evidence i.e. carbonisation, metallisation of capacitors, tracks, cracked insulators, burn marks etc. (This does not apply in the case of internal triode valve breakdown however).

- Insulation test can be performed on the H.T. transformer. Using an Insulation Meter check the resistance between primary and secondary windings. The reading must be Infinite. Check the resistance between the ends of each winding. The primary is less than 1ohm. The secondary is less than 50ohms
- flash overs within the electrode area will automatically activate the recycling mode, and unless serious tracking or burn damage has been sustained, recycling will restore the RF power within the time limit of the recycling routine.
- The PLC terminates the recycling routine after a number of attempts (set in the limit and trips screen) and activates an alarm. Visible marking can identify the cause of the problem, and thorough cleaning and replacement of any damaged items (i.e. feeder straps, insulators etc) should be made.

ALARM MESSAGES TROUBLESHOOTING

NON-RF ALARMS

Place the H.T. isolator in the OFF position. This permits the testing of all electro-mechanical services and their relevant controls. The machine can be operated as normal except The H.T. contactors are not operated. With HT Circuit Breaker in the off position the H.T power supply is not energised and hence no R.F. power is produced.

This tests the following circuits:

- 1 Supply & control of air circulation fans
- 2 Electrode drive motor
- 3 Heater fan & elements
- 4 Oscillator & cabinet cooling fan
- 5 Triode valve filament supply
- 6 Triode valve grid & anode protection alarms
- 7 Thermal protection

All Faults are indicated on the HMI with an appropriate message. Some generic faults are listed below along with troubleshooting steps

Alarm	Troubleshooting
Circuit Breaker off/trip	1. Verify there is no Short Circuit on the Load. 2. Verify that rated current is flowing through the Load.

3. Verify Current Setting on Circuit Breaker.
4. Verify Auxiliary Contact of Circuit Breaker is functioning.
5. Verify Circuit Breaker Function.
6. Check PLC input is receiving the correct signal

Contactor Coil Fault	1. Contactor Coil Open Circuit 2. Connections to the contactor coil are not loose 3. Contactor auxiliary Contact is functioning correctly 4. Supply to PLC output 5. Supply to auxiliary contact 6. Verify PLC Input is receiving the correct signal
Contactor Coil Stuck	1. Contactor welded closed 2. Wiring to contactor coil 3. Contactor auxiliary contactor
Demin Water Tank Low Level Trip (Optional)	1. Check if Demineralised Water Level is low 2. Check if the Level switch is functioning 3. Verify if the PLC Input is receiving the correct signal
Flow Switch Low (Optional)	1. Check the cooler pump 2. Check the flow sensors

3. Verify if the PLC Input is receiving the correct signal

Secondary Water Temperature High Trip (Optional)	1. Check that Cooler Circuit is functioning correctly and the water temperature is below the trip point 2. Verify that the RTD is functional 3. Check the PLC Input
Valve Thermal High	1. Check condition of valve thermal and the attached string as described in Triode valve Thermal Fuse Links on page 49
Emergency Stop Pressed	1. This alarm is displayed if the Emergency stop push buttons have been activated. Ensure that the E-stop buttons are in the safe operating position
Door Open/Door Switch Open	1. This alarm is displayed if the Door open switch have been activated. Ensure that the all the doors are closed and door switches are engaged.

RF ALARMS



Warning: Very High Voltages are present in the HT circuit of the machine. QMTI does not assume responsibility of the safety of any personnel attempting RF Troubleshooting unless directly deputed by QMTI.

Alarm	Troubleshooting
Anode Over Current Relay - B1	1. Check if OCR factory setting is disturbed. 2. Check OCR function by manually moving its lever. 3. Follow steps 2 to 4 for Anode over current trip alarm
*** Anode Current High ***	1. Check Anode 1st limit and 2nd limit, should are not too close to Ia high setpoint. 2. Check Incoming voltage is not abnormally high. 3. Test run the machine with reduced HT as described in Half-HT Configuration on page 126. If tripping observed on reduced HT, follow instructions detailed in Tests for Troubleshooting on page 123 4. Check filament voltage and current are as per Technical Description. If found abnormal, Triode tube may require replacement.
*** Grid Current Low ***	1. Check grid undercurrent setting in Trips and limits of service settings and set as per the Test Report. 2. Check KV meter reading, if it is too low or zero check rectifiers and spike capacitor. 3. Check Incoming voltage is not abnormally low. 4. If tripping occurs during build-up of anode current during a production run, check Grid Switch setting is as per the test report.

	5. Check grid resistors as explained in Grid Components Fault Diagnosis on page 126
*** Grid Current High ***	1. Check grid over current setting in Trips and limits of service settings and set as per the Test Report. 2. Check Incoming voltage is not abnormally high. 3. If tripping occurs during build-up of anode current during production run, check Grid Switch setting is as per the test report. 4. Check grid resistors as explained in Grid Components Fault Diagnosis on page 126
Arc Suppression Unit Trip. Arc in Oscillator	1. Check for arc marks in the oven and oscillator. 2. Clean any arc sports in the oven and oscillator. 3. Refer Oscillator Components Fault Diagnosis on page 125 4. Refer Oven Components Fault Diagnosis on page 123

RECYCLE LOCKOUT



WARNING: Always investigate the cause of repeated arcing and recycle lockouts. Ensure the problem is rectified before continuing production. Failure to investigate can cause irreparable damage to the machine.

A recycle lockout occurs when successive RF faults occur in the machine within a set time period. For details on Recycle Operation, refer the appropriate section in the Installation and Operation Manual

A recycle lockout is preceded in the Alarm History by one of the RF Alarms listed above. Refer the table above for troubleshooting.

TESTS FOR TROUBLESHOOTING



NOTE: Before proceeding with any tests described in this section, conduct an oven components fault diagnosis as described below

OVEN COMPONENTS FAULT DIAGNOSIS

- Follow steps 1 to 2 below.
- Check the Oven compartment visually for any signs of flash-over or arcing. Check the parts shown in *OVEN COMPARTMENT (Door A)* on page 90 for any signs of arcing damage.
- Clean the arc marks and ensure all connections are tight
- Replace suspect components
- Repeat steps 9 to 14 below

H.T. SUPPLY TEST

The high tension DC return (0 Volt supply) is permanently bonded to the machine earth or chassis by means of a stud bolt located inside the main contactor compartment.



Warning: Do not under any circumstance disconnect the safety earth. Always discharge capacitors using a grounding probe. Refer the section detailing `Electrical Compartment Access on page 87

Follow the procedure below to check for faults within the H.T.:

- 1 Follow the procedure noted in Open and padlock off Main circuit breaker and H.T. circuit Breaker.
- 2 Operate the door interlocks to open the doors and gain safe access to the components inside
- 3 Ensure all active components (e.g. capacitors) are discharged using a suitable grounding probe before working in the compartment
- 4 Connect a shorting link between C1 and a suitable earth point.
- 5 Disconnect the H.T. line by removing choke L1 within the oscillator compartment
- 6 REMOVE THE SHORTING LINK between C1 and earth.
- 7 Close and lock the compartment door. Ensure the door seals make contact with the frame around the entire perimeter of the door.

- 8 Lock the doors using the interlock system
- 9 Ensure no work is being carried out on the machine and all personnel are clear of the machine.
- 10 Remove the padlocks on Control Circuit Breaker and HT Circuit Breaker.
- 11 Close Control Circuit Breaker. Ensure the HMI has booted up
- 12 Reset any active alarms on the HMI and Close the HT Circuit Breaker
- 13 Go to the service settings and disable Grid Under current trip.
- 14 Press the RF ON push button
- 15 Check the kV Meter. Ensure the meter reads approx 11 kV.
- 16 Repeat steps 1 to 4; reconnect choke L3, then repeat steps 6 to 12, enable grid undercurrent in service settings.



NOTE: If a trip occurred during the test, then the fault may be associated with the H.T. Transformer, rectifier bank, spike filter or the HT filter.

Possible Causes of Breakdown

- Broken lead touching the grounded frame work of the cabinet.
- Failed spike filter capacitor.
- Faulty H.T. transformer.
- H.T. filter damaged.
- Faulty H.T. rectifier diodes

To access the H.T. compartments, repeat steps 1 to 6. The entire circuit must be checked for the points above both by visual inspection and using an insulation test meter. On completion steps 8 to 13.



NOTE: If no trip occurred during the test, then the fault may be associated with the oscillator circuit and associated components. Conduct the Grid Line Test described below

GRID LINE TEST

If a trip occurs, the fault is with the triode valve or an associated component.

1. Open and padlock off Main circuit breaker and H.T. circuit Breaker.
2. Operate the door interlocks to open the doors and gain safe access to the components inside

3. Ensure all active components (e.g. capacitors) are discharged using a suitable grounding probe before working in the compartment.
4. Connect a shorting link between C1 and a suitable earth point.
5. disconnect the grid line (wire no. 908) from the Capacitor C15 within the oscillator compartment
6. REMOVE THE SHORTING LINK between C1 and earth.
7. Close and lock the compartment door. Ensure the door seals make contact with the frame around the entire perimeter of the door.
8. Lock the doors using the interlock system
9. Ensure no work is being carried out on the machine and all personnel are clear of the machine.
10. Remove the padlocks on Control Circuit Breaker and HT Circuit Breaker.
11. Close Control Circuit Breaker. Ensure the HMI has booted up
12. Reset any active alarms on the HMI and Close the HT Circuit Breaker
13. Go to the service settings and disable Grid Under current trip.
14. Press the RF ON push button
15. Check the kV Meter. Ensure the meter reads approx 11 kV.



NOTE: If a trip occurred during the test, then the fault may be with the oscillator components or the triode valve

OSCILLATOR COMPONENTS FAULT DIAGNOSIS

Repeat steps 1 to 4 above.

- Check the oscillator compartment visually for any signs of flash-over or arcing.
- Check the circuit and all the associated components for possible loose connections. A poor connection causes a build-up of heat due to high resistance which leads to failure of the component.
- Visually inspect capacitors shown in *on page 92*
- Examine the physical spacing of the tank input/output coils labelled as part 3 in

the diagram referenced above; uneven spacing of the coils can lead to breakdown and arcing. Inspect the capacitors for evidence of over-heating or cracks in the ceramic. Check all coil windings for shorts between turns or contact with the grounded framework.

- Replace any suspect devices and repeat steps 8 to 15 and check if the analogue kV meter displays about 11kV
- If it still trips, repeat steps 1 to 4 and check the triode valve using an insulation tester as shown in **Illustration 27: Triode valve insulation test locations** on page 128



NOTE: If no trips occur in the Grid line test, repeat steps 1 to 6 to reconnect grid wire 412 and repeat steps 8 to 20. If a trip occurs then the fault may be with the grid components

GRID COMPONENTS FAULT DIAGNOSIS

Repeat steps 1 to 4:

- Carry out an inspection of each item shown in OSCILLATOR COMPARTMENT (DOOR B) on page 93 for any signs of arc marks or damage. Replace suspect parts.
- Using a resistance (ohm) meter, test the grid line for continuity between the disconnected grid lead through the grid resistors to ground. The resistance here should be 360 ohms.

If the reading obtained differs from this value, the following test must be made:-

- Test each grid resistor individually using the resistance meter. Look for open circuit or short circuit readings.
- Test the grid line for an open circuit; i.e. between the grid ring through choke L4, the wires 908 through to AG card/meter and finally to ground (via wire 910).
- Using the insulation tester, check for any short to ground between the first grid resistor and back through to the grid choke in the oscillator.
- Repeat steps 8 to 15

HALF-HT CONFIGURATION

This configuration allows the user to start the machine at half its rated High Voltage in order to troubleshoot the oscillator circuit at lower voltages.

- Repeat steps 1 to 4 above.
- Open the door to the main HT transformer compartment
- Connect the links on the HT transformer as shown in the image below. This is also indicated on the Transformer terminal board as 'Star Connection'

- Fully tighten all the nuts and ensure no product is present in the Oven

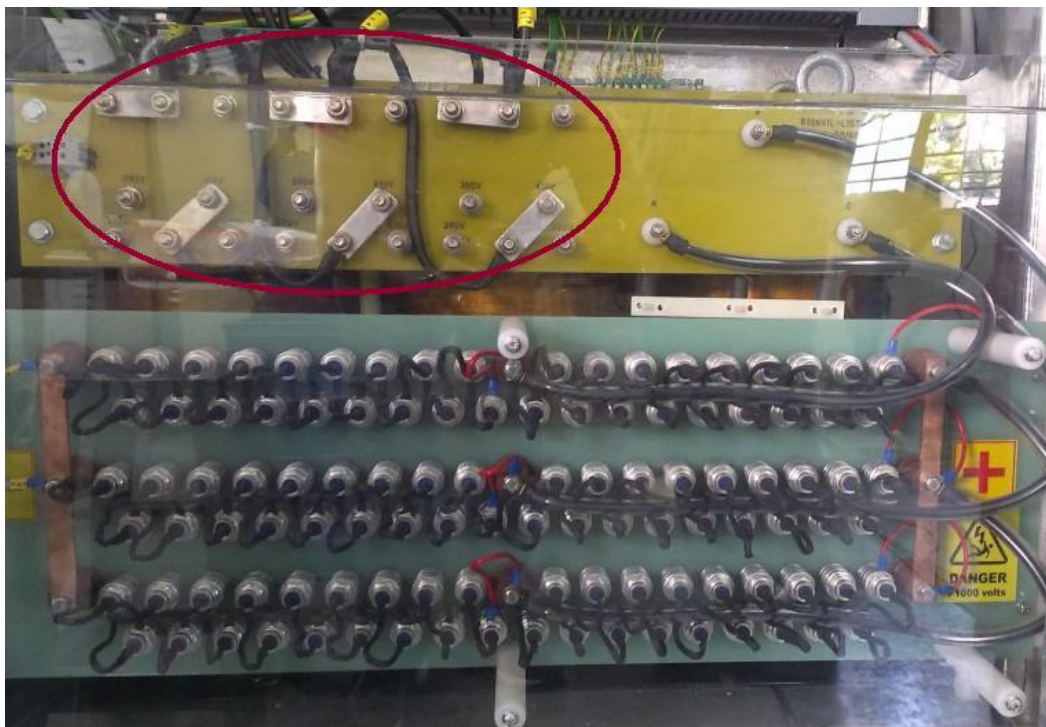


Illustration 26: Half HT Configuration (Star Connection)

TRIODE VALVE

If a new triode valve is received for the purpose of maintaining as a spare or for replacement the following checks are recommended

INSPECTING A TRIODE VALVE

Examine the packaging for any signs of damage and ensure that the new triode valve is the correct type by examination of the part number. This should be marked on the shipping carton and should match the part number given for the particular the GP™ machine in Appendix A at the rear of this manual.

Open the package carefully and remove any packing material. Examine the new triode valve for any signs of damage. Remove the triode valve (keeping it in a vertical position) The following passive checks are recommended to verify the state of the new triode valve and avoid the possibility of fitting a faulty component. Place the triode valve in its foam packing to ensure the triode valve is protected against mechanical shocks during the checks. Using a 1000V insulation tester, test for insulation breakdown as shown in the table below.-

	MEASURE	RANGE/READING
1	Anode - Grid	1000 volt - infinity
2	Anode - Cathode	1000 volt - infinity
3	Grid - Cathode	1000 volt - infinity
4	Filament	Ohm range - less than 1 Ohm

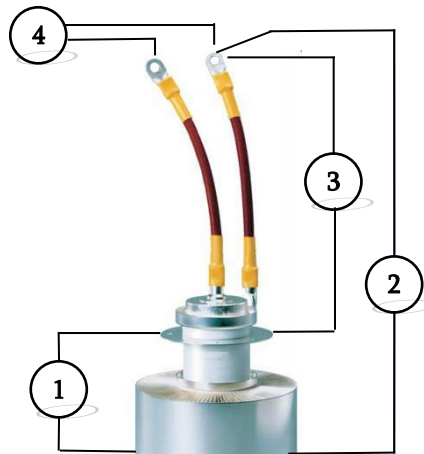


Illustration 27: Triode valve Insulation test locations

Any value less than infinity for measurements 1, 2 & 3 indicates a potential resistance breakdown between the electrodes concerned. The filament reading **(4)** is extremely low, being typically less than 1 Ohm.

If for any reason the new triode valve has suffered damage or is in any way suspect, refer to the warranty claim section in Appendix B of this manual and return the triode valve in its original packaging, together with the claim form, to QMTI following the instructions given.

For more information on installing new triode valve, please contact Gentle processing

TRIODE VALVE THERMAL FUSE LINKS

A non resettable thermal fuse link is located in the mounting assembly surrounding the triode valve. When the thermal fuse operates, the spring cord attached to the device is released which operates a micro switch. The fuse operates at approximately 145°C



WARNING: Under no circumstances can ordinary lead/tin solder be used in conjunction with these devices. DO NOT ATTEMPT TO REPAIR THE FUSE. REPLACE IT WITH A NEW ONE) Not doing so could damage the triode valve and void the warranty

Appendices

Appendix A Warranty

This warranty is for a period of 12 months from dispatch of machine, against faulty workmanship and materials and does not cover wear parts like rubber, glass and plastic components, indicating bulbs and other similar items.

This warranty does not cover defects brought about due to negligent handling, wilful damage, acts of nature, damage in transit and operation/maintenance and repairs by unauthorized and untrained personnel.

Triode Valve used in the machine is not covered by our warranty. The warranty provided by the manufacturer of the Triode Valve is to be availed by you directly.

This warranty covers only cost of components subject to these being sent to our works freight and packing paid to and fro. The Company reserves the right to either replace or repair such components at its sole discretion.

This warranty is void for claims made due to failure of components brought about by using the machine with un-stabilised electric supply. The electric supply should be as per specifications of the machine as mentioned in this manual. A three phase electric supply and dedicated earth is required. The supply must be maintained within $\pm 1\%$ of the nominal voltage (nominal values as accepted during the order confirmation) using a Servo stabilising transformer.

This warranty (for water cooled machines) is void for claims made due to failure of components brought about by using demineralised water [in the integral cooler circuit] which has a maximum solid contents exceeding 30 ppm & conductivity exceeding 100 μ siemens/cm.

The Company's liability/obligation does not cover any consequential losses whether to you or any other third party due to non running of the machine while warranty repairs are being undertaken.

Whether a claim falls under warranty will be at the sole discretion of QMTI GP™ machine. All/Any disputes arising out of warranty claims shall be subject to Pune jurisdiction only.

This warranty is limited to the extent of the text here-in-above and does not cover or include any other warranty, whether implied or otherwise, unless specifically separately agreed to in writing.

Appendix B Technical Description of the Machine

Electrical Power Supply

Mains Supply	600 Volts, three phase, 60Hz + 3 wire plus earth. 42 kVA (maximum voltage variation $\pm 5\%$)
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Electrical System

Osc. Cooling fan	12-0037-0080 (1.5 kw)
Electrode drive motor	12-0037-0081 (0.37 kw) Inverter Controlled
Extraction fan	12-0037-0080 (1.5 kw) Inverter Controlled
Conveyor drive motor	12-0037-0082 (0.75 kw), Inverter Controlled
Control circuit	24V dc and 110Vac single phase PLC digital I/O 24Vdc

Oscillator Circuit

Grid Resistors	6 elements of 200 ohms each wired in series total 1200 ohms
Triode valve	Type: Type 1.21.034 (12-0024-0013) Filament voltage 6.0 max, 5.2 min. AC Filament current 145A Approximately
Oscillator	Operating frequency 27.12MHz power output 15kW maximum No Load Standing Conditions (typical) anode current 0.4 A anode voltage 9.18 kV

grid current 0.75 A

Full Load Maximum Power (typical)

anode current 2.58 A

anode voltage 8.38 kV

grid current 0.55 A

Conveyor

Drive	Direct drive inverter controller 800mm modular link type 0.1-2 meter per Min (Typical)
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Cooling System

Medium	Air cooled
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Flow	1119 CFM
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Appendix C Maintenance & Service Record

All repair and/or maintenance work carried out on QMTI equipment may have a bearing on any subsequent guarantee claims; this applies in particular to the triode valve. It is therefore recommended that complete records are kept from the date of commissioning onwards.

[illegible]

Appendix D Engineering Drawings

MECHANICAL DRAWINGS

59-0001-0005_REV1...OUTLINE DRAWING FOR GP 15

52.0009.0008.... PNEUMATIC CIRCUIT

ELECTRICAL DRAWINGS

94-003-000 - 94-003-022

SUPPLEMENTARY INFORMATION

(If applicable & if available)

PLC MicroLogix 1400 : Part no : 12-0018-0028 Allen Bradley

HMI : Part no : 12-0019-0012Proface

Drive Part no: Part no : 1.50.212 (0.4 kw)Allen Bradly

Drives Part no: Part no : 1.50.213 (0.75 kw) Allen Bradley

Drive Part no: Part no : 1.50.214 (1.5 kw)Allen Bradly

Appendix E Test Report

Appendix F Drive Settings

Appendix G Spares

The recommended spare parts and quantities are listed in the table below.

Part Number	Description	Qty
12-0010-0015	CAPACITOR FEED THRU. 1000 PF 10 KV TYPE 09	1
12-0010-0000	CAPACITOR DISC 1000 PF 13 KV , CERAMIC	1
12-0010-0026	CAPACITOR FILTER 1.0 uF 12.5kV	1
12-0010-0034	CAPACITOR FEED THRU 200A	1
12-0009-0013	RESISTOR FIRE BAR 60 OHM/1KW	1
12-0009-0042	RESISTOR 3R3/60W	1
12-0031-0001	SWITCH LIMIT PC603 SNAP ACTION	1
1.17.014	AIR FILTER 610 (L) X 610 (W) X50 (T) mm	2
81.20.006	TEFLON BUSH 30 DIA X 53 mm LENGTH	4
10.01.107	VALVE THERMAL FUSE	1
1.21.034	OSCILLATOR TRIODE VALVE (12-0024-0013)	1
12-0042-0011	ENCODER 500ppr Electrode	1
12-0042-0010	ENCODER 500ppr Conveyor	1
10.00.241	COPPER CHOKE 22 TURNS	1
10.07.874	RF FEED STRIP 18" long	4
12-0047-0002	DIODE 17 AMP	10
12-0015-0022	ANODE GRID MONITOR PCB	1
5.28.002	LMP SOLDER 18swg. 1.2mm-145 deg	1
	BRASS HARDWARE	1

Appendix H Alarm List

Alarms fall under various categories based on what fault triggers it. These are listed below based on their category.

No.	Alarm Message	Alarm Code and Location
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EMERGENCY STOP ALARMS		
1	IN5: Emergency Safety Relay - K4: Press "RF OFF" Push Button To Reset	K4 -Control Panel
2	IN10: Emergency Stop - HMI Infeed Left Operated. S6	S6 -Conveyor Infeed HMI Panel
3	IN8: Emergency Stop Outfeed Left Operated. S4	S4 -Conveyor Outfeed LH Side
4	IN7: Emergency Stop Infeed Right Operated. S2	S2 -Conveyor Infeed RH Side
5	IN9: Emergency Stop Outfeed Right Operated. S5	S5 -Conveyor Outfeed RH Side
6	PLC Digital Output 2 +VDC Flt. Control Panel. Check safety Circuit	PLC -Control Panel
7	IN19: 110 VAC Control Circuit Breaker Off/Trip. F1 - Control Panel	F1 -Control Panel
8	IN4: Electronic Circuit Break Trip. T21	T21 -Control Panel
9	IN4: Door Switch Solenoid Guard Relay - K3	K3 -Control Panel
10	IN6: Door Switch Trojan Guard Relay - K5	K5 -Control Panel

CONVEYOR ALARMS		
1	IN9: Conveyor Air pressure Low. S30 - Infeed Left Hand Side	S30 - Conveyor Infeed

2	IN5: Conveyor Drive Circuit Breaker Off/Trip. Q21 - Control Panel	Q21 -Control Panel
3	IN6: Conveyor Motor Disconnect Switch Open. Q22 - Outfeed Left Hand Side	Q22- Conveyor Outfeed LH side
4	IN7: Conveyor Drive Not Ready. T12 - Control Panel	T12 -Control Panel
5	Conveyor Drive Faulted. T12 - Control Panel	T12 -Control Panel
6	Conveyor Encoder Speed Below Minimum. B4 - Conveyor	B4- Conveyor Outfeed LH side
7	Conveyor Stopped from VSD. T12 - Control Panel	T12 -Control Panel
8	Conveyor Drive Communication Lost. T12 - Control Panel	T12 -Control Panel
9	IN10, IN11: Tracking: Double Detection - Right & Left End Limit Switch: S31, S32 - Conveyor Infeed	S31/32 - Conveyor Infeed
10	IN12, IN13: Tracking: Double Detection - Optical Sensor: S55, S56 - Conveyor Infeed	S55/S56 - Conveyor Infeed
11	IN15: Tracking: Left Cylinder Fail To Extract: S61 - Conveyor Infeed	S61 - Conveyor Infeed
12	IN15: Tracking: Left Cylinder Fail To Retract: S61 - Conveyor Infeed	S61 - Conveyor Infeed
13	IN11: Tracking: Left End Limit Switch: S31 - Conveyor Infeed	S31 - Conveyor Infeed
14	IN14: Tracking: Right Cylinder Fail To Extract: S62 - Conveyor Infeed	S62 - Conveyor Infeed
15	IN14: Tracking: Right Cylinder Fail To Retract: S62 - Conveyor Infeed	S62 - Conveyor Infeed
16	IN10: Tracking: Right End Limit Switch: S32 - Conveyor Infeed	S32 - Conveyor Infeed

ELECTRODE ALARMS		
1	IN0: Electrode Drive Circuit Breaker Off/Trip. Q15 - Contactor Panel	Q15 -Control Panel
2	Electrode Drive Faulted. T13 - Control Panel	T13 -Control Panel
3	IN1: Electrode Drive Not Ready. T13 - Control Panel	T13 -Control Panel
4	Electrode Drive Encoder Direction Reverse. M5/B3 - Extraction Compartment	M5/B3 - Oven Roof Panel
5	Electrode at Home Position - Machine Stopped	Please set the electrode height lower than home position
6	Home Sequence Not Completed	Please Home Machine from Limits & Trips - I (Electrode Home PB)
7	Electrode Home Sequence Failed. Watch Dog Timer.Please Home Machine from Service Setup (Home PB)	Please Home Machine from Limits & Trips - I (Electrode Home PB)
8	IN3: Electrode Drive Lower Limit Switch. S15 - Extraction Compartment	S15 - Oven Roof Panel
9	OUT7: Electrode MPCB Shunt Coil Flt. Q15 - Control Panel	Q15 -Control Panel
10	Electrode Drive Stopped from Drive. T13 - Control Panel	T13 -Control Panel
11	Electrode MPCB Forced Tripped. Q15. Down Movement Detected by Encoder. M5/B3 - Extraction Compartment	Q15 -Control Panel
12	Electrode MPCB Forced Tripped. Q15. Up Movement Detected by Encoder. M5/B3 - Extraction Compartment	Q15 -Control Panel
13	IN2: Electrode Up Limit Switch Inhibits further Electrode Travel. S14 - Extraction Compartment	S14 - Oven Roof Panel
14	Electrode Drive Communication Lost. T13 - Control Panel	T13 -Control Panel

OSCILATOR ALARMS		
1	IN12: Generator Extraction Fan Circuit Breaker Off/Trip. Q26 - Control Panel	Q26 -Control Panel
2	IN13: Generator Extraction Fan Contactor Coil Flt. Q110 - Control Panel	Q110 -Control Panel
3	IN13: Generator Extraction Fan Contactor Stuck Flt. Q110 - Control Panel	Q110 -Control Panel
4	IN14: Harmonic Filter Cooling Fan Circuit Breaker Off/Trip. F12 - Control Panel	F12 -Control Panel
5	IN16: Anode Over Current Relay. B1 - OCR/HT Panel	B1 -Control Panel
6	Anode Over Current Trip. Software Set Trip	Software Set Trip - HMI.
7	IN8: Arc Suppression Unit Trip. Arc in Oscillator. Detected by B10 - Control Panel	B10 -Control Panel
8	IN18: Controls Transformer Circuit Breaker Off/Trip. Q4 - Control Panel	Q4 -Control Panel
9	IN10: Oscillator Cooling Fan Circuit Breaker Off/Trip. Q5 - Control Panel	Q5 -Control Panel
10	IN11: Oscillator Cooling Fan Contactor Coil Flt. Q103 - Control Panel	Q103 -Control Panel
11	IN11: Oscillator Cooling Fan Contactor Stuck Flt. Q103 - Control Panel	Q103 -Control Panel
12	IN14: Cubicle Temperature High Trip - S12	S12 -Control Panel
13	IN5: Filament Contactor-1 Coil Flt. Q113 - Control Panel	Q113 -Control Panel
14	IN5: Filament Contactor-1 Stuck Flt. Q113 - Control Panel	Q113 -Control Panel

15	IN6: Filament Contactor-2 Coil Flt. Q119 - Control Panel	Q119 -Control Panel
16	IN6: Filament Contactor-2 Stuck Flt. Q119 - Control Panel	Q119 -Control Panel
17	IN4: Filament Transformer Circuit Breaker Off/Trip. Q3 - Control Panel	Q3 -Control Panel
18	Grid Over Current Trip. Software Set Trip	Software Set Trip - HMI.
19	Grid Under Current Trip. Software Set Trip	Software Set Trip - HMI.
20	IN0: HT Circuit Breaker not Closed. Q2 - Isolator Panel	Q2 -Infeed Attn Duct
21	IN1: HT Circuit Breaker Trip. Q2 - Isolator Panel	Q2 -Infeed Attn Duct
22	IN2: 1st HT Contactor Coil Flt. Q101 - Isolator Panel	Q101 -Infeed Attn Duct
23	IN2: 1st HT Contactor Stuck Flt. Q101 - Isolator Panel	Q101 -Infeed Attn Duct
24	IN3: 2nd HT Contactor Coil Flt. Q102 - Isolator Panel	Q102 -Infeed Attn Duct
25	IN3: 2nd HT Contactor Stuck Flt. Q102 - Isolator Panel	Q102 -Infeed Attn Duct
26	IN15: HT Transformer Thermal High. S11 - HT Transformer	S11 -Control Panel
27	IN15: Matching Box Cooling Fan Circuit Breaker Off/Trip. F13 - Control Panel	F13 -Control Panel
28	IN17: Main Circuit Breaker Not Closed. Q1 - HT Panel	Q1 -Infeed Attn Duct
29	Anode Current Control - Anode Current High. Check Product Moisture	Check Product Moisture
30	Recycle Lockout (RF Faults). See Manual for Description	See Manual for Description.

31	IN13: Oscillator Valve Thermal High Temperature. Fuse Link on V6 operates S10 - Control Panel	Fuse Link on V6 operates S10 - Control Panel.
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EXTRACTION ALARMS		
1	IN8: Oven Extraction Fan Circuit Breaker Off/Trip. Q17 - Control Panel	Q17 -Control Panel
2	Oven Extraction Fan Drive Faulted. T10 - Control Panel	T10 -Control Panel
3	IN9: Oven Extraction Fan Drive Not Ready. T10 - Control Panel	T10 -Control Panel
4	Oven Extraction Fan Stopped from VSD. T10 - Control Panel	T10 -Control Panel
5	Oven Extraction Fan Drive Communication Lost. T10 - Control Panel	T10 -Control Panel

OVEN HEATER ALARMS		
1	IN2: Oven Heater-1 Circuit Breaker Off/Trip. Q9 - Control Panel	Q9 -Control Panel
2	IN3: Oven Heater-1 Contactor Coil Flt. Q105 - Control Panel	Q105 -Control Panel
3	IN3: Oven Heater-1 Contactor Stuck Flt. Q105 - Control Panel	Q105 -Control Panel
4	IN0: Oven Heater-1 Fan Circuit Breaker Off/Trip. Q7 - Control Panel	Q7 -Control Panel
5	IN6: Oven Heater-2 Circuit Breaker Off/Trip. Q12 - Control Panel	Q12 -Control Panel
6	IN7: Oven Heater-2 Contactor Coil Flt. Q107 - Control Panel	Q107 -Control Panel

7	IN7: Oven Heater-2 Contactor Stuck Flt. Q107 - Control Panel	Q107 -Control Panel
8	IN4: Oven Heater-2 Fan Circuit Breaker Off/Trip. Q10 - Control Panel	Q10 -Control Panel
9	Oven Air Temperature - Warning. S95 - Extraction	S95 - Oven Roof Panel
10	Variation in Outfeed Temperature: Please check Temperature Sensors	Comparison of all temp sensors-HMI.
11	IN1: Heater Fan #1 Contactor Fail To Start : Q104	Q104 -Control Panel
12	IN1: Heater Fan #1 Contactor Fail To Stop : Q104	Q104 -Control Panel
13	IN5: Heater Fan #2 Contactor Fail To Start : Q106	Q106 -Control Panel
14	IN5: Heater Fan #2 Contactor Fail To Stop : Q106	Q106 -Control Panel
15	Mint AI-8 Module Communication Lost. K51 - HT Compartment	K51 -Control Panel
16	Mint AI-8 : CH1 Open Circuit. Temperature #1 Optric - Outfeed	K51 -Control Panel
17	Mint AI-8 : CH2 Open Circuit. Temperature #2 Optric - Outfeed	K51 -Control Panel
18	Mint AI-8 : CH3 Open Circuit. Temperature #3 Optric - Outfeed	K51 -Control Panel
19	Mint AI-8 : CH4 Open Circuit. Temperature #4 Optric - Outfeed	K51 -Control Panel

MAINTENANCE ALARMS		
1	Clean Air Filters! Please Refer to Manual	Please Refer to Manual.
2	General Electrical Maintenance Required !!. Please Refer to Manual	Please Refer to Manual.
3	!!! Mechanical Lubrication Required. Please Refer to Manual	Please Refer to Manual.

Appendix I Glossary of Terms

The following are a list of common terms used in the process of radio frequency drying.

They might not be applicable to your equipment.

ANODE

The positive electrode of any electronic device; i.e. triode valve, diode etc. In a triode valve the positive charge applied to the anode, attracts electrons emitted by the *cathode* and releases a flow of positive ions, thus creating a flow of current in direct proportion to the positive voltage applied to the anode.

APPLICATOR

An alternative name for the electrode assembly (in the load circuit) which is used to transfer radio frequency energy in the form of heat directly to the product.

ATTENUATION DUCTS

Small apertures of specific dimensions located at the beginning and end of an RF heating or drying unit to prevent leakage of RF radiation outside the equipment.

CAPACITANCE

The effect obtained by placing two conducting surfaces (i.e. metal plates etc) of specific area and separated by a known distance, which when subjected to an alternating electrical field appear to store energy.

CATHODE

The negative electrode of any electronic device (i.e. triode valve, diode etc). In a triode valve the cathode is connected to ground, and when heated releases a stream of electrons (negative ions) which are attracted by the positive charge applied to the *anode*.

CAVITY OSCILLATOR

A mechanical screened construction surrounding the active components (capacitance, inductance and triode valve) which together form the RF oscillator.

CHOKE

A device having inductance which can be used to prevent the passage of specific radio frequency currents.

DIELECTRIC

Any non-conducting (i.e. insulating material) which may be polarised by the application of an RF field.

ELECTRODES

The use of two (or more) parallel metal plates (or other formed shapes) which when placed in proximity exhibit the effect of capacitance. The capacitance is used to form the high power RF field which then heats the product (a dielectric) placed between them.

EMU

Environment Management Unit.

FILAMENT

The live end of the negative electrode within a triode valve connected to a low voltage high current supply.

FINES TRAY

A removable metal tray beneath the conveyor outfeed which collects waste material from the conveyor.

FOOD DRAG

A horizontal rubber bar mounted at the infeed to the GP™ machine which prevents badly loaded product (which may be above the permitted height) from entering the GP™ machine.

GRID

In a triode valve the third (usually negative) electrode used to restrict the flow of electrons from the cathode to the anode, thereby forming a control.

H.T.

High tension - Usually taken to mean any high voltage dc supply used to supply parts of an electronic assembly. In the case of RF heaters, a high tension supply of at least 10,000 volts is generated from the ac mains voltage and fed to the triode valve anode.

INDUCTANCE

The property of a circuit (or component) to form a magnetic field when subjected to an alternating electric current.

INFEED

The area at the start of the conveyor belt where product enters the QMTI RF process.

INTERFERENCE

Radio frequency energy (from whatever source) which causes problems to other radio devices operating on the same or exact sub-multiples of a given frequency.

LDR

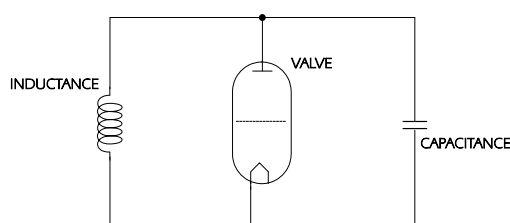
Light dependant resistor - A device being sensitive to visible (and electromagnetic) 'light' used to detect abnormal situations within RF heaters.

LOSS FACTOR

Used to quantify the amount of energy lost (dissipated) in any circuit.

OSCILLATOR CIRCUIT

An electronic circuit employing a triode valve and other components, which together form



a tuned circuit and thereby oscillates at a known design frequency. The frequency being determined by the value of the components chosen.

Figure G.1 Basic triode valve oscillator

OUTFEED

The area at the end of the conveyor belt where product leaves the QMTI RF process.

POWER DENSITY

The density of the power distribution over a given area within an RF field.

RADIO FREQUENCY

Abbreviated to RF and taken as covering all electromagnetic radiation (i.e. radio waves) from VLF (very low frequency 1kHz) to EHF (Extra high frequency up to 10GHz).

RF GENERATOR

The combination (using triode valve technology) of a high voltage power supply and associated components which together form an RF oscillator of known frequency. (Usually 27.12MHz for RF heaters).

TRIODE VALVE (VACUUM TUBE)

An electronic component having a glass outer construction and with various internal electrodes which operates on the principle of thermionic emission. i.e. the emission of electrons due to direct heating. The diagram shows a typical triode valve having an Anode, Grid and Cathode.

Appendix J Passwords

It is recommended this appendix is removed and kept for reference by a designated responsible person. The Passwords are fixed cannot be changed from the HMI

There are 4 levels of security used, each with a user name and password. The users and access are listed below

User	Login	Access	Password
Operator (DEFAULT)	No	Access to Product Menu and Status Screens Warning: No Manual operation access	No password required
SUPERVISOR S	Yes	Same menus as Operator. Full access to Pre-sets / Recipe menu and Alarms Manual operation access	578
ENGINEER E	Yes	Full access to all menus. Access to Service Setup, except some Trip Settings (view only access)	56
QMTI	Yes	Full Access without any restriction including all menus	Only available to QMTI engineers